“ADNET: An Improved Alzheimer’s Disease Recognition System Using Convolutional Neural Networks with Parallel Ensemble Strategy and Explainable AI”

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
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Declaration

It is hereby declared that

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2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.

3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.

4. We have acknowledged all main sources of help.

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Approval

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Abstract

In the evolving landscape of global health challenges, Alzheimer's disease (AD) stands as a significant and escalating concern, demanding effective diagnostic tools. This thesis navigates through the intricate dimensions of AD, emphasizing its pervasive global impact and delving into nuanced implications within Bangladesh. Against this backdrop, ADNET is introduced—a groundbreaking deep learning model leveraging Convolutional Neural Networks (CNN) and a parallel ensemble strategy for precise Alzheimer's disease detection from MRI images. Pioneering a novel noise removal approach on the imbalanced ADNI dataset from Kaggle enhances data quality, forming the foundation for ADNET's meticulous fine-tuning and development. Evaluated against diverse pre-trained models, including VGG19, InceptionV3, ResNet50, MobileNetV1, Xception, AlexNet, EfficientNetB0, and DenseNet 121, ADNET demonstrates exceptional metrics— 98.75% accuracy, 99.32% precision, 98.87% recall, 99.41% specificity, 99.09% F1 score, 99.97 AUC-ROC, and a Matthews correlation coefficient (MCC) of 0.8821 positioning it as a robust tool for accurate Alzheimer's disease diagnosis. Beyond outperforming existing models, the incorporation of eXplainable Artificial Intelligence (XAI), specifically Grad-CAM, augments ADNET's interpretability, contributing to enhanced understanding of its decision-making processes. This research not only signifies a significant contribution to Alzheimer's disease detection but also holds promise for advancing standards in clinical diagnosis and patient care on a global scale, with particular relevance in the context of Bangladesh.

Keywords: Alzheimer's disease, Deep learning, Convolutional Neural Networks (CNN), Ensemble strategy, MRI images, Pre-trained models, Evaluation metrics, eXplainable Artificial, Intelligence (XAI), Grad-CAM, Healthcare innovation.

Dedication

To our loving parents,
Your unwavering support fueled our journey. To the exceptional faculty, your guidance shaped our path. This thesis stands as a testament to your influence. Thank you for making this journey worthwhile.

With heartfelt gratitude,

Rimon Roy & Soshanto Chandra Das

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Nomenclature

The next list describes several symbols & abbreviation that will be later used within the body of the document

AD Alzheimer's Disease

$ADNI$ Alzheimer's Disease Neuroimaging Initiative

API Application Programming Interface

AUC Area Under the Curve

CNN Convolutional Neural Network

$GRAD-CAM$ Gradient-weighted Class Activation Mapping

$LIME$ Local Interpretable Model-agnostic Explanations

MCC Matthews Correlation Coefficient

MCI Mild Cognitive Impairment

MID Mild Demented

MOD Moderate Demented

MRI Magnetic Resonance Imaging

ND Non-Demented

PET Positron Emission Tomography

ROC Receiver Operating Characteristic

$SHAP$ SHapley Additive exPlanations

VGG Visual Geometry Group

VMD Very Mild Demented

XAI Explainable Artificial Intelligence

Chapter 1

Introduction

This chapter presents an in-depth overview of our thesis work, emphasizing its pivotal significance within the field. Our motivation and objectives underscore the fundamental drive behind this research endeavor. Furthermore, the thesis outline provides a structured roadmap elucidating the subsequent chapters and their contributions.

1.1 Overview

Alzheimer's disease, a progressive neurodegenerative disorder, prominently impacts memory and cognitive functions. Pathologically, it is characterized by the accumulation of aberrant protein deposits, including beta-amyloid plaques and tau tangles, within the brain [3]. The disease encompasses two main classes: Early-onset Alzheimer's, occurring before the age of 65, and Late-onset Alzheimer's, more common and typically developing after 65 [82].

Alzheimer's disease is a global health crisis, with the World Health Organization (WHO) projecting a tripling of dementia cases, where Alzheimer's plays a significant role, by 2050 [46]. In Bangladesh, as in other nations, the prevalence of Alzheimer's is rising due to factors such as an aging population and evolving lifestyle choices [10]. Timely diagnosis and intervention are imperative to address the escalating impact of the disease.

Failure to implement preventive measures against Alzheimer's could impose substantial burdens on healthcare systems, escalate caregiving responsibilities, and diminish the quality of life for affected individuals [14]. The economic and societal implications are profound, with a growing number of Alzheimer's cases posing challenges that emphasize the critical role of early detection and intervention [13].

The early diagnosis of Alzheimer's is hindered by the subtle onset of symptoms and the absence of definitive diagnostic tests [7]. Ongoing research to identify reliable and non-invasive biomarkers for early detection underscores the complexity of identifying the disease in its initial stages [52]. Furthermore, there is a pressing need for accurate and scalable diagnostic tools to effectively manage the rising number of cases [4].

In addressing these challenges, our project adopts a robust strategy utilizing a Convolutional Neural Network (CNN) with a parallel ensemble approach [20], [51]. The incorporation of Explainable Artificial Intelligence (XAI), particularly Grad-CAM (Gradient-weighted Class Activation Mapping), enhances the transparency of the model's decisions—a crucial factor in the medical field where interpretability is as pivotal as prediction accuracy [48].

The parallel ensemble strategy, combining predictions from multiple sub-models, contributes to improved overall performance and robustness [51]. This methodology aligns with recent advancements in deep learning for medical image analysis, presenting a promising avenue for accurate and interpretable Alzheimer's disease detection [41].

1.2 Importance of this study

In the realm of Alzheimer's disease research, our study introduces a groundbreaking model named **ADNET**, tailored for the intricacies of the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset—an imbalanced yet pivotal resource boasting 6400 images distributed across four classes[116].

ADNET represents a paradigm shift in Alzheimer's prediction models, uniquely designed to address the challenges posed by imbalanced datasets. Its innovative parallel ensemble strategy[76] and noise-removing techniques, particularly in brightness and outlier considerations, empower the model to predict all classes with unparalleled precision.

What sets ADNET apart is not just its predictive prowess but its adaptability and resilience in navigating the complexities of the ADNI dataset[116]. By strategically integrating a noise-removing approach[58], ADNET demonstrates a versatility that is crucial in the unpredictable landscape of neurodegenerative diseases.

To ascertain the superiority of ADNET, we conduct a rigorous comparative evaluation against established pretrained models. Spanning a spectrum of evaluation metrics, this analysis unequivocally showcases ADNET's exceptional performance, solidifying its status as a frontrunner in Alzheimer's prediction.

Beyond its predictive capabilities, ADNET prioritizes transparency and interpretability through the incorporation of eXplainable Artificial Intelligence (XAI), specifically leveraging Grad-CAM. This addition not only enhances the model's understanding but also empowers healthcare professionals and researchers, bridging the gap between model predictions and clinical insights.

In essence, our study champions ADNET as an unparalleled solution in the landscape of Alzheimer's prediction models. Its prowess in handling imbalanced datasets, coupled with its transparency through XAI, propels the field forward, offering a transformative tool for precise and interpretable Alzheimer's disease predictions.

1.3 Current Scenario and Motivation

Alzheimer's disease stands as a pervasive global health challenge, with the prevalence of dementia, including Alzheimer's, steadily increasing on a worldwide scale.Someone in the world develops dementia every 3 seconds[118], underlining the relentless pace at which this neurodegenerative condition is escalating globally. Alzheimer's disease, the predominant form, affects over 55 million individuals worldwide, with more than 60% in low- and middle-income countries. The prevalence is alarming, with nearly 10 million new cases reported annually, underscoring its increasing global burden. Alzheimer's disease, contributing to 60–70% of cases, stands as the most prevalent form and ranks as the seventh leading cause of death globally. The economic toll is staggering, with the cost of dementia care reaching a monumental 1.3 trillion USD in 2019. Informal caregivers, dedicating an average of 5 hours daily, bear half of these costs. Furthermore, women, who provide 70% of care hours, experience higher disability-adjusted life years and mortality rates, emphasizing the urgency for comprehensive global efforts. These efforts should encompass research, healthcare interventions, caregiver support, and policy development to address this pervasive and debilitating condition[117][118].

Turning our focus to Bangladesh, a recent study by the International Centre for Diarrhoeal Disease Research highlights a significant local concern. In this South Asian nation, one in 12 elderly individuals, or 8.0%, suffers from dementia, with a higher prevalence among women, those without education, and in the Rajshahi division. This marks the first national dementia survey in Bangladesh, emphasizing the urgent need for resources and policies to address the growing prevalence. The study projects a doubling of dementia cases by 2051, from 12 lakh in 2023 to 24 lakh. Currently, inadequate treatment facilities and the lack of a comprehensive national dementia care strategy pose considerable challenges in managing the increasing burden of dementia in Bangladesh[112].

In the context of Bangladesh, a nation experiencing a parallel rise in age-related diseases, including Alzheimer's, there is a unique set of challenges that demands tailored solutions. Limited resources and evolving demographics present hurdles in effectively managing the growing incidence of Alzheimer's. This local scenario serves as a poignant motivation for our thesis project. We aim to contribute not only to the broader global fight against Alzheimer's but also to address the specific needs of Bangladesh's healthcare landscape. By developing a model adept at early detection and precise classification, we aspire to provide insights that can assist healthcare professionals in navigating the complexities of Alzheimer's within the constraints of available resources.

The imbalanced nature of the ADNI dataset further propels our project's motivation, emphasizing the critical need to address complexities in imbalanced medical datasets for building robust predictive models. Innovative strategies, including a parallel ensemble approach and noise removal techniques, drive our commitment to overcoming these challenges, enhancing the adaptability of our model, ADNET.

A pivotal aspect of our project involves comparative evaluation and benchmarking against established pretrained models. Rigorous evaluation metrics aim to validate the effectiveness of ADNET, showcasing its superiority and making a valuable contribution to Alzheimer's prediction models.

Our dedication to transparency and interpretability fuels the integration of Explainable Artificial Intelligence (XAI), particularly Grad-CAM. Understanding ADNET's decisions is paramount for establishing trust and facilitating acceptance within the medical community. By incorporating XAI, we aspire to advance medical image analysis and empower healthcare professionals with insights into ADNET's classifications.

In summary, our thesis project is motivated by the dual objectives of contributing to the global fight against Alzheimer's and addressing the unique challenges faced by Bangladesh. Through innovative methodologies and a focus on transparency, we aim to provide a valuable tool for early detection and precise classification of Alzheimer's disease, significantly impacting healthcare practices.

1.4 Research Objectives

Alzheimer's disease (AD) stands as a critical global health concern, posing substantial risks and lacking definitive remedies. Its impact extends beyond individual health, intersecting with global health crises like COVID-19. In Bangladesh, AD not only affects health indices but also exerts economic repercussions. Our research objectives encompass diverse dimensions:

1. **Raising Awareness:** This study aims to elevate awareness about AD, facilitating early diagnosis to predict dementia's progression. Empowering individuals with knowledge about potential risks enables proactive measures to avert the advanced stages of AD.

2. **Addressing Misconceptions:** In Bangladesh, misconceptions surrounding AD often link it solely to old age. We strive to alter this perception, emphasizing the seriousness of the disease and advocating for focused attention to curb its prevalence.

3. **Early Prediction:** Focusing on early-stage prediction, our approach seeks to enable proactive measures, ensuring timely interventions to mitigate the severe consequences associated with AD.

4. **Enhancing Care:** Through improved accuracy and justifiable insights, our research aims to enhance the physical and mental well-being of AD patients and their families, fostering better caregiving strategies.

5. **Improved Communication:** Our study endeavors to improve communication channels, fostering informed dialogue among AD patients and relevant stakeholders, leading to more effective management strategies.

6. **Introducing a Novel Model:** We are introducing a novel model based on Convolutional Neural Networks (CNN), aiming to significantly contribute to the research field. This innovative approach seeks to advance the accuracy and efficiency of Alzheimer's disease prediction, offering new perspectives for future research in this domain.

1.5 Contributions

In our pursuit of unraveling the intricacies of Alzheimer's disease detection, we proudly present ADNET—a groundbreaking model meticulously crafted to navigate the complexities of the ADNI dataset. Rooted in Convolutional Neural Networks (CNN) and fortified by a parallel ensemble strategy, ADNET goes beyond convention. What sets it apart is the fusion of cutting-edge innovations, particularly in noise removal techniques, propelling it to redefine the landscape of medical image analysis.

Key Contributions:

1. **Navigating the Imbalances:** Rather than resorting to artificial balancing, we embraced the real-world intricacies of an imbalanced dataset. ADNET's deliberate focus on accuracy amid inherent imbalances directly addresses the pragmatic challenges inherent in medical image analysis.

2. **Pioneering Noise Removal Techniques:** ADNET stands as a trailblazer in the realm of noise removal for the ADNI dataset. Introducing avant-garde outlier removal and brightness adjustment techniques, it elevates the robustness of Alzheimer's disease detection, addressing unique challenges posed by medical images.

3. **ADNET's Artistry in Model Craftsmanship:** A novel contribution to the field, ADNET unfolds with a parallel ensemble strategy nested within a CNN framework. Tailored for nuanced Alzheimer's disease stage classification, its ambition extends beyond existing models, embracing the distinctive characteristics of medical imaging datasets.

4. **Comparative Brilliance:** Rigorous comparisons with eight pretrained models using seven distinct evaluation metrics underscore ADNET's brilliance. This in-depth analysis offers a panoramic view, illuminating the efficacy of ADNET in comparison to established models.

5. **Transparency through Explainable AI (XAI):** ADNET stands as a beacon of transparency, employing Explainable Artificial Intelligence (XAI) with the prowess of Grad-CAM. Beyond mere validation, it delves into interpretable insights, shedding light on crucial regions for Alzheimer's disease classification and fortifying the trustworthiness of its decisions.

6. **Holistic Context in Comparison:** Our endeavor extends beyond model development—it is contextualized through a comprehensive comparative analysis with existing Alzheimer's disease detection research. This holistic perspective positions ADNET as an unparalleled contribution, reshaping the ever-evolving landscape of Alzheimer's disease detection.

1.6 Thesis Outline

This thesis aims to construct a robust model for predicting various stages of Alzheimer's Disease (AD) and discerning associated risk factors in its diagnosis. The primary goal is to develop an accurate model, underscored by a focus on providing visual interpretations of model outputs.

In the introductory section(**Chapter 1**), the study provides a comprehensive overview, highlighting the significance of the research within the context of Alzheimer's Disease (AD). Emphasis is placed on the relevance of the study to the current situation of AD in Bangladesh, exploring the motivations that drive this research. The section delineates the research objectives and contributions, setting the stage for the subsequent chapters.

The second chapter delves into an exhaustive review of existing works in the realm of Alzheimer's Disease Classification and prediction. Significant achievements by researchers in the field are highlighted, and the chapter critically addresses the limitations inherent in current methodologies. This chapter lays the foundation for subsequent analyses, informed by the insights gained from prior research endeavors.

Chapter 3 provides a detailed background analysis of key components essential for understanding the study. Covering MRI imaging, Convolutional Neural Networks (CNNs), and a spectrum of pre-trained models, including AlexNet, VGG19, Inception V3, ResNet50, MobileNet, DenseNet121, Xception,and EfficientNet B0 , the chapter aims to establish a comprehensive understanding. Additionally, an exploration of evaluation metrics and Explainable AI (XAI) techniques enriches the reader's comprehension.

Chapter 4 unfolds the materials employed in the study and outlines the proposed methodology. Discussions encompass data acquisition, preprocessing steps such as noise removal and normalization, and the implementation of deep learning models, including ResNet50, VGG19, InceptionV3, and the bespoke model. The chapter offers detailed descriptions of the model architectures and individual layers.

Dedicated to presenting and analyzing experimental results, **Chapter 5** showcases evaluation metrics and visual representations. A comparative analysis with other models illuminates the superior performance of our proposed model. Further, an in-depth examination of Explainable AI (XAI) results sheds light on the interpretability of our model's decisions.

The final chapter, **Chapter-6** encapsulates conclusions drawn from our study, acknowledging inherent limitations and proposing directions for future research endeavors. Summarizing key findings, this chapter underscores the study's significance in the realm of Alzheimer's Disease detection, paving the way for continued exploration and refinement.

Chapter 2

Literature Review

This chapter comprehensively explores related works in the field, accentuating their relevance and significance. By synthesizing existing research, we aim to establish a comprehensive foundation for our thesis work while identifying gaps that necessitate further exploration.

2.1 Related Works

Alzheimer's disease (AD) remains an intractable, progressively debilitating neurological condition and represents the most prevalent cause of dementia on a global scale. The impact of AD is burgeoning, especially within the demographic of individuals aged 65 and older. Projections from esteemed institutions like the UN Aging Program and the US Centers for Disease Control and Prevention indicate a staggering rise in this demographic, reaching nearly 1 billion by 2030, up from 420 million in 2000 [15]. The implication is dire: without early detection, the implications of this disease can swiftly evolve into a significant public health concern.

Researchers have extensively explored statistical and machine learning models in the quest to detect Alzheimer's disease promptly. Early identification holds immense promise in mitigating the abnormal degradation of brain function, curbing soaring medical expenses, and enhancing treatment outcomes. Recent setbacks in Alzheimer's disease research underscore the pivotal role of early intervention and diagnosis in bolstering the efficacy of treatments [80]. The integration of diverse neuroimaging techniques, bolstered by advancements in machine learning, has significantly bolstered diagnostic accuracy across various dementia subtypes [55].

A multitude of modalities—ranging from Magnetic Resonance Imaging (MRI) to Positron Emission Tomography (PET) and genotype sequencing—have been pivotal in this pursuit. However, the time-intensive nature and potential side effects, particularly the radioactive impact of modalities like PET, necessitate a discerning choice. In this context, the MRI modality stands out due to its versatility in imaging, exceptional tissue contrast, absence of ionizing radiation, and its capacity to provide comprehensive insights into human brain anatomy [31], [32].

Deep Learning (DL) models, characterized by intricate multi-level architectures, have demonstrated exceptional prowess in extracting intricate information from images [34]. Convolutional Neural Networks (CNNs), leveraging the computational advantage of Graphics Processing Units (GPUs), have significantly expedited image analysis processes. This amalgamation of technology has extended its reach into various domains—be it Medical, Agriculture, or Communication—ushering in transformative progress [18]. In the medical arena, researchers have harnessed MRI scans and deep learning to discern patterns indicative of Alzheimer's disease, with a spate of publications delineating the classification capabilities of MRI-based models [18], [61].

The study tackled diagnostic limitations in AD through a deep learning approach featuring stacked auto-encoders, exhibiting significant performance gains and the ability to analyze multiple classes. Despite not specifying the dataset used, the method demonstrated enhanced diagnostic performance and reduced reliance on labeled samples [17]. The discussion underscored the importance of accurate early-stage diagnosis and addressed inherent limitations in existing models [17].

Addressing the automatic recognition of AD, Luo, Li, and Li (2017) focused on a deep learning algorithm based on Convolutional Neural Network (CNN) utilizing the three-dimensional topology of the brain. The study achieved high AD recognition accuracy with sensitivity and specificity values, showcasing the potential of deep learning in AD diagnosis [45]. Addressing multi-class AD detection, Islam and Zhang (2017) presented a deep convolutional network designed for Brain MRI data. The study showcased successful multi-class AD detection and classification using the proposed deep learning model, emphasizing the potential of deep learning techniques for improving AD diagnosis accuracy [39].

Introducing two distinct learning models, Dolph et al. (2017) proposed multiclass deep learning classification of Alzheimer's disease using texture and associated features extracted from structural MRI. The models, involving subcortical area-specific feature extraction, feature selection, and stacked auto-encoder deep neural network, achieved competitive classification accuracies and demonstrated potential for rapid, objective, and non-invasive assessment of AD [36].

Addressing early diagnosis challenges, Ebrahimighahnavieh et al. (2018) developed a hybrid classification strategy utilizing Random Forest for feature selection and Deep Neural Network for classification. The study achieved the third position for overall accuracy (38.8%) in the Kaggle challenge, highlighting the significantly higher accuracy of Deep Neural Networks compared to other machine learning strategies. The proposed hybrid strategy contributed to the automated prediction of Alzheimer's Disease [53]. Liu, Cai, Pujol, and Kikinis (2019) addressed the challenge of predicting the progression from Mild Cognitive Impairment (MCI) to Alzheimer's disease (AD) dementia. The authors developed and validated a deep learning method based on MRI scans, achieving a concordance index of 0.762-0.781. The proposed model effectively predicted individual subjects' progression and offered a cost-effective means for prognosis. Combining the deep learning-based progression risk with baseline clinical measures improved overall performance [63].

In their work on early AD diagnosis, Ji, Liu, Yan, and Klette (2019) utilized Convolutional Neural Networks (ConvNets) with MRI image slices of gray and white matter. Employing ensemble learning methods, the authors achieved high accuracy rates, specifically 97.65% for AD/mild cognitive impairment and 88.37% for mild cognitive impairment/normal control [59]. Ding, Sohn, Kawczynski, and Trivedi (2019) developed a deep learning algorithm for early AD prediction using 18F-FDG PET. The InceptionV3 architecture achieved an area under the ROC curve of 0.98, with 82% specificity at 100% sensitivity, predicting AD an average of 75.8 months prior to the final diagnosis [56].

Toschi and the Alzheimer's Disease Neuroimaging Initiative (2019) presented a deep learning architecture for identifying high-risk MCI patients. The model, based on dual learning and 3D separable convolutions, achieved high performance with an AUC of 0.925 and a 10-fold cross-validated accuracy of 86% for predicting MCI patients developing AD within 3 years [70]. The systematic review by Jo, Nho, and Saykin (2019) explored deep learning approaches for the diagnostic classification of AD using neuroimaging data. The study reviewed 16 relevant studies, outlined key findings, and emphasized the use of multimodal neuroimaging and fluid biomarkers for improved classification performance. The authors acknowledged the evolving nature of AD research using deep learning [60].

Applying a CNN and LeNet-5 architecture, Shen et al. (2019) achieved high accuracy in distinguishing Alzheimer's brains from normal, healthy brains using functional MRI (fMRI) data. The study emphasized the effectiveness of CNNs in extracting shift and scale invariant features for distinguishing clinical data from healthy data in fMRI [67].Proposing a classification method based on hippocampal texture, So, Madusanka, Choi, and Choi (2019) achieved an 85% accuracy in AD-NC classification. The method, developed using image processing, texture analyses, and deep learning, demonstrated the accuracy of the proposed approach in diagnosing early Alzheimer's disease [68].

Applying a multimodal recurrent neural network, Lee et al. (2019) predicted conversion from Mild Cognitive Impairment (MCI) to Alzheimer's disease. Integrating cross-sectional neuroimaging biomarkers, longitudinal cerebrospinal fluid, and cognitive performance biomarkers, the study achieved high accuracy and highlighted the advantage of incorporating longitudinal multi-domain data in prediction models [62]. Conducting an exploratory data analysis, Martinez-Murcia et al. (2019) utilized deep convolutional autoencoders to extract features from MRI images. The study successfully extracted imaging-derived markers predicting clinical variables with high correlations and achieved over 80% accuracy for the diagnosis of Alzheimer's disease [64].

A recent paper in October 2020 demonstrated the use of neuroimaging techniques leveraging sMRIs, capitalizing on cerebral shrinkage dynamics to achieve 91% accuracy in AD diagnosis without patient history records [8]. Deep learning models that forego extensive preprocessing have delivered notable accuracies, up to 96.0% for AD classification and 84.2% for MCI conversion prediction [65]. Employing spectral convolution-based ChebNet 8, another system classified subjects in the ADNI dataset with 92.44% accuracy [26].

A proposed 12-layer CNN model introduced an innovative approach, yielding an accuracy of 97.75%. The study explored various image segmentation methods, emphasizing the best-performing approach, likely achieving accuracy closer to 97.75% [105]. CNN coupled with SVM detected AD from structural MRI data with around 96% accuracy [57]. Amin-Naji et al. advocated for Siamese CNNs, specifically using axial sMRI views, resulting in an exceptional accuracy of 98.72% on the OASIS dataset, considered the highest to date [43].

Investigating early detection of Alzheimer's disease (AD), Puente-Castro, Fernandez-Blanco, and Pazos (2020) developed a system utilizing sagittal magnetic resonance images (MRI) and Transfer Learning (TL). The study demonstrated the effectiveness of sagittal MRI in identifying AD in its early stages, achieving comparable results to commonly used orientations. The use of TL highlighted the potential of pre-trained models for feature extraction and model training [77]. In their systematic literature review, the authors explored over 100 articles on AD detection using deep learning. Ebrahimighahnavieh, Luo, and Chiong (2020) outlined the most recent findings and trends, including useful biomarkers, pre-processing steps, and various deep learning models. The review discussed considerations for training parameters and highlighted unresolved issues and challenges in AD detection with deep learning [74].

The comprehensive review by Salehi, Baglat, and Gupta (2020) covered various deep learning algorithms for AD diagnosis using MRI brain images. The authors emphasized the application of Convolutional Neural Networks (CNN) and advocated for advanced deep learning techniques and the combination of datasets for early prediction of AD. The review concluded by recommending the use of advanced deep learning techniques in different datasets to increase efficiency in AD prediction [78].

Salehi et al. (2020) introduced a deep learning-based framework for AD classification using tau PET scans. Leveraging a 3D CNN and layer-wise relevance propagation (LRP), the model achieved high accuracy. The study identified informative features and associated brain regions, shedding light on regions crucial for AD classification. The correlation between AD probability scores and tau deposition in the medial temporal lobe in Mild Cognitive Impairment participants further emphasized the framework's relevance and potential clinical impact [79].

Bari Antor, Jamil, and Mamtaz [94] conducted a comparative analysis of machine learning algorithms for Alzheimer's disease prediction in 2021, with support vector machine providing the best results. Tanveer, Richhariya, and Khan [73] provided a comprehensive review of machine learning techniques for Alzheimer's disease diagnosis in 2021. Addressing the critical need for early diagnosis of Alzheimer's disease (AD), Raees and Thomas (2021) proposed an automated deep learning-based system. The study classified a large MRI dataset into Mild Cognitive Impairment (MCI), AD, and Normal classes, achieving high accuracy (80-90%) in AD prediction. The authors emphasized the clinical, social, and economic importance of early AD diagnosis and advocated for computational-automated machine learning tools [90].

Balne and Elumalai (2021) contributed to AD diagnosis through a survey comparing machine learning models. The study highlighted the significance of multimodal studies and combining features from different structural MRI analysis techniques for improved accuracy. The authors provided a comprehensive overview of machine learning algorithms for disease diagnosis [83]

In their pursuit of comprehensive Alzheimer's Disease (AD) staging, Venugopalan, Tong, and Hassanzadeh (2021) utilized multi-modal data, including imaging (MRI), genetic (SNPs), and clinical test data. Employing stacked denoising auto-encoders for clinical and genetic data and 3D-convolutional neural networks (CNNs) for imaging data, the authors demonstrated the superior performance of deep models over shallow models. The study highlighted the significance of integrating multi-modality data, showing improved classification metrics [95].

In their work on AD detection, Buvaneswari and Gayathri (2021) presented a deep learning-based segmenting method using SegNet for structural MRI. Classifying AD and dementia conditions with ResNet-101, the authors achieved high sensitivity and accuracy over 240 ADNI sMRI. The study demonstrated the effectiveness of deep learning in detecting subtle brain morphological changes due to AD [84].

The study by the Alzheimer's Disease Neuroimaging Initiative focused on addressing continuous progression in AD and developing prognostic biomarker signatures. Using deep learning algorithms to differentiate patients with AD, mild cognitive impairment (MCI), or no evidence of dementia, the authors integrated features with other baseline biomarkers for developing an AD prognostic signature [86].

Utilizing 2D CNN and VGG16, Mggadadi et al. (2021) achieved accuracies of 67.5% and 70.3% for Alzheimer's disease prediction on MRI images. The study concluded that VGG16 outperformed 2D CNN, emphasizing the potential of deep learning, particularly VGG16, in improving Alzheimer's disease prediction [88].

Presenting a new method based on deep learning and image processing, Saratxaga et al. (2021) achieved high balanced accuracy for image-based automated Alzheimer's disease diagnosis. Surpassing state-of-the-art proposals, the study demonstrated the efficacy of the proposed solution using the OASIS collection [92]. In 2021, Suriya and Chandran proposed DEMNET, a model adept at discerning four types of dementia from an imbalanced dataset of 6400 MR Images sourced from Kaggle. Their use of the SMOTE technique for dataset balancing culminated in a 95.23% accuracy [80]. Similarly, Masud and Manowarul devised a DL model combining Inception V3 and a Custom CNN Model, achieving an accuracy of 97.31% while classifying the four stages of dementia [34].

A collaborative research effort between the UK and China implemented a deep convolutional neural network (DCNN) to extract crucial features from sMRIs. They utilized a strictly processed pipeline and resliced each volume before input into the DCNN. The study achieved significant results, identifying four stages of Alzheimer's with an average accuracy of 94.8% for LMCI versus EMCI, 94.5% for NC versus LMCI, 97.81% for EMCI versus AD, 97.2% for LMCI and AD, and 96.9% for NC versus AD [105].

Acknowledging limited datasets and their impact on CNN training accuracy, the paper proposed a two-part unsupervised deep learning method. This approach involves an unsupervised ML method followed by unsupervised classification employing PCANet, a CNN that takes input from three orthogonal MRI views, and a k-means clustering algorithm, resulting in an average accuracy of 92.5% [47].

In their pursuit of early AD detection, Helaly, Badawy, and Haikal (2022) employed CNNs, presenting an end-to-end framework utilizing simple CNN architectures achieving 93.61% and 95.17% accuracy for 2D and 3D multi-class AD stage classifications. Transfer learning with the VGG19 model yielded 97% accuracy [helaly2022deep]. The study emphasized the practicality of remote AD checks during the pandemic and the efficiency of CNN architectures [100].

Hamdi, Bourouis, and Rastislav (2022) developed a computer-aided diagnosis system based on a Convolutional Neural Network (CNN) for AD using 18FDG-PET images. The proposed CNN achieved high accuracy (96%), sensitivity (96%), and specificity (94%) in discriminating normal control from AD patients [99].

Hojjat, Tofighi, and the Alzheimer's Disease Neuroimaging Initiative (2016) explored challenges in distinguishing AD from healthy brain data in older adults using neuroimaging data. Employing deep learning pipelines on a high-performance computing platform, the study achieved high and reproducible accuracy rates for distinguishing Alzheimer's MRI and fMRI from normal controls [27].

Bustamante and the Alzheimer's Disease Neuroimaging Initiative (2022) investigated the application of a voxel-based deep learning approach using structural MRI to detect neurodegeneration in prodromal AD. The study outperformed other neuroimaging biomarkers, showcasing the practical advantages of using deep learning to extract biomarker information from conventional MRIs [98].

In their extensive survey, Khojaste-Sarakhsi and Haghighi (2022) conducted a comparative analysis of approximately 100 papers published since 2019 on deep learning-based architectures for Alzheimer's Disease (AD) diagnosis. They explored various models, including CNNs, RNNs, and generative models, and delved into emerging topics such as explainable models, normalizing flows, graph-based architectures, self-supervised learning, and attention mechanisms. The work addressed challenges in data, methodology, and clinical adoption, providing valuable insights and future recommendations for AD diagnosis [101].

Presenting a novel approach, Liu et al. (2022) developed a 3D deep convolutional neural network for accurately diagnosing mild Alzheimer's disease dementia. They compared this model with a reference model based on brain region volumes and thickness. Validated on internal and external cohorts, the deep-learning model exhibited superior accuracy and speed, particularly in forecasting progression from mild cognitive impairment. The study contributes to the advancement of accurate and efficient AD diagnosis using structural MRIs [102].

Proposing an innovative end-to-end approach, Saleem et al. (2022) systematically evaluated CNN models for AD diagnosis. Their Two-stage Random RandAugment data augmentation strategy and the use of Grad-CAM++ for heatmap generation enhanced classification performance and transparency. The approach outperformed state-of-the-art methods, providing a robust solution for automated AD diagnosis using structural MRIs [106].

Presenting a deep learning algorithm, Xu et al. (2022) achieved high accuracy in predicting individual diagnoses of Alzheimer's disease and mild cognitive impairment based on a single structural MRI scan. The algorithm's robustness across imaging protocols and scanners highlighted its potential for individualized AD diagnosis [107].

Investigating deep learning models for computer-aided diagnosis (CAD) of Alzheimer's disease, Loddo, Buttau, and Di Ruberto (2022) developed a fully automated deep-ensemble approach. The approach demonstrated high accuracy in binary and multi-class classification tasks across multiple datasets, showcasing its potential for robust and reliable CAD systems [103].

Miah and Rashid [89] proposed a CNN-based model with effective dimensionality reduction, achieving a 93% accuracy using Random Forest with Random Projection (RFRP). Abed, Fatema, and Nabil [71] explored AD prediction using CNN with pre-existing architectures and transfer learning, achieving notable accuracies with VGG19, Inception v3, and ResNet50. Hussain, Hasan, and Hassan [75] introduced a deep learning binary classification model with a 12-layer CNN, achieving an outstanding accuracy of 97.75% and outperforming pre-trained CNN models.

Kaur, Gupta, Singh, and Gupta [72] applied a deep convolutional neural network to detect Alzheimer's disease in 2022, achieving a high accuracy of 98.9% for the moderate demented class.Karthiga, Sountharrajan, and Nandhini [105] utilized AdaBoost with SVM for Alzheimer's disease prediction in 2022, achieving an accuracy of 95.66%. Kavitha, Mani, Srividhya, and Khalaf [109] proposed a model for early-stage Alzheimer's disease prediction in 2022, achieving a test accuracy score of 83%.

Aruchamy, Haridasan, and Verma [28] proposed a method using machine learning classifiers on 3D MR images in 2022, achieving an accuracy of 90.9%.Kumar, Azath, and Velmurugan [93] utilized a hybrid AI approach integrating SVM with CNN for Alzheimer's disease prediction in 2022, focusing on different disease stages.Joshi, Shenoy, and Simha [96] conducted a study on the classification of Alzheimer's and Parkinson's disease, identifying influencing risk factors. Uddin, Alam, and Aryal [113] developed a machine learning model in 2022, with the voting classifier attaining the highest validation accuracy of 96% for the AD dataset.

El-Sappagh et al. [91] focused on AD progression detection with early fusion of multimodal data, demonstrating the superiority of a random forest model. Ali, Islam, Haque, and Das [81] proposed the Modified Random Forest (m-RF) algorithm in 2023, achieving a 96.43% accuracy and surpassing other traditional algorithms. Mamun, Shawkat, and Ahammed [104] applied four deep learning models in 2024, with CNN outperforming others and achieving an accuracy of 97.60%, recall of 97%, and AUC of 99.26%.

Mahmud, Ali, and Shahriar [111] utilized CNN classifiers with preprocessing steps in 2023, achieving 88.20% accuracy and demonstrating high potential for early AD detection. Joshi, Shenoy, and Simha [5] proposed a model for the classification of Alzheimer's and Parkinson's disease, identifying influencing risk factors.

Rangaswamy, Dharshini, and Yesudhas [87] developed VEPAD in 2023, predicting the effect of variants associated with Alzheimer's disease, achieving an accuracy of 81.21% on cross-validation. Lokesh, Challa, and Satwik [85] developed a deep learning model for early Alzheimer's disease detection in 2023, achieving a maximum accuracy of 99.96%. Salunkhe, Bachute, Gite, and Vyas [66] focused on the classification of Alzheimer's disease patients using texture analysis and machine learning in 2023, achieving an accuracy of 90.2%.

Kapoor, Kapoor, and Shukla [97] applied machine learning algorithms on ADNI-Longitudinal data in 2023, with Random Forest exhibiting the highest accuracy of 99.8%.Tanveer, Richhariya, and Khan [40] provided a comprehensive review of machine learning techniques for Alzheimer's disease diagnosis in 2021.

These studies collectively highlight the significant strides in utilizing deep learning and CNNs for Alzheimer's diagnosis, showcasing diverse approaches, datasets, and methodologies, advancing the field's diagnostic accuracy and potential.

Chapter 3

Background Analysis

This chapter delves into the background analysis essential to comprehend the core components of our thesis work. It provides detailed insights into MRI technology, as well as comprehensive explanations of CNN architectures such as Alexnet, VGG19, InceptionV3, ResNet50, MobileNet, DenseNet121, Xception and EfficientNet. Additionally, this section elucidates the evaluation matrices utilized, offering a robust framework for assessing and validating the outcomes of our research.

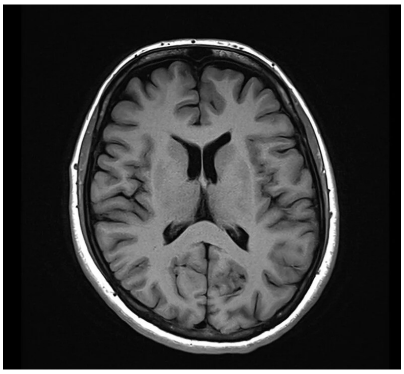
3.1 MRI

A non-invasive imaging method that creates three-dimensional, finely detailed anatomical images is called magnetic resonance imaging (MRI). It is frequently employed in the diagnosis, monitoring, and detection of diseases. It is based on advanced technology that rotates protons in the water that makes up living tissues, stimulates them, and detects when their direction changes[119].

At the heart of Magnetic Resonance Imaging (MRI) lies the intricate dance of charged particles, particularly hydrogen nuclei, in response to a magnetic field. Nuclei with an odd number of protons and/or neutrons exhibit characteristic precession, creating a minute magnetic moment owing to their charge. As the human body enters a robust magnetic field, free hydrogen nuclei align themselves with this field, engaging in Larmor precession—a motion akin to gyroscopes. The frequency of this Larmor precession, termed the Larmor frequency, correlates directly with the strength of the applied magnetic field (B), determined by the gyromagnetic ratio (γ), a nuclei-specific constant. For hydrogen nuclei, the Larmor frequency is succinctly expressed as

$$\text{Larmor frequency} = \gamma \times B.$$

This underlying principle not only forms the essence of MRI but also fuels its technological prowess. By manipulating and detecting the magnetic moments generated by aligned hydrogen nuclei, MRI achieves unparalleled non-invasive and high-resolution imaging, making it an indispensable tool in medical diagnostics[115].



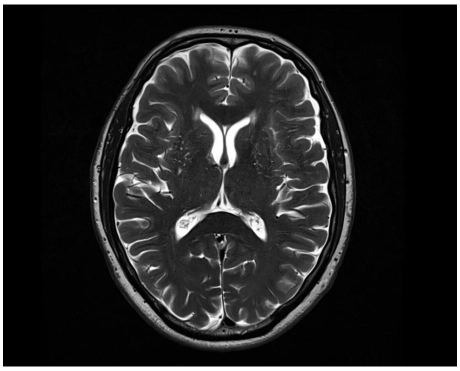


Figure 3.1: T1 and T2 Weighted Images Comparison

In the realm of clinical application, MRI stands as an invaluable tool, proven to be a pivotal method for both qualitative and quantitative diagnosis. Its effectiveness extends across a spectrum of organs, including the brain, spine, head and neck, breast, liver, kidney, prostate, cervix, and bones and joints. Beyond diagnosis, MRI plays a crucial role in monitoring the treatment efficacy of various diseases. Its versatility and precision make it a cornerstone in medical practice, providing detailed insights into diverse anatomical structures and contributing significantly to comprehensive patient care[119].

At the forefront of Magnetic Resonance Imaging (MRI) sequences are the ubiquitous T1-weighted and T2-weighted scans, each offering a distinctive perspective. T1-weighted images, crafted with short echo time (TE) and repetition time (TR), derive their contrast and brightness primarily from the tissue's T1 properties. Conversely, T2-weighted images come to life through extended TE and TR times, where the interplay of contrast and brightness is predominantly influenced by the T2 properties of the tissue. This nuanced orchestration of timing and tissue characteristics underscores the artistry of MRI, allowing clinicians to unveil rich details and insights across diverse medical scenarios with precision and diagnostic finesse[114]

T1-weighted images, characterized by shorter TR and TE times, present contrasting tissue features compared to T2-weighted images, which employ longer TE and TR times. The subtle differences in tissue appearance play a crucial role in distinguishing brain structures and anomalies.

Tissue	**T1-weighted**	**T2-weighted**
Cerebrospinal Fluid (CSF)	Dark	Light
White Matter	Light	Dark Gray
Cortex	Gray	Light Gray
Fat	Bright	Light
Inflammation	Dark	Bright

Table 3.1: Contrasting Tissues in T1 and T2 Weighted Images

Our exploration delves into a Kaggle dataset housing T1-weighted MRI images categorized across multiple dementia stages. Samples representing each class are exhibited below:

These images are invaluable, providing a visual narrative of various dementia stages, enabling in-depth analysis and diagnostic insights into brain conditions. The intricate details captured by MRI technology serve as a gateway to understanding and addressing neurological health concerns.

3.2 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs), introduced by LeCun and colleagues[9], have revolutionized medical image analysis, enabling automated feature extraction directly from raw pixel values. These networks, initially developed for handwritten digit recognition[2], are now indispensable in various medical imaging tasks[42]. Krizhevsky et al.'s work[9] on ImageNet classification further propelled CNNs into mainstream applications. This paper explores the foundational aspects, architecture, and applications of CNNs, particularly in medical image classification and disease detection[42].

3.2.1 CNN Architecture

The architecture of a CNN comprises layers for input representation, convolutional operations for feature extraction[2], and pooling layers to reduce spatial dimensions[9]. The Max Pooling operation aids in computational efficiency[9]. Fully connected layers provide a global understanding of learned features, forming a comprehensive framework for image analysis[42].

Input Layer: Represents the raw input image.

Convolutional Layers: These layers utilize convolutional operations to detect intricate patterns within the input image[2]. The convolution operation is defined as:

$$S(i,j) = (K * I)(i,j) = \sum_m \sum_n K(m,n) I(i-m, j-n) \tag{3.1}$$

Where S is the output, I is the input image, and K is the kernel or filter.

Pooling Layers: Following convolution, pooling layers, such as max pooling, reduce the spatial dimensions, aiding in computational efficiency. The Max Pooling operation is defined as:

$$P(i,j) = \max_{m,n} I(i \cdot s + m, j \cdot s + n) \tag{3.2}$$

Where P is the output of the pooling layer, I is the input, and s is the stride.

Fully Connected Layers: These layers connect every neuron in one layer to every neuron in the next layer, providing a global understanding of the learned features.

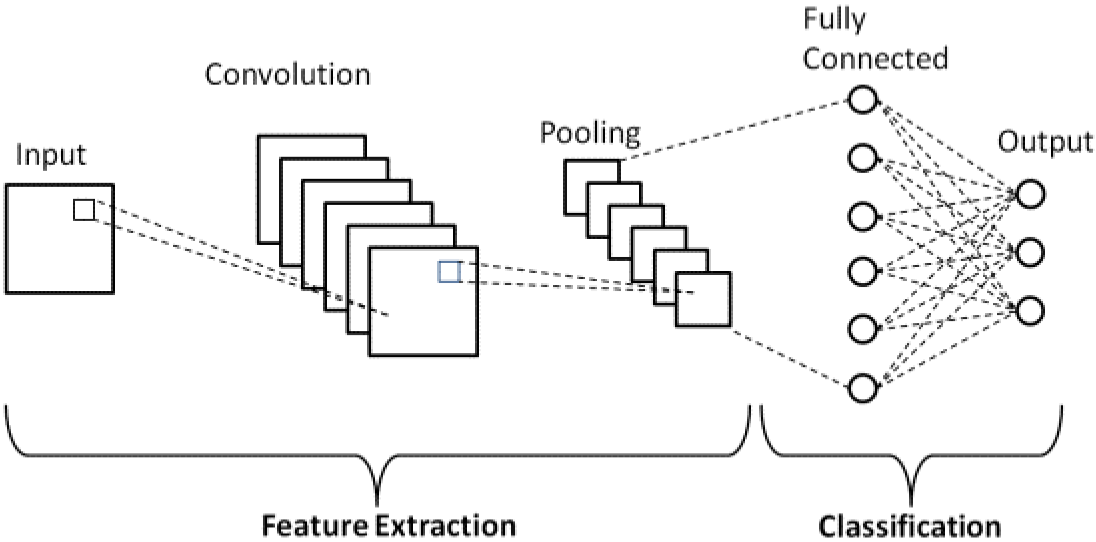


Figure 3.2: Simple architecture of a Convolutional Neural Networks.

3.3 AlexNet

AlexNet, a groundbreaking convolutional neural network (CNN) architecture, was introduced by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton in 2012, marking a significant milestone in deep learning [9]. This architecture not only secured victory in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) but also demonstrated its potential for image classification tasks[9]. In the context of medical image recognition, AlexNet has proven to be a valuable asset, leveraging its ability to extract intricate features from large datasets for tasks such as tumor classification and disease diagnosis[9][110].

3.3.1 Architecture of AlexNet

AlexNet's architecture is characterized by five convolutional layers followed by three fully connected layers. The convolutional layers are interspersed with max-pooling layers, contributing to the extraction of hierarchical features. The first convolutional layer processes raw pixel values, while subsequent layers capture increasingly abstract and complex features. The inclusion of rectified linear units (ReLU) as activation functions and dropout regularization enhances the model's ability to generalize[19].In our context, figure-3.3 denotes the visual representation specifically illustrating the architecture of AlexNet

3.3.2 Layers of AlexNet

The layer-wise composition of AlexNet involves the following key elements:

Input Layer

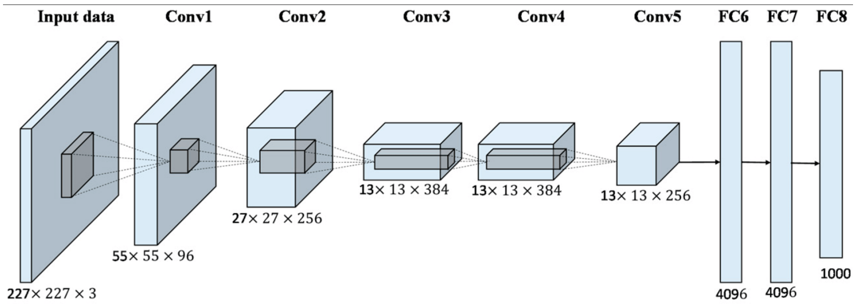


Figure 3.3: Architecture of AlexNet

Convolutional Layers - Five convolutional layers capturing hierarchical features.

Max-Pooling Layers - Interspersed to downsample spatial dimensions.

Fully Connected Layers - Three fully connected layers for high-level reasoning.

Output Layer - Produces the final classification output.

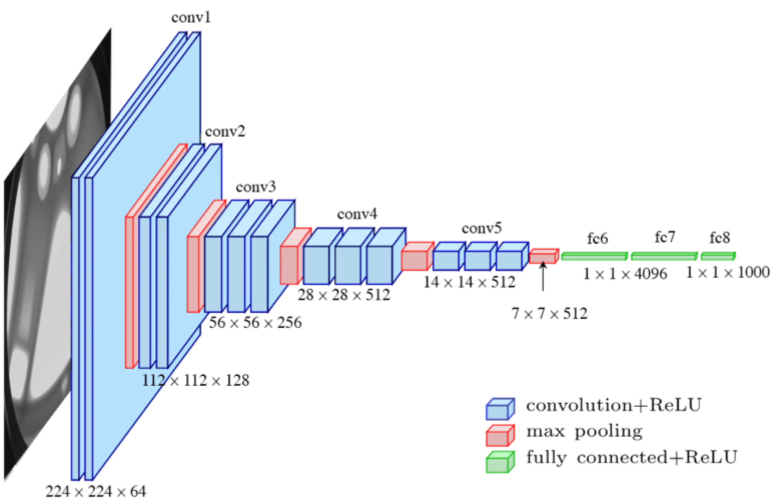


Figure 3.4: Layers of AlexNet

These figures illustrate the overall architecture (figure-3.3) and layer-wise structure (figure-3.4) of AlexNet, providing a visual representation of the model's design [9][110]. In the domain of medical image recognition, AlexNet's historical significance, versatile architecture, and successful applications underscore its continued relevance in advancing diagnostic capabilities [9].

3.4 VGG19

In 2014, Karen Simonyan and Andrew Zisserman introduced the VGG19 architecture, representing a significant advancement in deep learning for image recognition. Evolving from its predecessor, VGG16, VGG19 was meticulously designed to explore the impact of increased depth on performance. With 19 layers, including 16 convolutional layers and 3 fully connected layers, VGG19 utilizes 3x3 filters and max-pooling to capture intricate features and provide spatial abstraction. The architecture's thoughtful design strikes a balance between model depth and computational efficiency, making it successful in various image recognition tasks. Beyond traditional image classification, VGG19 finds valuable applications in medical image processing, demonstrating efficacy in tasks such as image segmentation, disease detection, and classification within medical imaging modalities, including X-rays, MRI scans, and histopathological images [19] [21] [22]. The history of VGG19, emerging in 2014, marked a pivotal moment in deep learning, emphasizing the role of deep neural networks in complex visual tasks. Motivated by the quest to understand the impact of increased network depth, VGG19 expanded the architecture to 19 layers, contributing to insights that shaped subsequent advancements in deep learning research. Simonyan and Zisserman's work on VGG16 and VGG19 has become foundational, influencing research directions and contributing to broader advancements in artificial intelligence and computer vision [19][21] . The legacy of VGG19 extends into medical image processing, where its deep architecture proves effective in tasks such as image segmentation, disease detection, and classification. Studies demonstrate its utility in identifying abnormalities associated with Alzheimer's disease, showcasing its ability to learn and distinguish subtle structural and textural differences in medical images [110] [22]. VGG19's application in diverse medical imaging modalities reflects its versatility and potential to advance diagnostic tools and methodologies in the medical field [41].

3.4.1 Architecture of VGG19:

VGG19, an extension of VGG16, features a meticulously designed architecture with 19 layers, including 16 convolutional and 3 fully connected layers. Employing 3x3 filters and max-pooling with a 2x2 window and stride of 2, the network captures both local and global features effectively. Introduced by Simonyan and Zisserman, this architectural choice strikes a balance between model complexity and computational efficiency [19]. The strategic use of max-pooling contributes to spatial downsampling, aiding in hierarchical feature extraction for discerning patterns of varying complexities [19][21]. Fully connected layers process high-level features, leading to a softmax activation for diverse image classification applications [19]. Figure-3.5 illustrates the architecture, highlighting VGG19's depth and thoughtful design principles. Overall, VGG19's architectural intricacies emphasize a balanced approach for optimal performance and interpretability in image classification [19].

3.4.2 Layers of VGG19:

VGG19, an influential convolutional neural network designed by Simonyan and Zisserman [1], exhibits a meticulous layer composition for hierarchical feature extraction and pattern recognition. The network begins with two consecutive convolutional layers, each employing 64 filters with a 3x3 kernel, followed by max-pooling for spatial dimension reduction. Subsequent blocks of convolutional layers with increasing filters (128, 256, 512) deepen the feature hierarchy, interspersed with max-pooling for

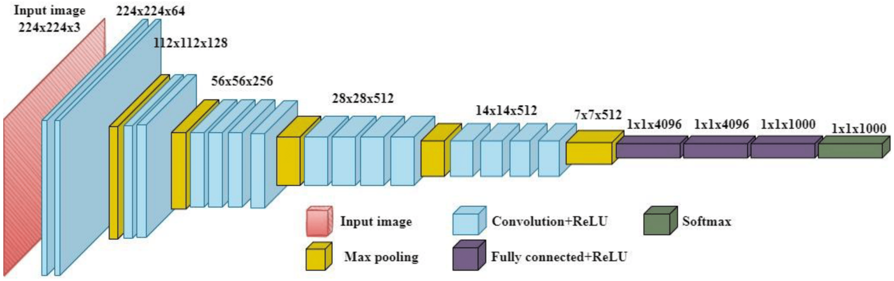


Figure 3.5: VGG19 Architecture Overview.

spatial abstraction. Two fully connected layers process high-level features, culminating in an output layer with softmax activation for multi-class classification. This balanced architecture, as illustrated in Figure-3.6, demonstrates VGG19's prowess in diverse image recognition tasks, striking a harmonious balance between depth and computational efficiency [19].

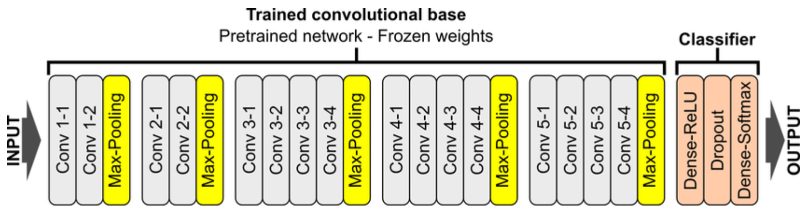


Figure 3.6: Layer-by-Layer Insight into VGG19.

3.5 Inception V3

Introduced by Google researchers in 2015, InceptionV3 represents a crucial advancement in deep learning architectures, specifically tailored for computer vision tasks. Designed by Christian Szegedy, Vincent Vanhoucke, and Sergey Ioffe, InceptionV3 balances model depth and computational efficiency through its innovative architecture, featuring inception modules with parallelized convolutions of varying sizes. This design allows the network to efficiently capture multi-scale features, enhancing its adaptability across diverse datasets. Beyond traditional computer vision tasks, InceptionV3's versatility extends to medical image processing, proving effective in image segmentation, disease detection, and classification within medical imaging [33].

The history of InceptionV3, introduced in 2015, marks a milestone in deep learning architecture evolution. Addressing the trade-off between model depth and computational efficiency, InceptionV3 incorporates parallelized convolutions of varying sizes, achieving a balance that captures both local and global features effectively. The architecture, building upon its predecessors, introduces novel features that enhance accuracy and adaptability. In medical image processing, InceptionV3's ability to discern intricate patterns proves effective, contributing to enhanced diagnostic accuracy in tasks such as image segmentation, disease detection, and classification across different scales [110]. The adoption of InceptionV3 showcases its potential to significantly advance diagnostic tools and methodologies in the medical field, making it a standout in the realm of deep learning for diverse medical imaging scenarios [33].

3.5.1 Architecture of InceptionV3:

InceptionV3, known for its intricate architecture [33], features a stem network for basic feature extraction and multiple inception modules for efficient capture of multi-scale features. These modules incorporate 1x1, 3x3, and 5x5 convolutions, along with max-pooling operations, enabling adaptability to diverse input patterns. The strategic use of 1x1 convolutions reduces computational complexity while preserving essential information [33]. The network's design, illustrated in Figure-3.7, emphasizes its ability to efficiently capture features at different scales.

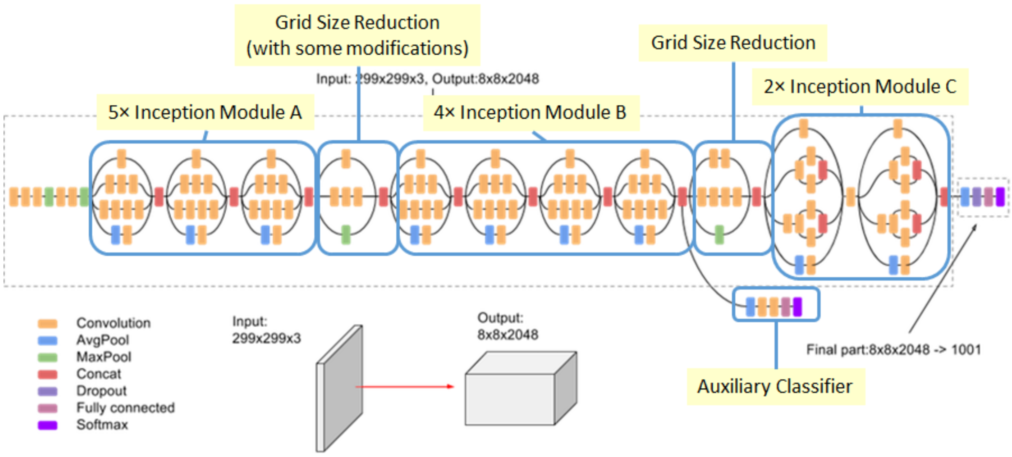


Figure 3.7: Architecture of InceptionV3

3.5.2 Layers of InceptionV3:

InceptionV3's layer composition comprises a stem network for initial feature extraction, followed by inception modules featuring parallelized 1x1, 3x3, and 5x5 convolutions, and max-pooling operations. A Global Average Pooling Layer condenses spatial features, leading to a Fully Connected Layer for classification, and an Output Layer with softmax activation for multi-class classification.

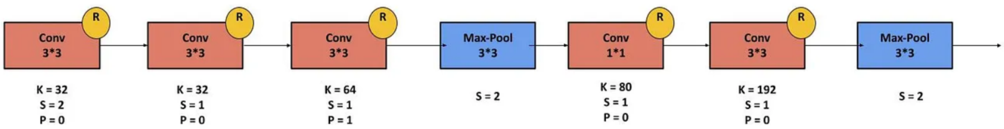


Figure 3.8: Layers of InceptionV3

This modular and parallelized design, as highlighted by its architecture figure-3.8 shows that, underscores InceptionV3's capability to effectively capture diverse features across different scales, making it well-suited for a broad spectrum of computer vision tasks.

3.6 ResNet50

ResNet50, introduced in 2015 by Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, stands as a pivotal advancement in deep learning architectures within the ResNet (Residual Network) family. Specifically designed to address challenges associated with training very deep neural networks, ResNet50's innovation lies in the incorporation of residual blocks(figure-3.9. These blocks, featuring shortcut connections, enable the training of exceptionally deep networks without succumbing to vanishing gradient issues. Comprising 48 convolutional layers, 1 global average pooling layer, and 1 fully connected layer, ResNet50 mitigates degradation problems linked to network depth, resulting in improved accuracy and generalization [25].

ResNet50's impact extends into medical image processing, showcasing its proficiency in handling complex visual data for diverse applications. In tasks such as disease diagnosis, image segmentation, and anomaly detection within medical imaging datasets, ResNet50 has demonstrated effectiveness. In a 2023 study by Iqbal et al., titled "On the Analyses of Medical Images Using Traditional Machine Learning Techniques and Convolutional Neural Networks" [110], ResNet50 exhibited exceptional performance in the classification of medical images, emphasizing its potential for accurate disease identification and localization. As we further explore ResNet50's architecture, layers, and applications, its significance in the realms of deep learning and medical image processing becomes increasingly evident.

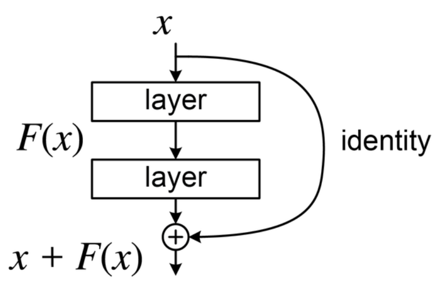


Figure 3.9: A residual block (for ResNet 50).

3.6.1 Architecture of ResNet50:

ResNet50's architecture centers on residual blocks, featuring multiple convolutional layers and shortcut connections that facilitate the direct flow of information, crucial for training deep models. The stacking of these blocks enables the learning of intricate hierarchical features.

The use of global average pooling condenses spatial dimensions, enhancing model interpretability and efficiency, leading to a final fully connected layer for classification output.

This design, as proposed by He et al. [25], highlights ResNet50's ability to overcome training challenges in deep networks. Figure-3.10 indicates the graphic depiction that expressly shows ResNet50's architecture.

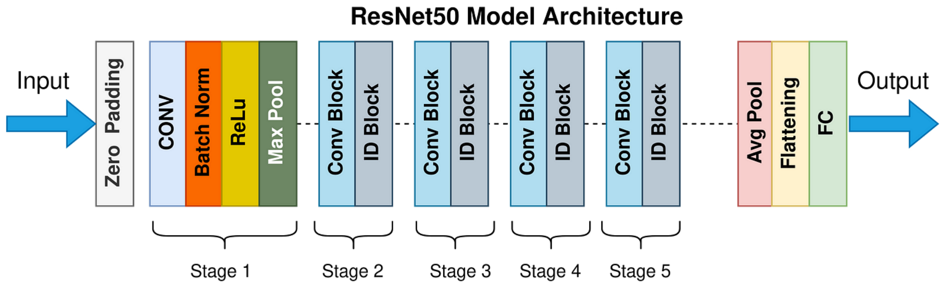


Figure 3.10: Architecture of ResNet-50.

3.6.2 Layers of ResNet50:

The layer composition of ResNet50 involves the following key elements:

Input Layer

Convolutional Layers within Residual Blocks - Stacking of residual blocks containing convolutional layers.

Global Average Pooling Layer - Condenses the spatial dimensions of the features.

Fully Connected Layer - Produces the final classification output. In this context, Figure-3.11 provides a graphical overview of the intricate layer structure inherent in a ResNet50 model.

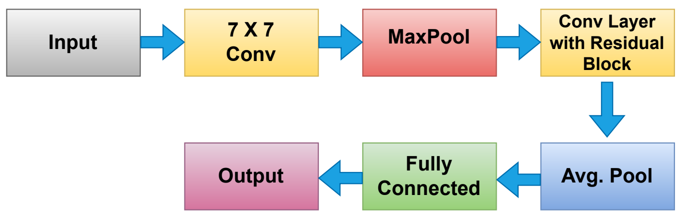


Figure 3.11: Layers of ResNet50

This design, as proposed by He et al., highlights ResNet50's ability to overcome training challenges in deep neural networks[25].

3.7 MobileNet

MobileNet, renowned for its efficiency and lightweight architecture, finds valuable applications in medical image processing, presenting a pragmatic solution for on-device analysis and diagnostics. The network's effectiveness extends to tasks such as image segmentation, disease classification, and point-of-care diagnostics within the medical imaging domain.

In a 2023 study by Iqbal et al., titled "On the Analyses of Medical Images Using Traditional Machine Learning Techniques and Convolutional Neural Networks" [110], MobileNet showcased its potential for accurate and real-time analysis of medical images. The research underscored MobileNet's effectiveness as a valuable tool for point-of-care applications, emphasizing its capabilities in on-device medical image analysis [8]. As we delve into MobileNet's architecture, layers, and applications, its significance at the intersection of deep learning and medical image processing becomes increasingly apparent.

3.7.1 Architecture of MobileNet:

MobileNet's architecture is distinguished by depthwise separable convolutions, employing depthwise convolution followed by pointwise convolution, a strategy that significantly reduces the number of parameters and computational complexity.

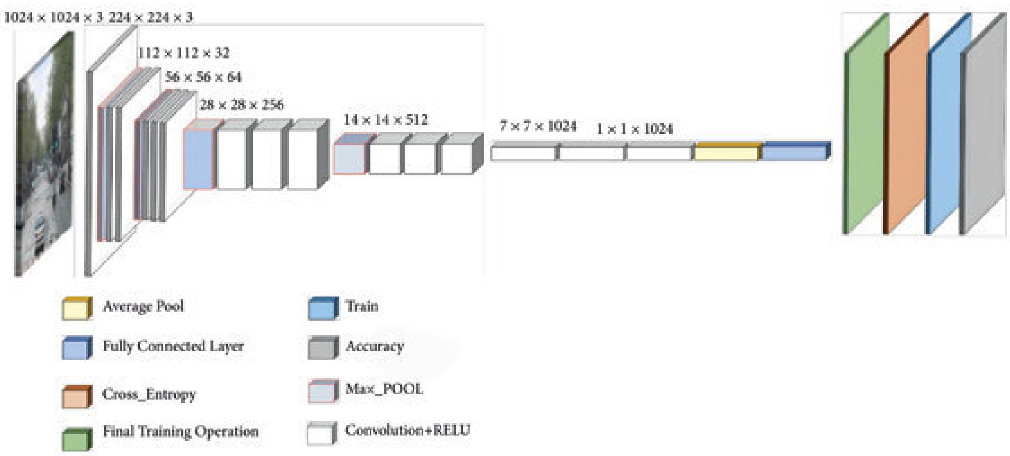


Figure 3.12: Architecture of MobileNet

This modular and lightweight design, as introduced by Howard et al. [37], makes MobileNet highly efficient for diverse computer vision tasks, demonstrating an optimal balance between performance and demand on hardware resources.

3.7.2 Layers of MobileNet:

The layer composition of MobileNet involves the following key elements:

Input Layer

Depthwise Separable Convolutional Layers - Stacking of depthwise separable convolutions.

Global Average Pooling Layer - Condenses the spatial dimensions of the features.

Fully Connected Layer - Produces the final classification output.

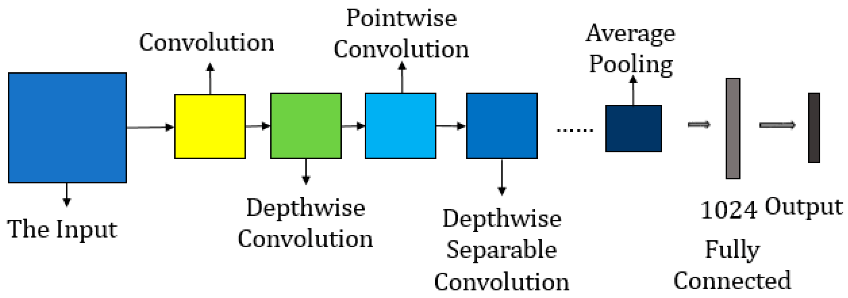


Figure 3.13: Layers of MobileNet

In this context, Figure-3.13 presents a graphical picture of the sophisticated layer structure inherent in a MobileNet model.

3.8 DenseNet121

Introduced in 2017 by Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger, DenseNet121 stands as a notable advancement in convolutional neural network (CNN) architectures. As part of the Densely Connected Convolutional Networks (DenseNet) family, it introduces a dense connectivity pattern, where each layer receives direct input from all preceding layers. The innovative dense block, fostering a feed-forward connectivity pattern, promotes feature reuse, mitigates the vanishing gradient problem, and optimizes parameter utilization efficiently [38].

DenseNet121's unique connectivity pattern and efficient parameter sharing render it well-suited for various medical image processing tasks. Its applications extend to disease classification, lesion detection, and medical image segmentation. In a 2017 study by Huang et al., titled "Densely Connected Convolutional Networks," DenseNet121 exhibited superior performance in classifying medical images, showcasing its potential for accurate and robust diagnosis [38]. Furthermore, a more recent study by Tao Zhou et al. in 2022, titled "Dense Convolutional Network and Its Application in Medical Image Analysis," further validated DenseNet121's effectiveness in medical image analysis [108]. The adoption of DenseNet121 in these studies highlights its significance and impact in the realm of deep learning and medical image processing.

3.8.1 Architecture of DenseNet121

DenseNet121's architecture is characterized by densely connected blocks, comprising multiple dense layers. Each dense layer receives input from all preceding layers within the block, creating a dense feature reuse pattern. Transition layers with pooling and convolution operations control the growth of feature maps and contribute to the network's compactness. Figure-3.14 signifies the visual representation of the architecture of DenseNet121 model.

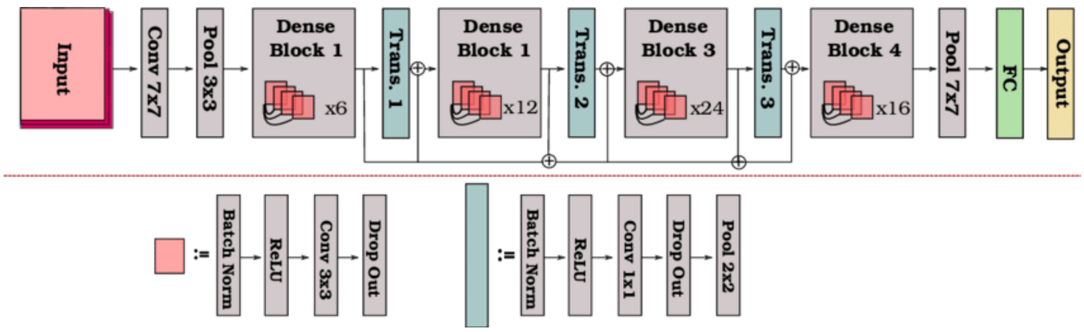


Figure 3.14: Architecture of DenseNet121

The dense connectivity and efficient parameter sharing in DenseNet121 contribute to improved model training and generalization. This architectural design, proposed by Huang et al. (2017) in "Densely Connected Convolutional Networks" [38], facilitates superior performance in image classification tasks, laying the foundation for its applications in medical image processing.

3.8.2 Layers of DenseNet121

The layer composition of DenseNet121 unfolds as follows:

Input Layer

Dense Blocks - Stacking of densely connected blocks.

Transition Layers - Introducing compression to control feature map size.

Global Average Pooling Layer - Condenses the spatial dimensions of the features.

Fully Connected Layer - Produces the final classification output.

The efficiency of DenseNet121 in medical image processing, as highlighted in studies like "Dense Convolutional Network and Its Application in Medical Image Analysis" by Tao Zhou et al. (2022) [108], is intricately tied to the interconnected and densely packed layers, showcasing its adaptability and efficacy in handling complex medical imaging data.

3.9 Xception

Introduced by François Chollet in 2017, Xception is a unique convolutional neural network architecture that extends the Inception model by implementing depthwise separable convolutions. This innovation enhances the network's ability to capture complex patterns while reducing parameters, leading to improved efficiency. In medical image processing, Xception's emphasis on intricate pattern capture makes it suitable for tasks like disease classification and image segmentation. A 2023 study by Iqbal et al. demonstrated Xception's promising results in accurately analyzing medical images, showcasing its potential to advance diagnostic capabilities [110]. The versatility and efficacy of Xception in medical image processing become evident as we explore its architecture, layers, and applications.

3.9.1 Architecture of Xception

Xception's architecture is characterized by a series of depthwise separable convolutions, creating a highly efficient and expressive feature extractor. The network comprises multiple blocks of depthwise separable convolutions, allowing for rich hierarchical feature learning. Figure-3.15 signifies the visual representation of the architecture of Xception model.

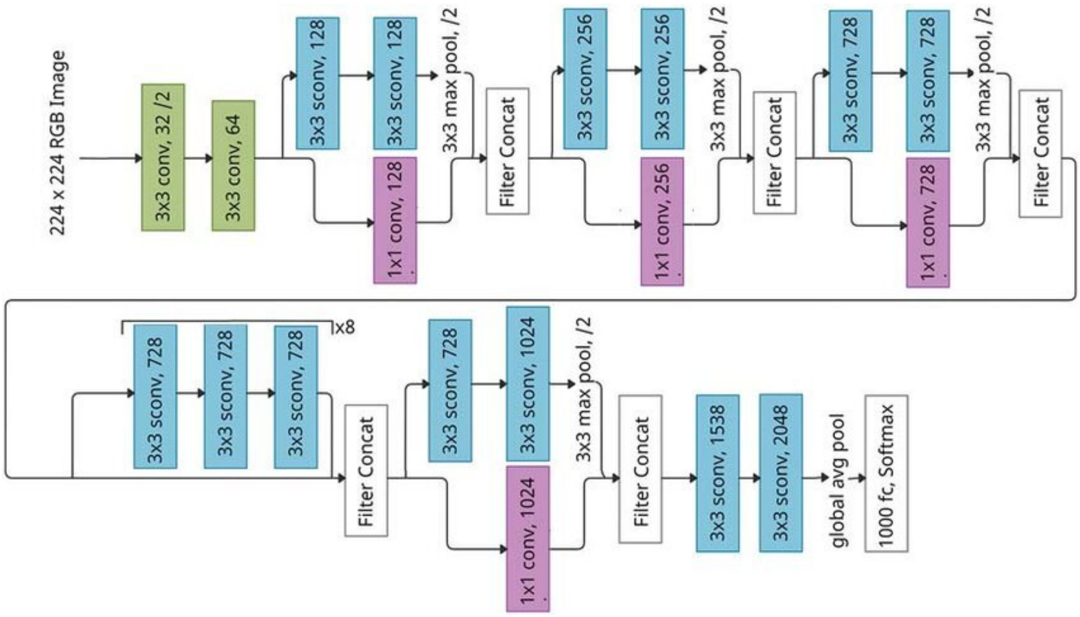


Figure 3.15: Architecture of Xception

The modular design and depthwise separable convolutions in Xception, introduced by Chollet (2017) in "Xception: Deep Learning with Depthwise Separable Convolutions" [35], contribute to its success in various computer vision tasks.

3.9.2 Layers of Xception

The layer composition of Xception(figure-3.16) involves the following key elements:

Entry Flow - Initial convolutional and depthwise separable convolution blocks.

Middle Flow - Stacks of depthwise separable convolutions for feature extraction.

Exit Flow - Global average pooling and fully connected layers for classification.

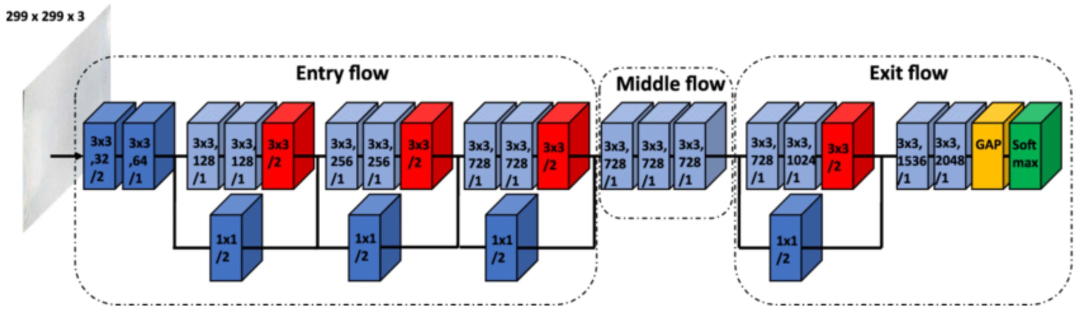


Figure 3.16: Layers of Xception

The efficient use of depthwise separable convolutions in Xception enhances its representational power while maintaining computational efficiency.

3.10 EfficientNet B0

EfficientNet B0, introduced by Mingxing Tan and Quoc V. Le in 2019, represents a notable advancement in convolutional neural network (CNN) architectures [19]. As part of the EfficientNet family, it incorporates compound scaling principles to achieve optimal performance with efficient model sizes. This innovative approach automates the scaling process across multiple dimensions, leading to improved efficiency in terms of both accuracy and computational cost [19].

3.10.1 Architecture of EfficientNet B0

EfficientNet B0's architecture is based on a compound scaling factor, ensuring a balanced increase in network dimensions. It consists of a baseline network with inverted bottleneck structures and mobile inverted residuals, achieving a judicious tradeoff between computational efficiency and model accuracy. Squeeze-and-excitation blocks are employed to enhance efficiency by emphasizing essential features for improved representation learning [19] [21].Figure-3.17 denotes the visual representation specifically illustrating the architecture of EfficientNet B0.

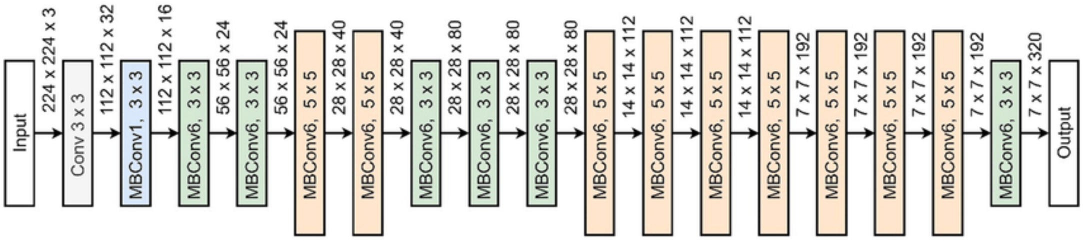


Figure 3.17: Architecture of EfficientNet B0

3.10.2 Layers of EfficientNet B0

The layer-wise composition of EfficientNet B0 involves a stack of inverted bottleneck blocks, each featuring lightweight depthwise separable convolutions. Depthwise separable convolutions reduce computation, enabling efficient learning. Skip connections are incorporated to enhance gradient flow and mitigate vanishing gradient issues. The final layers include global average pooling and a fully connected layer for classification output [69].

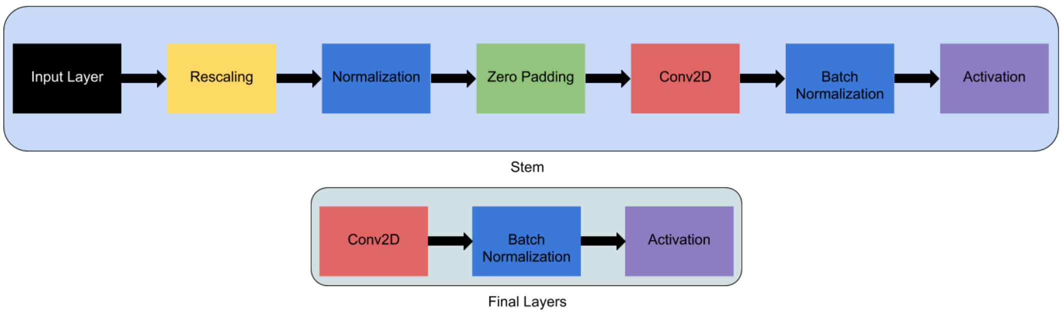


Figure 3.18: Layers of EfficientNet B0

These visual representations depict the overall architecture and layer-wise structure of EfficientNet B0, providing insights into the model's design [69] [54]. In the domain of medical image recognition, EfficientNet B0 has exhibited promising results, making it a compelling choice for tasks such as disease diagnosis and anomaly detection [69][54]. Its adaptability to diverse medical imaging datasets underscores its significance in advancing diagnostic capabilities.

3.11 Evaluation Metrics

3.11.1 Confusion Matrix

Certainly! The confusion matrix is a crucial tool in the field of machine learning for evaluating classification algorithms [1]. Its primary purpose lies in assessing the accuracy of a model by summarizing classification results and categorizing them into four key metrics: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

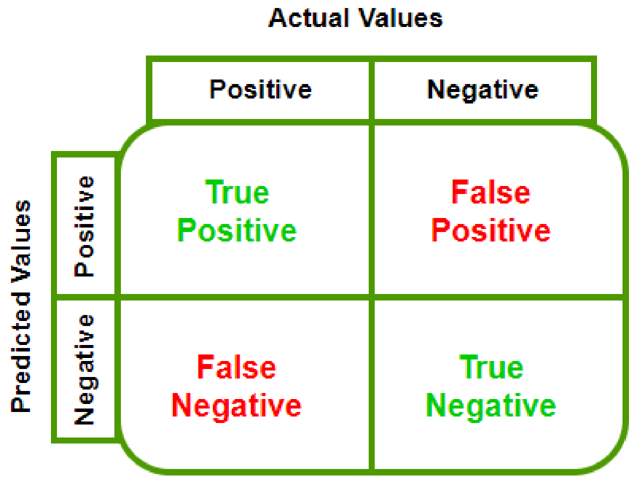


Figure 3.19: Confusion Matrix

In this matrix:

- True Positives (TP) represent instances where the model correctly predicts the positive class.

- True Negatives (TN) denote instances where the model accurately predicts the negative class.

- False Positives (FP) occur when the model predicts the positive class, but the actual class is negative.

- False Negatives (FN) arise when the model predicts the negative class, but the actual class is positive.

This tabular representation facilitates a comprehensive evaluation of a classification model's performance by providing insights into its predictive accuracy and potential areas for improvement [2]

3.11.2 Precision

Precision is a vital metric in assessing the accuracy of positive predictions made by a classification model. It is calculated using the formula:

$$Precision = \frac{TP}{TP + FP} \tag{3.3}$$

where:

TP (True Positives): Instances correctly predicted as positive.

FP (False Positives): Instances predicted as positive but actually negative.

Precision measures the model's ability to avoid false positives, indicating the proportion of accurately identified positive instances among all predicted positives. It is particularly valuable in contexts where the cost of false positives is high, such as medical diagnostics [3].

3.11.3 Recall

Recall, or sensitivity, gauges a classification model's ability to correctly capture all positive instances. It is calculated as:

$$Recall = \frac{TP}{TP + FN} \tag{3.4}$$

where TP represents True Positives and FN represents False Negatives.

A high recall indicates the model effectively identifies a large portion of positive instances, making it crucial in scenarios where missing positive cases is costly, such as in medical diagnoses [4].

3.11.4 Specificity

Specificity, also known as the true negative rate, is a metric that assesses a classification model's ability to correctly identify negative instances. It is particularly relevant when the cost of false positives is high.

Specificity is calculated using the following formula:

$$Specificity = \frac{TrueNegatives(TN)}{TrueNegatives(TN) + FalsePositives(FP)} \tag{3.5}$$

Specificity provides insight into how well a model can avoid misclassifying negative instances. A higher specificity indicates fewer false positives, making it a crucial metric in scenarios where the consequences of misidentifying negative cases are significant.

While sensitivity (recall) focuses on positive instances, specificity provides a complementary view, offering a balanced assessment of a model's performance across both positive and negative classes [4].

3.11.5 Accuracy

Accuracy is a key metric measuring the overall correctness of a classification model. It is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.6}$$

where TP is True Positives, TN is True Negatives, and Total Instances is the overall dataset size. Despite its simplicity, accuracy offers a quick assessment of a model's effectiveness in correct classification across all classes [4].

3.11.6 Matthews Correlation Coefficient (MCC)

The Matthews Correlation Coefficient (MCC) is a metric that provides a balanced measure of classification performance, especially useful in imbalanced datasets. It takes into account true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) to calculate a value between -1 and 1.

The MCC is calculated using the following formula:

$$\mathrm{MCC} = \frac{\mathrm{TP} \times (\mathrm{TN} - \mathrm{FP} \times \mathrm{FN})}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}} \tag{3.7}$$

In this equation,

- **True Positives (TP):** Instances correctly predicted as positive.
- **True Negatives (TN):** Instances correctly predicted as negative.
- **False Positives (FP):** Instances predicted as positive but actually negative.
- **False Negatives (FN):** Instances predicted as negative but actually positive.

The MCC ranges from -1 (indicating total disagreement between predictions and actual outcomes) to 1 (indicating perfect classification), with 0 representing random performance. A higher MCC suggests better model performance, considering both sensitivity and specificity.

MCC is particularly useful when dealing with imbalanced datasets, providing a comprehensive assessment of a model's overall performance [5].

3.11.7 F1-score

The F1 score is a metric that balances precision and recall, providing a comprehensive measure of a classification model's performance. It is especially useful when there is an uneven class distribution in the dataset.

The F1 score is calculated using the following formula:

$$F1\ score = \frac{2 * Recall * Precision}{Recall + Precision} \tag{3.8}$$

The F1 score ranges from 0 to 1, with 1 indicating perfect precision and recall and 0 representing poor performance. It is particularly effective in scenarios where both false positives and false negatives are critical and need to be minimized.

The F1 score serves as a robust metric for assessing a model's ability to balance precision and recall, providing a single value that captures both aspects of classification performance [6].

3.11.8 AUC-ROC (Area Under the ROC Curve)

The Area Under the Receiver Operating Characteristic (ROC) Curve, denoted as AUC-ROC, quantifies the discriminative ability of a classification model by assessing its performance across varying classification thresholds. The ROC curve is a graphical representation plotting the true positive rate (sensitivity) against the false positive rate (1-specificity) for different threshold values. The AUC-ROC, calculated as the area under this curve, ranges from 0 to 1, with higher values indicating superior discrimination. The AUC-ROC serves as a comprehensive metric for model evaluation, particularly valuable in scenarios with imbalanced class distributions or when both sensitivity and specificity are critical [7].

$$AUC\text{-}ROC = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d\mathrm{FPR} \tag{3.9}$$

Here, TPR represents the true positive rate (sensitivity), and FPR represents the false positive rate (1-specificity). The AUC-ROC computation involves numerical integration to calculate the area under the ROC curve.

3.12 Normalization

Normalization in machine learning is a preprocessing technique that focuses on scaling and standardizing input features, ensuring they share a consistent scale [29]. The primary objective is to promote uniformity among feature values, which can

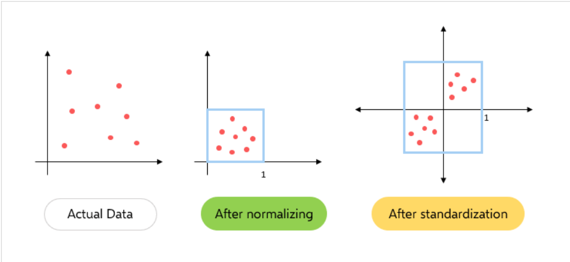


Figure 3.20: Normalization Process.

enhance the training process and boost the overall performance of machine learning models. Figure 3.20 illustrates the process of normalization.

In many cases, a common normalization method is min-max scaling, expressed by the equation:

$$X_{\text{normalized}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

This equation scales feature values to the range $[0, 1]$. For implementing our model, we have employed two distinct types of normalization. They are:

3.12.1 Feature-wise Standardization

Feature-wise standardization, often referred to as whitening, is a normalization technique applied independently to each feature (pixel) across all samples in a dataset [29]. It involves subtracting the mean and dividing by the standard deviation for each feature. The standardized value for a given feature $X_{i,j}$ in the dataset is computed as:

$$\text{Standardized value}_{i,j} = \frac{X_{i,j} - \text{mean}_j}{\text{std}_j}$$

Here, $X_{i,j}$ is the original value of feature j in sample i, mean_j is the mean of feature j across all samples, and std_j is the standard deviation of feature j across all samples.

3.12.2 Sample-wise Standardization

Sample-wise standardization, another type of normalization, is applied independently to each sample (image) in a dataset [29]. This technique involves subtracting the mean and dividing by the standard deviation for each sample. The standardized value for a given sample X_i is given by:

$$\text{Standardized value}_i = \frac{X_i - \text{mean}_i}{\text{std}_i}$$

Here, X_i represents the original values of all features for sample i, mean_i is the mean of all features for sample i, and std_i is the standard deviation of all features for sample i.

3.13 Optimizer

In the context of machine learning, optimizers play a critical role in minimizing the error or loss function to guide models towards optimal parameters. Optimizers, such as Stochastic Gradient Descent (SGD), RMSprop, Adagrad, Adam, Adamax, Nadam, and Adadelta, serve to reduce the disparity between predicted and target outputs, facilitating convergence during training. These optimizers vary in characteristics and are chosen based on factors like problem nature, dataset characteristics, and computational resources [23].

3.13.1 Adam Optimizer

Adam (Adaptive Moment Estimation) is a widely used optimization algorithm in machine learning, efficiently updating model parameters with adaptive learning rates and robustness across neural network architectures. It maintains moving averages for the first and second moments, combining momentum and RMSprop ideas. The update rule adjusts parameters based on bias-corrected moments, contributing to its effectiveness [16].

Equations

$$\begin{aligned}
m_t &= \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t \\
v_t &= \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2 \\
\hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \\
\hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \\
\theta_t &= \theta_{t-1} - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot \hat{m}_t
\end{aligned}$$

Here, α is the learning rate, β_1 and β_2 are exponential decay rates for the first and second moments, g_t is the gradient, and ϵ is a small constant to prevent division by zero [16].

3.13.2 Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent (SGD) is a fundamental optimization algorithm, updating model parameters iteratively in the opposite direction of the loss gradient. It employs a stochastic approach by using a random subset (mini-batch) of the training data for each parameter update, making it computationally efficient [1].

Equation

$$\theta_t = \theta_{t-1} - \alpha \cdot g_t$$

Here, α is the learning rate, g_t is the gradient, and ϵ is a small constant to prevent division by zero [1].

3.13.3 RMSprop Optimizer

RMSprop (Root Mean Square Propagation) is an adaptive learning rate optimization algorithm designed to address issues with constant learning rates in stochastic gradient descent. It adapts learning rates for each parameter based on the rolling average of squared gradients, introducing a dampening effect to prevent oscillations [11].

Equations

$$\theta_t = \theta_{t-1} - \frac{\alpha}{\sqrt{v_t} + \epsilon} \cdot g_t$$
$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$$

Here, α is the learning rate, β_1 and β_2 are exponential decay rates for the first and second moments, g_t is the gradient, and ϵ is a small constant to prevent division by zero [11].

3.13.4 Adagrad Optimizer

Adagrad (Adaptive Gradient Algorithm) is an optimization algorithm that adapts the learning rates of individual parameters based on historical squared gradients. It is particularly effective for problems with sparse data [6].

Equation

$$\theta_t = \theta_{t-1} - \frac{\alpha}{\sqrt{G_t} + \epsilon} \cdot g_t$$

Here, α is the learning rate, g_t is the gradient, and ϵ is a small constant to prevent division by zero [6].

3.13.5 Adamax Optimizer

Adamax is a variant of the Adam optimizer that utilizes the infinity norm (maximum) of past gradients. It provides stability in the update rule, especially when gradients have large variations [16].

Equation

$$\theta_t = \theta_{t-1} - \frac{\alpha}{\max(\beta_2 \cdot v_{t-1}, |g_t|) + \epsilon} \cdot \hat{m}_t$$

Here, α is the learning rate, β_1 and β_2 are exponential decay rates for the first and second moments, g_t is the gradient, and ϵ is a small constant to prevent division by zero [16].

3.13.6 Nadam Optimizer

Nadam (Nesterov-accelerated Adam) is an Adam variant that incorporates Nesterov accelerated gradient into the update rule. It aims to provide faster convergence by adjusting the momentum term [24].

Equation

$$\theta_t = \theta_{t-1} - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot (\beta_1 \cdot \hat{m}_t + (1 - \beta_1) \cdot g_t)$$

Here, α is the learning rate, β_1 and β_2 are exponential decay rates for the first and second moments, g_t is the gradient, and ϵ is a small constant to prevent division by zero [24].

3.13.7 Adadelta Optimizer

Adadelta is an adaptive learning rate method designed to address the diminishing learning rate problem of Adagrad. It uses a moving window of past gradients to adjust the learning rates [12].

Equation

$$\theta_t = \theta_{t-1} - \frac{\sqrt{\sum_{i=1}^{t-1} \delta g_i^2 + \epsilon}}{\sqrt{\sum_{i=1}^{t} \hat{v}_i + \epsilon}} \cdot g_t$$

Here, α is the learning rate, g_t is the gradient, and ϵ is a small constant to prevent division by zero [12].

3.14 Explainable AI (XAI)

Explainable Artificial Intelligence (XAI) addresses the challenge of understanding and interpreting the decisions made by machine learning models. As machine learning models, especially deep neural networks, become increasingly complex, they often operate as "black boxes," making it difficult for users to comprehend the rationale behind their predictions or classifications. XAI aims to bridge this gap by providing methods and techniques that make the decision-making process of AI systems transparent and interpretable.

XAI is used in applications where the stakes are high and human understanding of AI decisions is crucial. In healthcare, for instance, XAI can help medical professionals understand the factors contributing to a diagnostic decision. Similarly, in finance, XAI can provide insights into the features influencing credit scoring or investment recommendations. By enhancing transparency, XAI promotes trust in AI systems, aids in uncovering biases, and ensures that AI technologies are deployed ethically and responsibly [30].

3.14.1 XAI Techniques

XAI techniques can be broadly classified into two categories: model-agnostic and model-specific methods.

Model-Agnostic Techniques

These methods aim to explain the predictions of any machine learning model, irrespective of its underlying architecture. Examples include LIME, SHAP, and Integrated Gradients.

Model-Specific Techniques

These approaches are tailored to a particular type of model. GRAD-CAM, for instance, is designed specifically for convolutional neural networks (CNNs) used in computer vision tasks.

3.14.2 GRAD-CAM (Gradient-weighted Class Activation Mapping)

GRAD-CAM is a visualization technique that provides insights into the decision-making process of convolutional neural networks. It highlights the important regions of an input image that contribute most to the final prediction. The formula for computing the GRAD-CAM heatmap involves taking the weighted sum of the gradients of the predicted class score with respect to the feature maps of the last convolutional layer. This helps identify which parts of the input image are crucial for the network's decision-making process [49].

$$(\text{GRAD-CAM}(i,j)) = \text{ReLU}\left(\sum_k \alpha_k \cdot A_k(i,j)\right) \tag{3.10}$$

Here, α_k represents the importance of the k^{th} feature map, and $A_k(i,j)$ denotes the activation of the k^{th} feature map at position (i,j).

3.14.3 SHAP (SHapley Additive exPlanations)

SHAP values provide a comprehensive and fair way to attribute the contribution of each feature to a model's prediction. Based on cooperative game theory, SHAP values consider all possible feature combinations and calculate the average contribution of each feature. The formula for the SHAP value of a specific feature i is expressed as:

$$(\phi_i(f)) = \sum_{S \subseteq N \setminus \{i\}} \frac{\binom{|S|}{|N|-1}}{\binom{|N|}{|N|-1}} [f(S \cup \{i\}) - f(S)] \tag{3.11}$$

Where $f(S)$ is the model prediction when considering the subset of features S [44].

3.14.4 LIME (Local Interpretable Model-agnostic Explanations)

LIME is a technique that focuses on explaining individual predictions of complex models by approximating their behavior with interpretable local models. It generates perturbed instances of the input data, observes the model's response, and fits a locally interpretable model. The explanation provided by LIME is determined by minimizing a loss function and a complexity penalty:

$$(\hat{g}(x)) = \arg\min_g \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{3.12}$$

Here, f is the complex model, g is the interpretable local model, $\mathcal{L}$ is the loss function, and Ω is the complexity penalty [30].

3.14.5 Integrated Gradients

Integrated Gradients is a technique that attributes the model's output to its input features by integrating the gradients of the model's predictions with respect to the input features along a straight path from a baseline or reference point to the input. The method provides a comprehensive understanding of feature importance by considering the entire input space.

The formula for calculating the attribution for feature i using Integrated Gradients is given by:

$$\text{IntegratedGrads}_i(f) = (x_i - x'_i) \times \int_{\alpha=0}^{1} \frac{\partial f(\mathbf{x}' + \alpha \times (\mathbf{x} - \mathbf{x}'))}{\partial x_i} \, d\alpha \tag{3.13}$$

Here, f is the model, $\mathbf{x}$ is the input, $\mathbf{x}'$ is the baseline, and $\frac{\partial f(\mathbf{x}' + \alpha \times (\mathbf{x} - \mathbf{x}'))}{\partial x_i}$ is the gradient of the model's prediction with respect to the i^{th} feature [50].

Integrated Gradients provides a path-sensitive approach, ensuring that the attribution is distributed fairly among features, even in the presence of interactions between them. This technique is particularly useful for understanding complex, non-linear models.

Including Integrated Gradients along with SHAP, LIME, and GRAD-CAM provides a comprehensive overview of various XAI techniques.

Chapter 4

Materials and Proposed Methodology

This chapter provides a brief idea regarding our workflow. In flowchart, we have described our working process from beginning to end. At first, we collected our dataset from the official Kaggle website. After fetching data we came to know that there are images of different size. Here comes our data pre-processing part. There are two parts in our image preprocessing: image resizing and image denoising. After pre-processing we labeled our data for binary classification and fixed our sample size. After that, we split our data into train data, validation data and test data. We moved to our next part which is preparing our self-designed model and other deep learning models for training. After that, we checked results like accuracy, precision, recall, MCC, F1-score, and area under ROC curve for different models. The workflow of our proposed system is given in the Figure 4.1.

4.1 Data acquisition

Data acquisition is the foundational step in collecting pertinent information for subsequent storage, cleaning, and preprocessing. It involves sourcing and transforming data into a machine-readable format, providing the basis for training models and decision-making. Despite challenges in obtaining data from Neurosciences Hospitals, an alternative approach involves utilizing a valuable dataset from Kaggle named ADNI (Alzheimer's Disease Neuroimaging Initiative).

The ADNI dataset is a valuable resource for Alzheimer's research due to its recentness and integration of diverse sources. This dataset combines multiple Alzheimer's Disease datasets into a comprehensive repository, categorizing them into four classes Non-demented, Mild Demented, Very-mild-demented, and Moderate-demented. It encompasses a total of 6400 jpg-format images. The distinctiveness of ADNI lies in its up-to-date nature and amalgamation of varied sources, making it a rich reservoir for in-depth analysis and model development in Alzheimer's research.

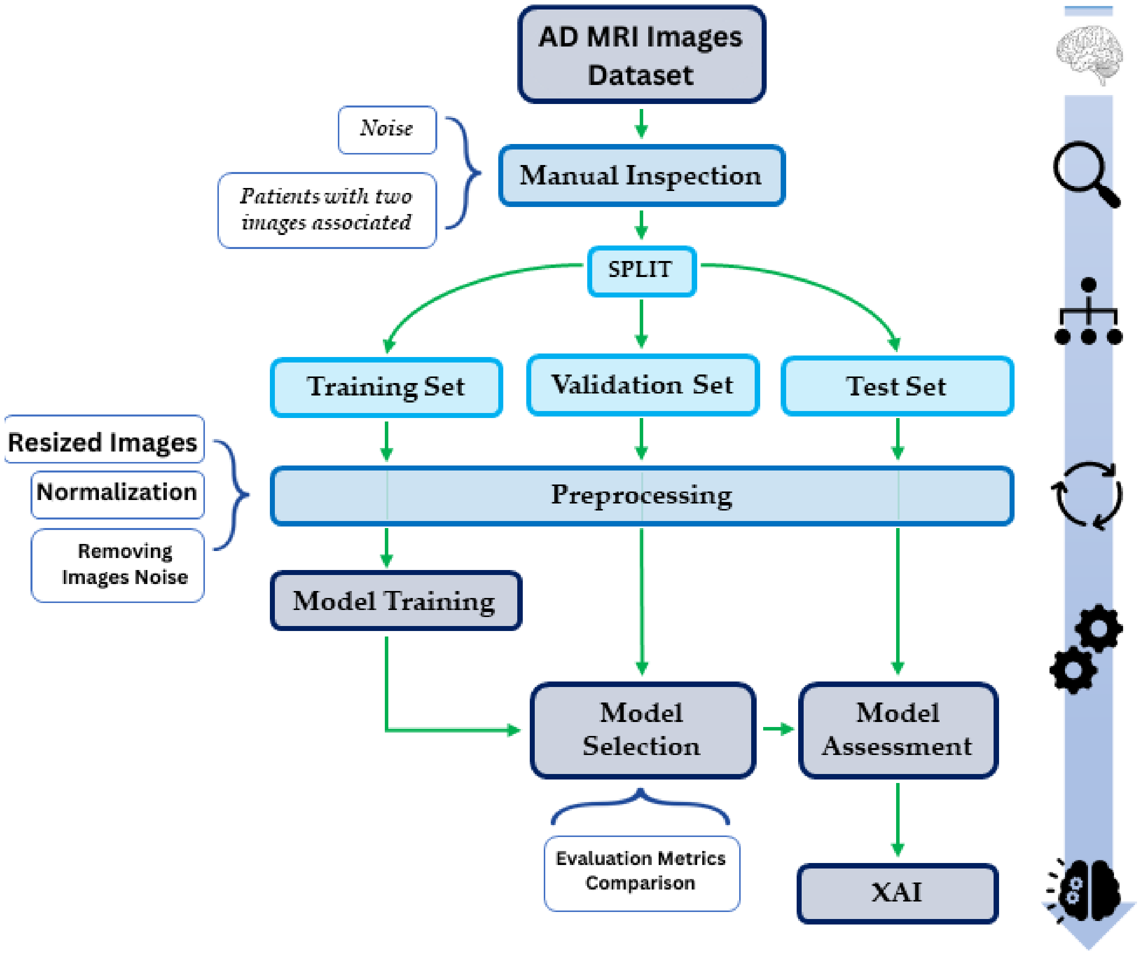


Figure 4.1: Workflow of our Proposed System

Class	No of Image
Mild Demented (MID)	896
Moderate Demented (MOD)	64
Non-Demented (ND)	3200
Very Mild Demented (VMD)	2240

Table 4.1: Distribution for each image classes of our utilized dataset

Figure 4.2: Mild Demented (MID) (left) and Moderate Demented (MOD) (right)

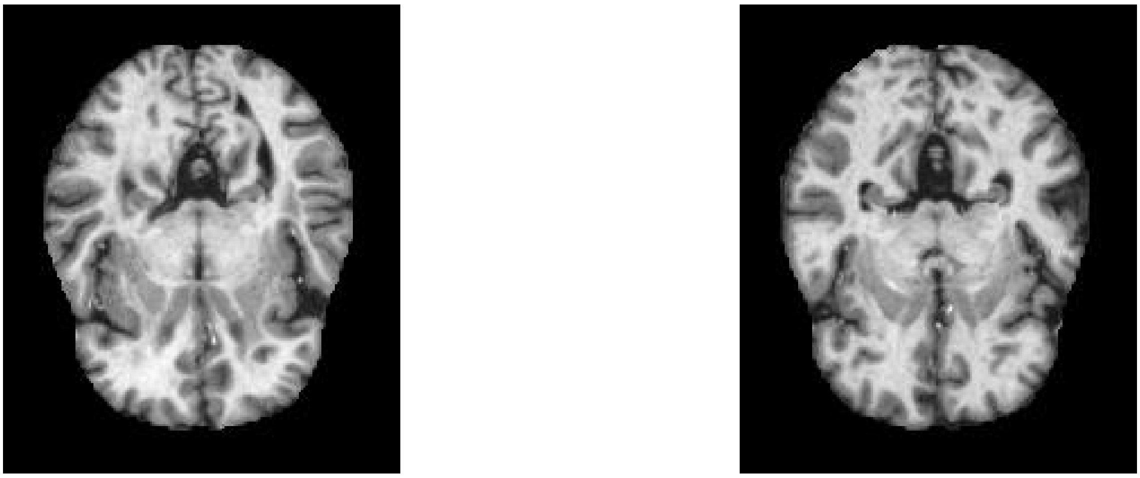


Figure 4.3: Non-Demented (ND) (left) and Very Mild Demented (VMD) (right)

4.2 Importing Libraries

To initialize our code, we imported crucial libraries and modules, including os, numpy (as np), scipy (as sp), pandas (as pd), seaborn (as sns), PIL (Image), tensorflow (as tf), keras_cv, AUTOTUNE (from tensorflow.data), Adam optimizer, and keras.preprocessing.image.ImageDataGenerator. Additionally, we incorporated tools from Keras, such as Input, Conv2D, MaxPooling2D, Flatten, Dropout, Dense, BatchNormalization, concatenate, Model, and callbacks like ReduceLROnPlateau, TensorBoard, and ModelCheckpoint. The sklearn.model_selection module was utilized for train-test splitting.

For data preprocessing, the trio of os, cv2 (represented as CV2), and numpy (as NumPy) played pivotal roles. OS was utilized to check the dataset directory, CV2 facilitated the reading of images from various dataset folders, converting them into arrays. NumPy was then employed for array-to-matrix conversion. The dataset itself consists of diverse Alzheimer's disease images, which underwent grayscale transformation using Python's cmap functions.

In the final steps, these processed images were organized into four distinct classes, each representing different types of Alzheimer's disease images. This systematic approach, combined with the integration of TensorFlow and Keras tools, establishes a robust foundation for subsequent stages in our code, ensuring comprehensive data handling, model training, and evaluation.

4.3 Dataset preprocessing

After assembling a comprehensive collection of raw data, we embarked on the pivotal phase of preprocessing to enhance the efficacy of our neural networks. This crucial step involves two fundamental methods: transformation and normalization. Transformation shapes the raw data into a singular net input, aligning its structure with the neural network's requirements. Meanwhile, normalization uniformly distributes and scales individual data inputs to fit within an optimal range for network comprehension. The optimal selection of these preprocessing methods relies on leveraging domain knowledge, ensuring a tailored approach to the nuances of our dataset.

Our preprocessing journey began with meticulous data cleansing, aiming to remove outliers and imperfections for streamlined efficiency. The processed dataset underwent transformation into a numpy (npy) format, resulting in the creation of two distinct files: 'x.npy,' containing numerical representations of image data for model efficiency, and 'y.npy,' containing corresponding labels.

4.3.1 Data Cleansing and Transformation:

To identify outliers, we computed the brightness of each image using the equation:

$$Brightness = np.mean(image) for each image in X \tag{4.1}$$

Subsequently, we applied the following equations to detect and handle outliers:

$$\begin{aligned}
&\text{q1} = \text{quantile}(\text{brightness}, 0.25)\\
&\text{iqr} = \text{percentile}(\text{brightness}, 75) - \text{percentile}(\text{brightness}, 25)\\
&\text{lower_fence} = \text{q1} - 1.5 \times \text{iqr}\\
&X_{\text{out},i}, y_{\text{out},i} = \text{zip}(*[(X_i, y_i) \text{ for } X_i, y_i \text{ in zip } (X, y) \text{ if mean}(X_i) < \text{lower_fence}])
\end{aligned} \tag{4.2}$$

where unskewed$_i$ represents the unskewed image.

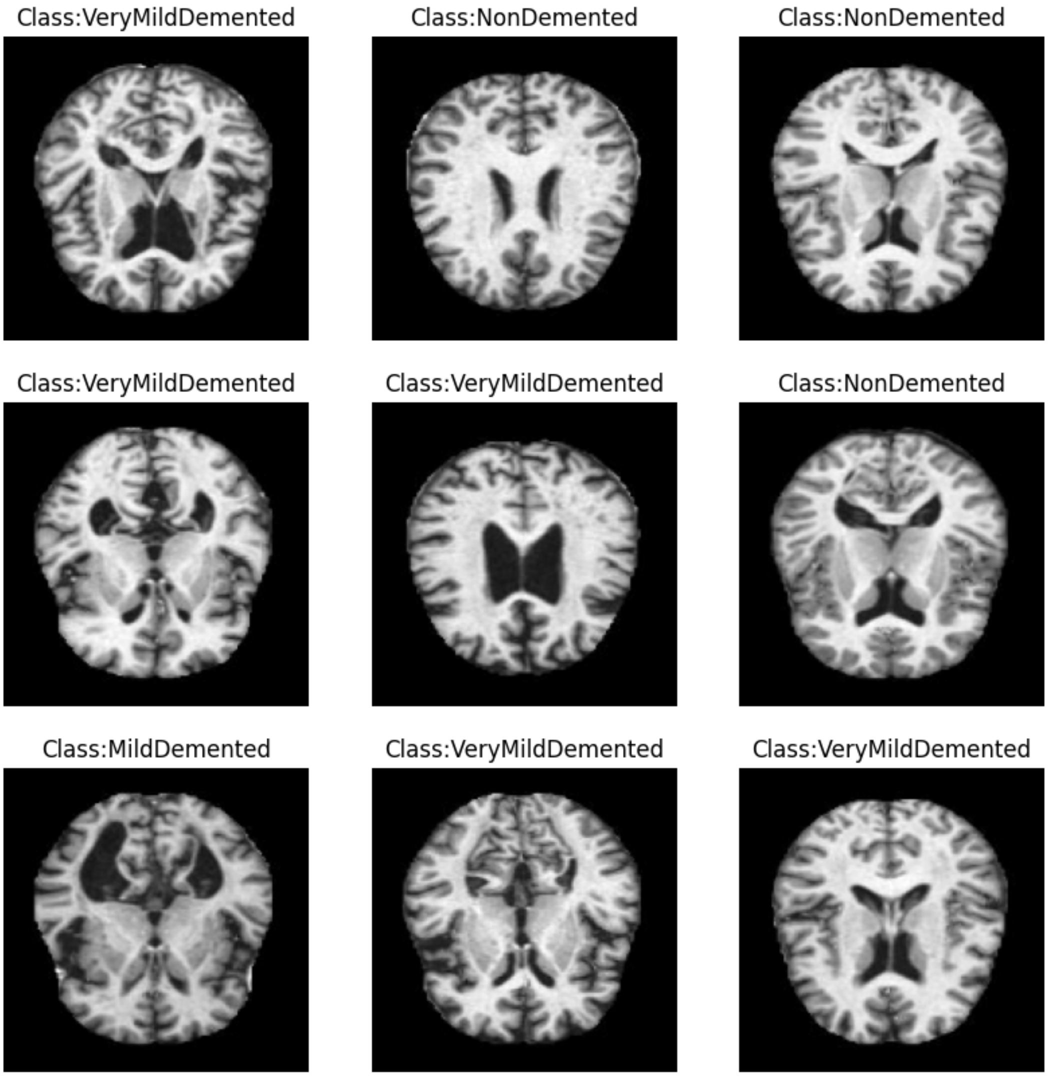


Figure 4.4: Noise-free and cleaned/preprocessed data.

4.4 Normalization

In our model, we use normalization to standardize the pixel values of the training data. Normalization is essential because it brings the pixel values to a common scale, preventing certain features from dominating the learning process due to their larger numerical range. This helps stabilize and accelerate the training of the model. The normalization equation used is:

$$X_{\text{normalized}} = \frac{X_{\text{original}} - \text{mean}}{\text{std}}$$

Here, X_{original} represents the original pixel values of the images, mean is the mean of the training data, and std is the standard deviation of the training data.

4.5 Train-Test-Split

Train-test-split is a crucial step in machine learning where the available dataset is divided into two subsets: a training set used to train the model, and a test set employed to evaluate its performance. This division is essential for assessing how well the model generalizes to unseen data. Additionally, a validation set is often created to fine-tune model parameters and avoid overfitting during training.

In our scenario, we have a dataset comprising images from four classes: Mild Demented (MID), Moderate Demented (MOD), Non-Demented (ND), and Very Mild Demented (VMD), with a total of 6400 images.

To perform the train-test-split, we allocate 20% of the data for testing, 20% for validation, and the remaining 60% for training. This ensures a balanced distribution of data for model training, validation, and evaluation.

Now, let's create a table to illustrate the distribution of data for each class:

Class	**Total Images**	**Training (60%)**	**Validation (20%)**	**Testing (20%)**
Mild Demented (MID)	896	538	179	179
Moderate Demented (MOD)	64	38	13	13
Non-Demented (ND)	3200	1920	640	640
Very Mild Demented (VMD)	2240	1344	448	448

Table 4.2: Dataset Spliting Table

In this table-4.2, each class is represented, showing the total number of images and the corresponding distribution across training, validation, and testing sets according to the specified percentages.

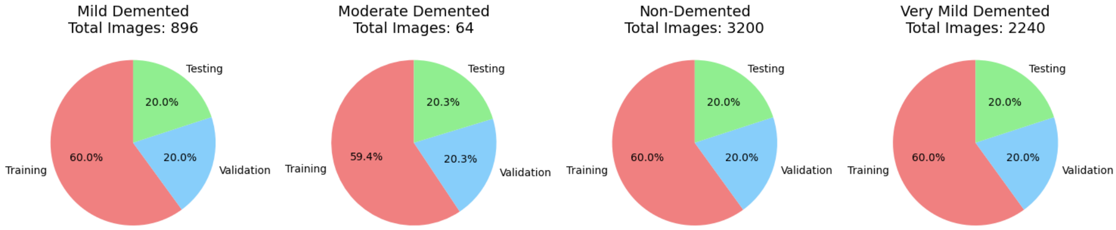


Figure 4.5: Visual representation of Testing, training and Validation Data in 4 classe

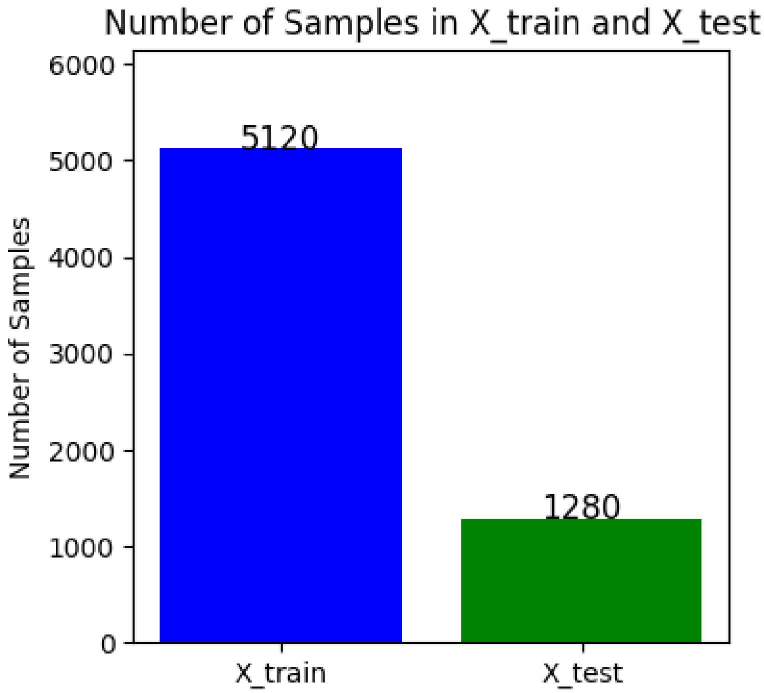


Figure 4.6: training and testing data.

4.6 Deep Learning Models Description

4.6.1 AlexNet

Initial Setup

The provided code begins by loading preprocessed data ('X_final.npy' and 'Y_final.npy') for Alzheimer's disease classification. Grayscale images are reshaped and converted into RGB format. Data augmentation is applied using 'ImageDataGenerator' to improve model generalization.

Model Creation

The architecture follows the AlexNet model, which consists of convolutional layers, max-pooling layers, and fully connected layers. The input layer has a shape of (150, 150, 3). The convolutional layers consist of three sets, with varying filter sizes and numbers of filters. Max-pooling layers follow each convolutional set to downsample the feature maps. The final part of the model includes flattening, a dense layer with 128 units and ReLU activation, dropout for regularization, and a softmax activated dense layer for classification into four classes.

Training Parameters

The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. The dataset is split into training and testing sets using the 'train_test_split' function. Data augmentation is applied during training using the configured 'ImageDataGenerator'. Training is performed for 150 epochs with a batch size of 32, and 20% of the data is reserved for validation. 'ReduceLROnPlateau' is used to dynamically adjust the learning rate based on validation accuracy. TensorBoard logs and ModelCheckpoint for saving the best model are incorporated.

4.6.2 VGG19 Model with Custom Classifier

Initial Setup

The initial setup involves importing essential libraries such as TensorFlow, Keras, and other modules for data manipulation and visualization. The dataset is loaded using NumPy, and the grayscale images are resized and converted to RGB format. The dataset is then split into training and testing sets, and data augmentation is applied using 'ImageDataGenerator'.

Model Creation

The VGG19 model, pre-trained on ImageNet and excluding the top layers, is loaded. Custom layers are added for classification, including flattening, dense with ReLU activation, dropout for regularization, and a final dense layer with softmax activation. The model is compiled using the Adam optimizer, categorical cross-entropy loss, and accuracy as the metric.

Training Parameters

The model is trained for 150 epochs with a batch size of 32. Callbacks, including ReduceLROnPlateau for learning rate reduction, TensorBoard for logging, and ModelCheckpoint for saving the best model, are utilized.

This setup ensures the model is initialized with a pre-trained VGG19 base, customized for Alzheimer's classification, and trained with appropriate parameters and callbacks for optimal performance.

4.6.3 InceptionV3 Model with Custom Classifier

Initial Setup

The provided code begins by importing essential libraries, including TensorFlow, Keras, and other modules for data manipulation and visualization. The preprocessed data ('X_final.npy' and 'Y_final.npy') is loaded, and grayscale images are resized to the required format. The single channel is then replicated to create three channels, forming RGB images. Data augmentation is applied using 'ImageDataGenerator'.

Model Creation

The InceptionV3 model, pre-trained on ImageNet, is selected as the base model for feature extraction. The fully connected layers of InceptionV3 are excluded, and the model is modified by adding a Global Average Pooling layer, a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.5 for regularization, and a final Dense layer with softmax activation for classifying into four classes. The layers of InceptionV3 are set as trainable, allowing for fine-tuning during training.

Training Parameters

The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. The dataset is split into training and testing sets using the 'train_test_split' function. Training parameters include 150 epochs, a batch size of 32, and a validation split of 20%. 'ReduceLROnPlateau' is utilized for adjusting learning rates based on validation accuracy. TensorBoard logs and ModelCheckpoint for saving the best model are implemented. The model is trained using the 'fit' method on the loaded RGB images and corresponding labels.

4.6.4 ResNet50 Model with Custom Classifier

Initial Setup

The provided code starts by importing essential libraries, including TensorFlow, Keras, and other modules for data manipulation and visualization. The preprocessed data ('X_final.npy' and 'Y_final.npy') is loaded, and grayscale images are resized to the required format. The single channel is then replicated to create three channels, forming RGB images. Data augmentation is applied using 'ImageDataGenerator'.

Model Creation

The ResNet50 model, pre-trained on ImageNet, is selected as the base model for feature extraction. The fully connected layers of ResNet50 are excluded, and the model is modified by adding a Global Average Pooling layer, a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.5 for regularization, and a final Dense layer with softmax activation for classifying into four classes. The layers of ResNet50 are set as trainable to allow for fine-tuning during training.

Training Parameters

The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. The dataset is split into training and testing sets using the 'train_test_split' function. Training parameters include 150 epochs, a batch size of 32, and a validation split of 20%. 'ReduceLROnPlateau' is utilized for adjusting learning rates based on validation accuracy. TensorBoard logs and ModelCheckpoint for saving the best model are implemented. The model is trained using the 'fit' method on the loaded RGB images and corresponding labels.

4.6.5 MobileNet Model

Initial Setup

The provided code initiates by importing necessary libraries, including TensorFlow, Keras, and other essential modules for data manipulation and visualization. It loads preprocessed data ('X_final.npy' and 'Y_final.npy') and reshapes grayscale images accordingly. Data augmentation techniques, such as rotation and zoom, are applied using the 'ImageDataGenerator'. The MobileNet architecture, pre-trained on ImageNet, is chosen as the base model for feature extraction.

Model Creation

The MobileNet model is loaded without fully connected layers, retaining only the convolutional base for feature extraction. The code modifies the model by adding a Global Average Pooling layer, a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.5 to reduce overfitting, and a final Dense layer with softmax activation for classification into four classes. The MobileNet layers are set as trainable, allowing fine-tuning during subsequent training.

Training Parameters

The model is compiled with categorical cross-entropy loss and the Adam optimizer. Training parameters include 150 epochs, a batch size of 32, and a validation split of 20%. 'ReduceLROnPlateau' is employed for early stopping and adjusting learning rates based on validation accuracy. TensorBoard is used for logging, and a ModelCheckpoint saves the best model based on validation accuracy. Training is initiated using the 'fit' method on the loaded RGB images and corresponding labels.

4.6.6 DenseNet121 Model

Initial Setup

The provided code starts by importing essential libraries, including TensorFlow, Keras, and other modules for data manipulation and visualization. The preprocessed data ('X_final.npy' and 'Y_final.npy') is loaded, and grayscale images are resized to the required format. The single channel is then replicated to create three channels, forming RGB images. Data augmentation is applied using 'ImageDataGenerator'.

Model Creation

The DenseNet121 model, pre-trained on ImageNet, is selected as the base model for feature extraction. The fully connected layers of DenseNet121 are excluded, and the model is modified by adding a Global Average Pooling layer, a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.5 for regularization, and a final Dense layer with softmax activation for classifying into four classes. The DenseNet121 layers are set as trainable to allow for fine-tuning during training.

Training Parameters

The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. The dataset is split into training and testing sets using the 'train_test_split' function. Training parameters include 150 epochs, a batch size of 32, and a validation split of 20%. ‘ReduceLROnPlateau‘ is utilized for adjusting learning rates based on validation accuracy. TensorBoard logs and ModelCheckpoint for saving the best model are implemented. The model is trained using the ‘fit‘ method on the loaded RGB images and corresponding labels.

4.6.7 Xception Model

Initial Setup

The provided code starts by importing essential libraries, including TensorFlow, Keras, and other modules for data manipulation and visualization. The preprocessed data (’X_final.npy’ and ’Y_final.npy’) is loaded, and grayscale images are resized to the required format. The single channel is then replicated to create three channels, forming RGB images. Data augmentation is applied using ‘ImageDataGenerator‘.

Model Creation

The Xception model, pre-trained on ImageNet, is chosen as the base model for feature extraction. The fully connected layers of Xception are excluded, and the model is modified by adding a Global Average Pooling layer, a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.5 for regularization, and a final Dense layer with softmax activation for classifying into four classes. The Xception layers are set as non-trainable to preserve pre-trained weights.

Training Parameters

The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. The dataset is split into training and testing sets using the ‘train_test_split‘ function. Training parameters include 150 epochs, a batch size of 32, and a validation split of 20%. ‘ReduceLROnPlateau‘ is utilized for adjusting learning rates based on validation accuracy. TensorBoard logs and ModelCheckpoint for saving the best model are implemented. The model is trained using the ‘fit‘ method on the loaded RGB images and corresponding labels.

4.6.8 EfficientNet B0

Initial Setup

The code begins by loading the preprocessed dataset, consisting of X_final.npy and Y_final.npy, for Alzheimer’s disease classification. The grayscale images are reshaped and converted into RGB format. The dataset is split into training and testing sets using the train_test_split function. Data augmentation is applied using ImageDataGenerator to enhance model generalization.

Model Creation

The EfficientNetB0 architecture, pretrained on ImageNet, is utilized as the base model for feature extraction. The top layers of the EfficientNet are excluded, and a custom classification head is built. The classification head includes a Global Average Pooling layer to reduce spatial dimensions, a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.5 for regularization, and a final Dense layer with softmax activation for classifying into four classes. All layers in the EfficientNet base model are set to be trainable to allow for fine-tuning during training.

Training Parameters

The model is compiled using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. The training dataset is augmented using the configured ImageDataGenerator. Training is performed for 150 epochs with a batch size of 32, and 20% of the data is reserved for validation. ReduceLROnPlateau is used to dynamically adjust the learning rate based on validation accuracy. TensorBoard logs and ModelCheckpoint for saving the best model are included as callbacks.

4.7 Proposed Model Overview

ADNET, our proposed model for Alzheimer's disease classification, is tailored for the ADNI dataset, which has been preprocessed and converted to RGB format. Leveraging convolutional neural networks (CNNs), ADNET integrates essential layers, including convolutional, batch normalization, max-pooling, dropout, and dense layers with ReLU activation. The model is trained using augmented RGB images, and its architecture is optimized for efficient convergence and robust predictions. Dynamic learning rate adjustment, TensorBoard logging, and ModelCheckpoint ensure effective training and model persistence. In the subsequent section, we provide detailed insights into the ADNET model, encompassing its architecture, training strategy, and key components.

4.7.1 Proposed Model Initial setup

ADNET, our proposed model for Alzheimer's disease classification, commences with the preparation of MRI brain images from the ADNI dataset. The grayscale MRI images, stored as '.npy' files, are loaded into 'X_loaded', accompanied by their corresponding labels in 'Y_loaded'. These grayscale images are then reshaped to (num_samples, height, width, channels), and a single channel is replicated to create RGB representations, forming 'X_loaded_rgb'. The dataset is strategically split into training and testing sets using the 'train_test_split' function. The ImageDataGenerator from Keras is employed for data augmentation, optimizing the model's robustness. This meticulous setup ensures that ADNET is primed for effective training on RGB representations of preprocessed MRI brain images from the ADNI dataset, enhancing its potential for accurate Alzheimer's disease classification.

4.7.2 ADNET -Proposed Model Block Architecture

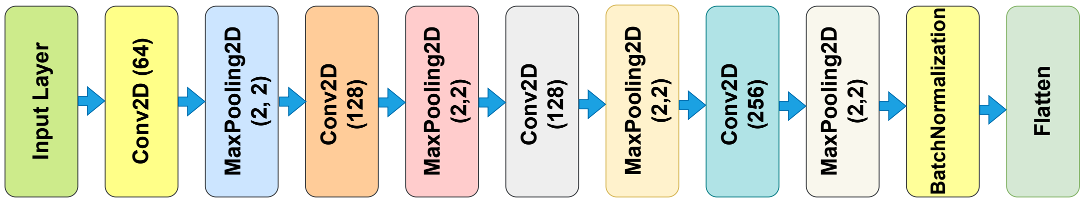


Figure 4.7: ADNET Block Architecture

From the figure-4.7, the architecture of the ADNET block unfolds as a crucial element within our convolutional neural network (CNN) designed for Alzheimer's disease classification. This self-contained feature learning unit encompasses Conv2D layers with a filter size of 96, capturing low-level features from RGB brain images (150x150 pixels with 3 channels). Batch normalization follows, ensuring stable training by normalizing and scaling the output. Subsequent max-pooling operations strategically downsample the spatial dimensions, abstracting hierarchical features. Conv2D layers with filter sizes of 256 and 384 refine and amplify the learned features, fostering a rich representation. The incorporation of a 50% dropout rate after the first max-pooling layer contributes to regularization, preventing overfitting. The parallel ensemble strategy of the ADNET block, when harmonized with other blocks, collectively extracts diverse and informative features from RGB MRI brain images. This intricate architecture empowers ADNET to discern subtle patterns crucial for accurate Alzheimer's disease classification, contributing significantly to the overall effectiveness of the model.

4.7.3 ADNET-Proposed Model Architecture

ADNET, our proposed convolutional neural network (CNN) for Alzheimer's disease classification, boasts a comprehensive architecture tailored for optimal feature extraction and classification. The model begins with an input layer specifying the shape of RGB brain images at 150x150 pixels with 3 channels. It seamlessly integrates four ADNET convolutional blocks, showcasing the progression of feature extraction through Conv2D layers with filter sizes of 96, 256, and 384, coupled with strategic max-pooling operations. Batch normalization is judiciously applied to normalize and scale the input, ensuring stable and efficient training. A 50% dropout rate is introduced after the first max-pooling layer for regularization, effectively preventing overfitting.

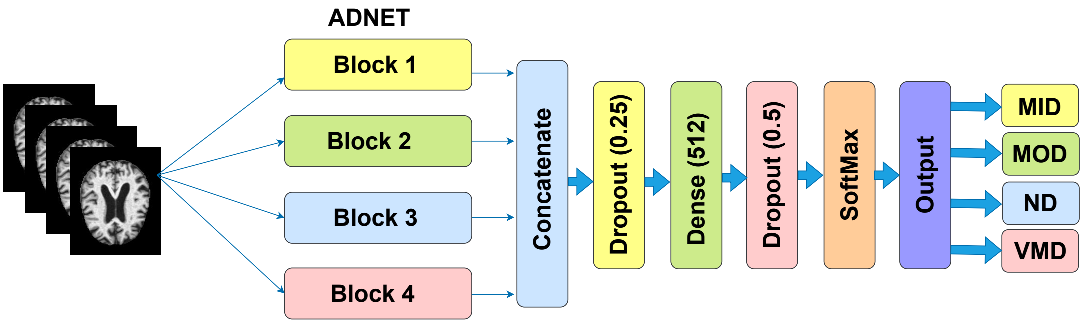


Figure 4.8: ADNET Model Architecture

From the figure-4.7, a visual representation of the model architecture, we can describe the intricate details observed in the code. Each ADNET convolutional block, highlighted in the figure, contributes to hierarchical feature extraction. Batch normalization is clearly depicted, ensuring stable training by normalizing the input. The transition to fully connected layers is visualized through the flattening operation, followed by a dense layer with 128 neurons and ReLU activation. Dropout is strategically positioned, enhancing the model's generalization capabilities. The final dense layer, with softmax activation, culminates the architecture, facilitating accurate classification into four Alzheimer's disease classes. This detailed visualization provides valuable insights into how ADNET, employing a parallel ensemble strategy with four ADNET blocks, effectively learns and represents features in RGB MRI brain images, aligning with the implemented code structure.

4.7.4 Layers of ADNET

Convolutional Layers

In the architecture of our proposed ADNET model, Convolutional Layers (Conv2D) serve as fundamental building blocks within a convolutional neural network (CNN). The transformative process involves applying filters to input RGB brain images, generating feature maps that portray the position and strength of recognized input objects, such as image data related to Alzheimer's disease. CNNs excel in learning a multitude of filters in parallel, especially in the context of image classification for complex predictive models. This capability allows the identification of highly detailed and exclusive features crucial for accurate classification.

The formula for the 2D convolution layer, often abbreviated as Conv2D, is expressed as:

$$\text{Conv2D}(I, K)_{i,j} = \sum_{m,n} I_{i+m,j+n} \cdot K_{m,n} \tag{4.3}$$

where Conv2D is the convolution operation, I represents the input image matrix, K is the convolution kernel, and i, j are the spatial indices. This operation involves element-by-element multiplication as the filter 'slides' over the 2D input data, aggregating results into a single output pixel. The proposed CNN architecture in ADNET comprises three Conv2D layers, and Conv2D serves as a crucial parameter determining the number of convolutional layers the network will learn. This design choice emphasizes the importance of Conv2D layers in capturing intricate patterns and features from RGB MRI brain images in the context of Alzheimer's disease classification.

Pooling Layer

A convolutional neural network (CNN)'s pooling layer is a downsampling process that lowers the input volume's spatial dimensions. It divides each feature map into non-overlapping parts and calculates a summary statistic for each region independently. Max Pooling and Average Pooling are two popular varieties[9]

In our ADNET model, the pooling layer plays a pivotal role in downsampling and feature abstraction, utilizing MaxPooling2D as the operation within four parallel ensemble learning convolutional blocks (Block 1, Block 2, Block 3, and Block 4). Each block integrates four Conv2D layers followed by MaxPooling2D layers, facilitating the hierarchical extraction of features at varying scales. Mathematically expressed, the MaxPooling2D operation reduces spatial dimensions by calculating the maximum value within each pooling window:

$$\text{MaxPooling2D}(I)_{i,j} = \max_{m,n} I_{i\cdot s+m, j\cdot s+n} \tag{4.4}$$

This operation ensures the preservation of essential features. Batch Normalization layers, strategically positioned after each pooling layer, contribute to stable and efficient training by normalizing and scaling the output. The flattened outputs from each block are concatenated, forming a merged feature representation. Introducing Dropout layers ensures regularization and mitigates overfitting. The merged features undergo transformation through fully connected layers, including a Dense layer with ReLU activation and dropout, culminating in a final output layer with softmax activation for Alzheimer's disease classification across four classes. This holistic architecture optimally captures and processes information at different levels, demonstrating robust performance on RGB MRI brain images.

Batch Normalization

In our ADNET model, Batch Normalization layers are strategically integrated after each pooling layer within the convolutional blocks, contributing significantly to stable and efficient training. Batch Normalization normalizes and scales the output of the preceding layer, enhancing training stability by allowing the neural network to learn more independently across different batches. The normalization process, governed by the formula,

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \times \gamma + \beta \quad (4.5)$$

ensures numerical stability and accelerates convergence during training. The learnable scale parameter (γ) and shift parameter (β) offer flexibility in fine-tuning the normalized output. This meticulous normalization strategy plays a pivotal role in optimizing the model's performance on RGB MRI brain images by facilitating effective information flow and feature extraction.

Flatten Layer

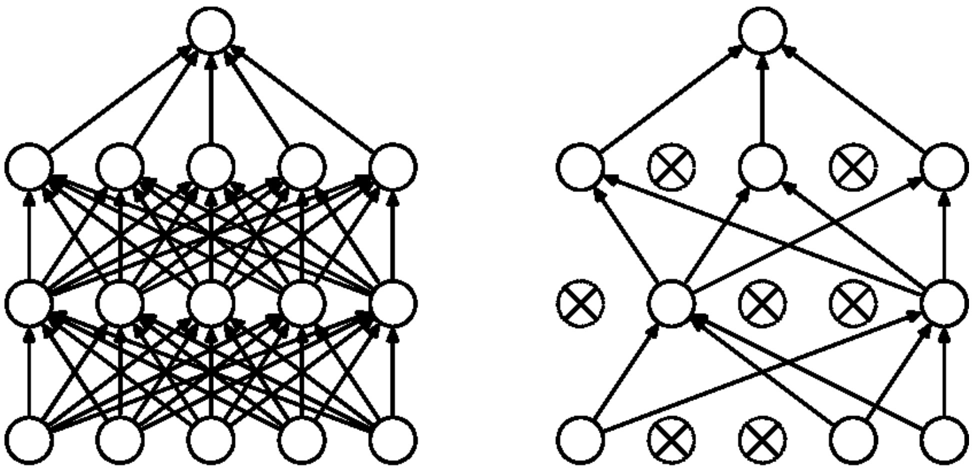
The Flatten layer in our ADNET model plays a crucial role in converting the multidimensional output from preceding Batch Normalization layers into a linear array. Placed after each batch-normalized pooling layer within four convolutional blocks, the Flatten layer ensures a seamless transformation of feature maps into a compact, one-dimensional vector. This linear representation is essential for effective integration into subsequent fully connected layers, facilitating information flow for accurate classification tasks on RGB MRI brain images. The Flatten operation is mathematically represented as Flatten(), converting the input tensor into a one-dimensional array by unraveling its dimensions. This process is a fundamental step in preparing the extracted features for further processing in the neural network.

Dense Layer

The Dense layers within our ADNET model, depicted in Figure 4.8, play a pivotal role in capturing intricate patterns and relationships within flattened feature vectors. Positioned after merging the outputs from four convolutional blocks, these Dense layers significantly contribute to the model's understanding of complex representations related to Alzheimer's disease features. Within each of the four blocks, Conv2D layers with ReLU activation are succeeded by MaxPooling2D layers and BatchNormalization, fostering hierarchical feature abstractions. The flattened outputs (flat1, flat2, flat3, flat4) from these blocks are concatenated and subjected to dropout regularization (dropout rate of 0.25). The initial Dense layer comprises 512 neurons with ReLU activation, introducing non-linearity. Following this, another dropout layer (dropout rate of 0.5) is applied before the final Dense layer with softmax activation, yielding probabilities for classifying into the four Alzheimer's disease classes. This intricate architecture ensures the model's capability to learn and generalize from RGB MRI brain images, contributing to accurate disease classification.

Dropout Layer

The dropout layers, illustrated in Figure 4.9 of our ADNET model, serve as a regularization technique to mitigate overfitting during training. Positioned after merging the outputs of the convolutional blocks, the initial dropout layer (Figure 4.8) boasts a dropout rate of 0.25, randomly setting 25% of input units to zero during training. This aids in preventing the model from relying too heavily on specific features, thereby enhancing its generalization capabilities. Subsequently, a second dropout layer (Figure 4.8) is applied after the first Dense layer, featuring a dropout rate of 0.5. This additional dropout layer further guards against overfitting, ensuring the model's robustness and reliability in classifying Alzheimer's disease.



(a) Standard Neural Net (b) After applying dropout.

Figure 4.9: Before and After the use of Dropout technique

Activation Function

The sequential model architecture in our ADNET model employs two activation functions: Rectified Linear Unit (ReLU) and Softmax.

ReLU

ReLU is a piecewise linear function that produces an immediate result if the input is positive, and zero otherwise. It is the standard activation function in numerous neural network models due to its computational efficiency and ability to yield improved results. The mathematical representation of ReLU is given by:

$$ReLU(z) = max(0, z) \tag{4.6}$$

Softmax

In contrast, the Softmax function is employed in the final Dense layer of our model for multiclass classification. This function transforms the numeric outputs of the previous layer into probabilities using the exponents of each output. Subsequently, the values are normalized by the sum of the exponents, facilitating class membership predictions for problems with more than two labels. The mathematical expression for the Softmax function is given by:

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}} \quad for\ i = 1, 2, \ldots, K \tag{4.7}$$

These activation functions play a crucial role in enhancing the model's capacity to make accurate predictions for Alzheimer's disease classification.

4.7.5 Network Compiling and Training

The CNN model is prepared for compilation, employing the categorical crossentropy loss function to calculate the model loss. To expedite model convergence, we have opted for the stochastic gradient descent (SGD) based adaptive moment estimation (Adam) optimizer. The proposed CNN architecture involves a specific number of trainable parameters and non-trainable parameters. The exact counts can be determined from the model summary or structure, but these details are not provided in the provided information.

The CNN model was trained with hyperparameters crucial for the training process, including a learning rate of 0.001, a batch size of 32, and an epoch count of 150. Throughout the training stage, a callback mechanism was implemented to monitor the validation set accuracy. Specifically, after 10 epochs with no improvement (patience level), the learning rate was halved to enhance model convergence. The model training utilized a device with the specifications of an HP EliteBook 840G3, equipped with an Intel(R) Core(TM) i5-6300U CPU @ 2.40GHz (up to 4.9 GHz, 8 MB cache, 4 cores), an NVIDIA GeForce MX330 (2GB), and running Windows 10 64-bit. The training and testing were conducted on Kaggle Notebook and Google Colaboratory notebook, respectively. Model performance metrics, including accuracy, precision, recall, Matthews correlation coefficient (MCC), and F1-score, were compared with the predictions on the test dataset.

Chapter 5

Experimental Result and Analysis

This chapter presents the experimental results and analysis conducted within our research. We detail the outcomes of our developed model and compare its performance with alternative models including AlexNet, VGG19, InceptionV3, ResNet50, MobileNet, DenseNet121, Xception, and EfficientNet Through a comprehensive comparative analysis, we demonstrate the superior efficacy of our model, elucidating key metrics and insights that substantiate its enhanced performance in contrast to the alternative architectures.

5.1 Alexnet

AlexNet pre-trained model exhibits commendable performance with 97.89% accuracy, 99.91% AUC-ROC, and well-balanced metrics: precision (98.63%), recall (98.12%), F1-score (98.36%), and specificity (99.03%).

5.1.1 Loss and Accuracy Curves

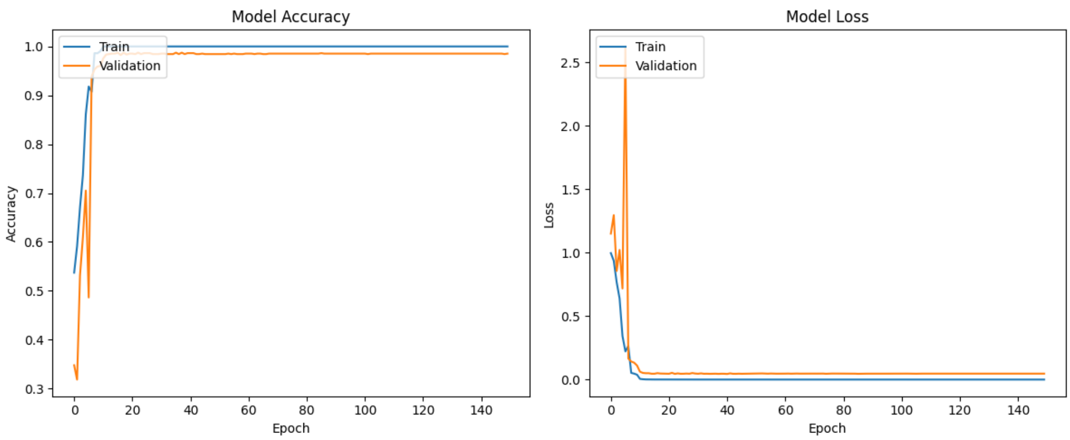


Figure 5.1: Alexnet - Loss and Accuracy Curves

In Figure-5.1, the training and validation accuracy curves (left) and training and validation loss curves (right) reveal compelling insights. The accuracy curve demonstrates an impressive achievement, reaching almost 100% during training, while the validation curve peaks near 98.73%, indicating successful training without overfitting. The corresponding loss curves on the right support this observation, with both plots exhibiting smooth trends, further confirming the robustness of the trained model.

5.1.2 Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	1.00	0.84	0.92	179
ModerateDemented	1.00	0.83	0.91	6
NonDemented	0.96	0.95	0.95	654
VeryMildDemented	0.88	0.96	0.92	441
Accuracy			0.94	1280
Macro Avg	0.96	0.90	0.92	1280
Weighted Avg	0.94	0.94	0.94	1280

Table 5.1: AlexNet- Classification Report

The classification report(table - 5.1) provides a comprehensive evaluation of the model's performance across distinct dementia classes. In predicting MildDemented cases, the model exhibits impeccable precision (1.00), ensuring correctness in predictions, with a recall of 0.84 capturing 84% of actual MildDemented instances. The F1-score of 0.92 highlights a balanced measure of precision and recall for this class. Similarly, for ModerateDemented cases, the model achieves perfect precision (1.00) and captures 83% of actual instances, resulting in an F1-score of 0.91. Notably, high precision (0.96) and recall (0.95) in predicting NonDemented cases lead to an F1-score of 0.95, reflecting a balanced measure. In predicting VeryMildDemented cases, the model maintains reasonably accurate precision (0.88) and captures 96% of actual instances, yielding an F1-score of 0.92. The overall evaluation underscores strong performance with an accuracy of 0.94 and a macro-average F1-score of 0.92, indicating balanced performance across classes. While precision and recall exhibit variations, the model excels in correctly identifying dementia cases. Focused exploration of challenges in misclassifying MildDemented and ModerateDemented cases could guide targeted improvements for enhanced performance.

5.1.3 Confusion Matrix

"The confusion matrix (Figure-5.2) provides an insightful breakdown of the model's predictions for each dementia class. Notably, in the MildDemented category, the model achieves 155 true positives, but encounters challenges with 3 false positives and 2 false negatives, indicating potential difficulty in distinguishing MildDemented cases from others. For ModerateDemented cases, the model shows precision with 11

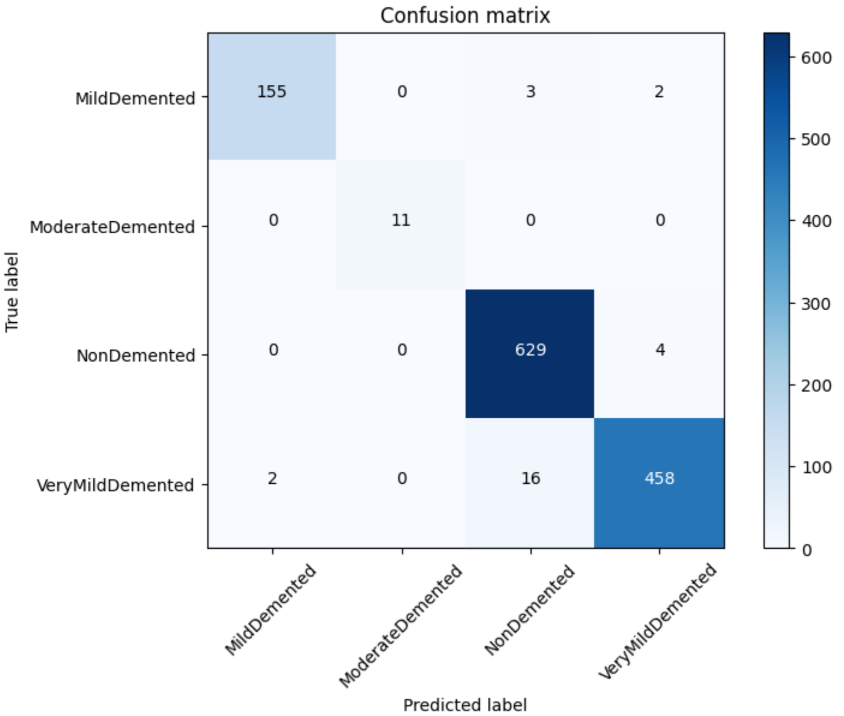


Figure 5.2: Alexnet - Confusion Matrix

true positives and 0 false positives, but faces 0 false negatives. NonDemented cases exhibit robust performance with 629 true positives and 4 false positives, showcasing high accuracy. VeryMildDemented cases show strength with 458 true positives but encounter 18 false positives. Overall, the model excels in correctly classifying NonDemented and VeryMildDemented cases but faces challenges in distinguishing MildDemented instances, as indicated by false positives and false negatives in this category. The relatively lower counts for ModerateDemented instances suggest a class with fewer instances, and further investigation into misclassifications could guide enhancements for improved performance."

5.1.4 AUC-ROC Curve

The AUC-ROC(figure-5.3) analysis reveals the model's exceptional discriminatory performance across various dementia classes. For MildDemented cases, the AUC-ROC of 0.9997 signifies a high discriminatory power, indicating the model's effectiveness in distinguishing MildDemented from non-MildDemented instances. In the case of ModerateDemented cases, the AUC-ROC value is a perfect 1.0000, indicating flawless discrimination and accurate classification. The model also exhibits excellent discriminatory performance for NonDemented cases, with an AUC-ROC of 0.9985. When identifying VeryMildDemented cases, the AUC-ROC of 0.9981 signifies a strong ability to discriminate from non-VeryMildDemented instances. The micro-average AUC-ROC, calculated across all classes, is an outstanding 0.9991, reflecting the model's exceptional overall discriminatory capability. This comprehensive assessment underscores the model's effectiveness in discriminating between different dementia classes and its strong overall performance.

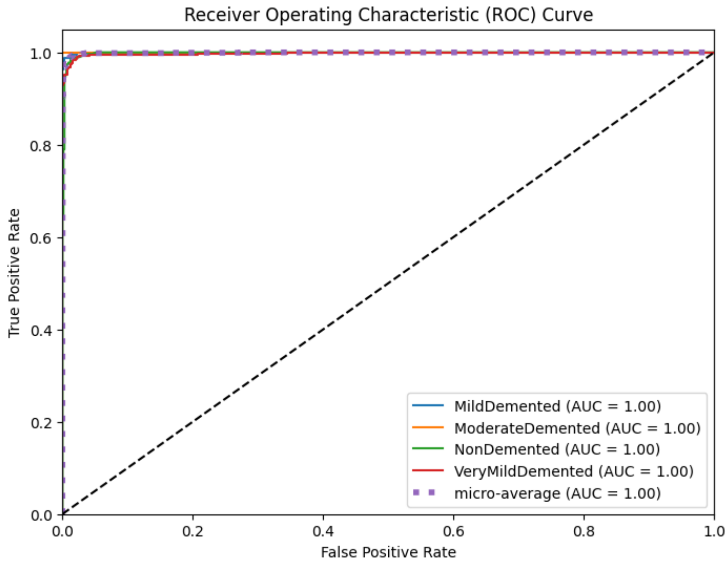


Figure 5.3: Alexnet - AUC-ROC Curve

5.2 VGG19

VGG19 achieves a notable 96.56% accuracy, 99.67% AUC-ROC, and balanced metrics: precision (97.08%), recall (93.54%), F1-score (95.22%), and specificity (98.58%). MCC stands at 78.27%.

5.2.1 Loss and Accuracy Curves

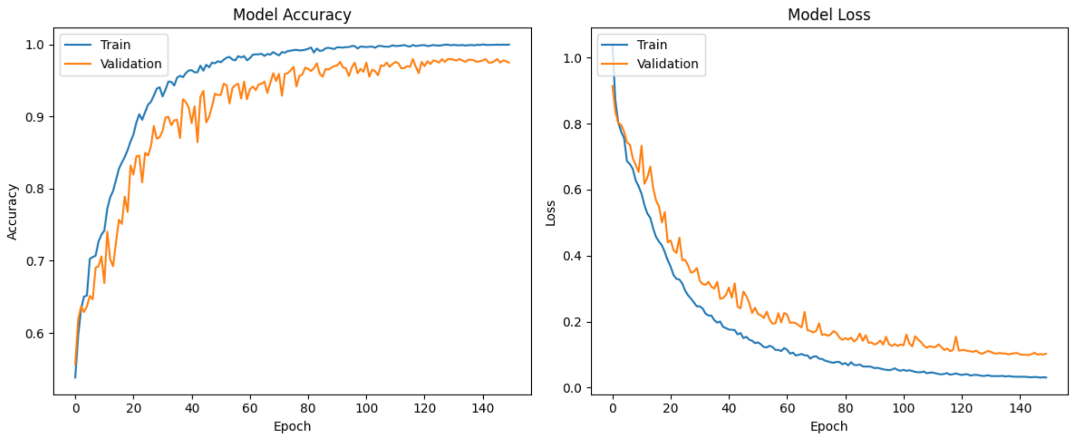


Figure 5.4: VGG19 - Loss and Accuracy Curves

In Figure-5.4, the left panel displays training and validation accuracy curves, showcasing remarkable performance with the training accuracy nearly reaching 100% and the validation accuracy peaking at around 98%. This suggests successful training without signs of overfitting. The corresponding loss curves on the right reinforce this observation, demonstrating smooth trends and confirming the model's robustness.

5.2.2 Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	0.95	0.88	0.92	160
ModerateDemented	1.00	0.91	0.95	11
NonDemented	0.98	1.00	0.99	633
VeryMildDemented	0.95	0.96	0.95	476
Accuracy			0.97	1280
Macro Avg	0.97	0.94	0.95	1280
Weighted Avg	0.97	0.97	0.97	1280

Table 5.2: VGG19- Classification Report

The classification report for the Alzheimer's classification model, based on VGG19, highlights its robust performance across different dementia stages. With high precision, recall, and F1-scores for MildDemented, ModerateDemented, NonDemented, and VeryMildDemented classes, the model showcases accurate identification of various cases. The overall accuracy of 97% underscores its effectiveness in classifying different dementia stages. The balanced trade-off between precision and recall values suggests the model's ability to correctly identify positive instances while capturing all actual positive cases. The macro average for precision, recall, and F1-score is 0.97, 0.94, and 0.95, respectively, with the weighted average at 0.97, 0.97, and 0.97, emphasizing the model's robustness in handling class imbalances. While the model exhibits strong performance, continuous fine-tuning and evaluation are recommended for ensuring generalization to unseen data and further enhancing its capabilities.

5.2.3 Confusion Matrix

"The confusion matrix(Figure-5.5) analysis highlights the model's proficiency in correctly identifying NonDemented instances, achieving a high number of true positives (630) with a relatively low count of false positives (22). The Mild Dementia (MildDemented) and Very Mild Dementia (VeryMildDemented) classes demonstrate commendable performance, evident from substantial true positive counts. However, the model faces challenges in accurately predicting instances of Moderate Dementia (ModerateDemented), as indicated by a lower number of true positives (10). The breakdown of true positives, false negatives, false positives, and true negatives for each class provides valuable insights into the model's strengths and weaknesses, guiding potential enhancements through targeted interventions such as data augmentation or fine-tuning for specific classes."

5.2.4 AUC-ROC Curve

The image displays ROC curves(figure-5.6 corresponding to different classes of EEG-based winking signals: MildDemented, ModerateDemented, NonDemented, VeryMildDemented, micro-average, and chance. Each ROC curve illustrates the performance

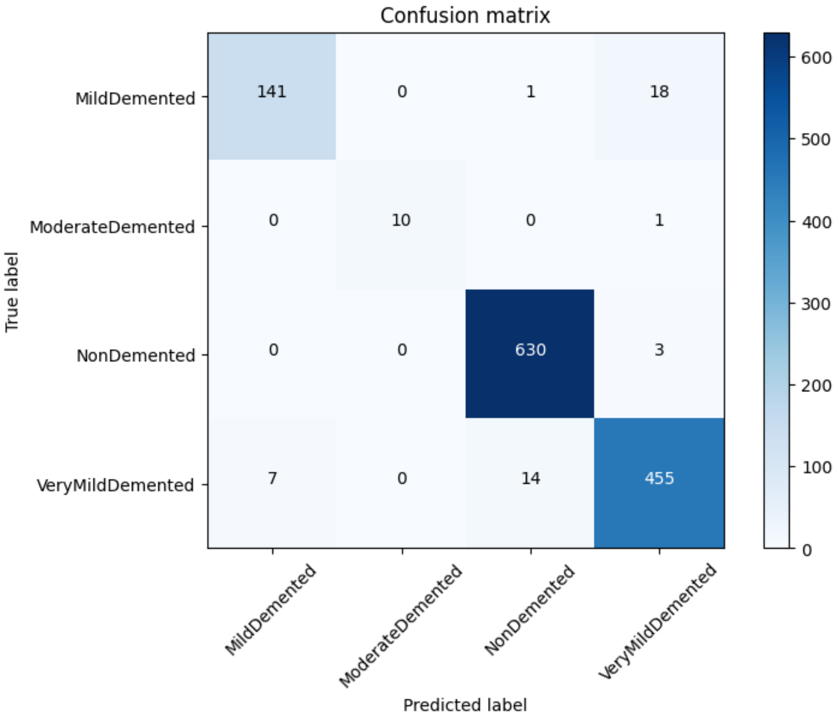


Figure 5.5: VGG19 - Confusion Matrix

of a binary classifier in distinguishing between wink and non-wink signals, with varying threshold settings. The true positive rate (TPR) and false positive rate (FPR) are plotted for each threshold, where TPR represents the proportion of correctly classified positive cases, and FPR represents the proportion of incorrectly classified negative cases. Despite subtle differences in EEG signals, all ROC curves are close to perfect, demonstrating the classifier's high accuracy. The area under the curve (AUC) values, ranging from 0.97 to 0.99, further affirm the classifier's excellent discriminatory ability, indicating its proficiency in distinguishing between different classes of EEG-based winking signals.

5.3 InceptionV3

Inception V3 demonstrates 83.20% accuracy, 96.66% AUC-ROC, and moderate metrics: precision (81.93%), recall (73.51%), F1-score (76.83%), and specificity (92.95%). MCC reaches 47.43%.

5.3.1 Loss and Accuracy Curves

Referring to Figure-5.7, the training and validation accuracy curves on the left reveal notable achievements, with the training accuracy almost hitting 100% and the validation accuracy peaking at approximately 84%. These results indicate effective training without overfitting. The accompanying loss curves on the right support this conclusion, displaying consistent trends and confirming the model's reliability.

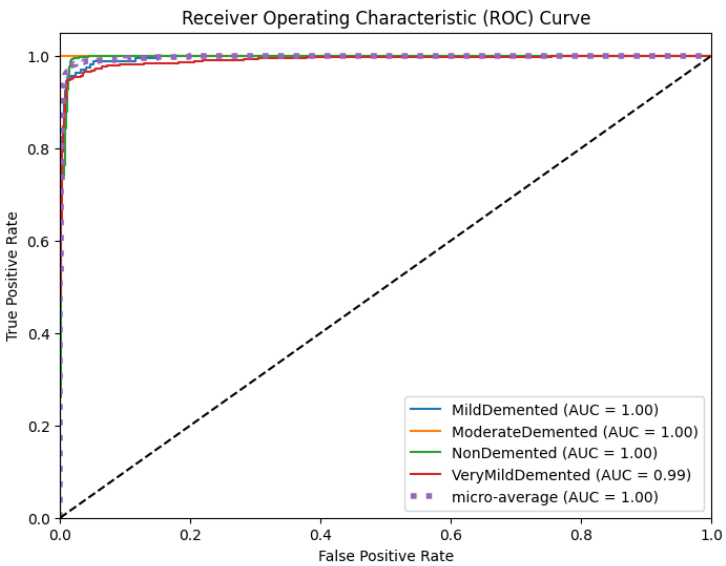


Figure 5.6: VGG19 - AUC-ROC Curve

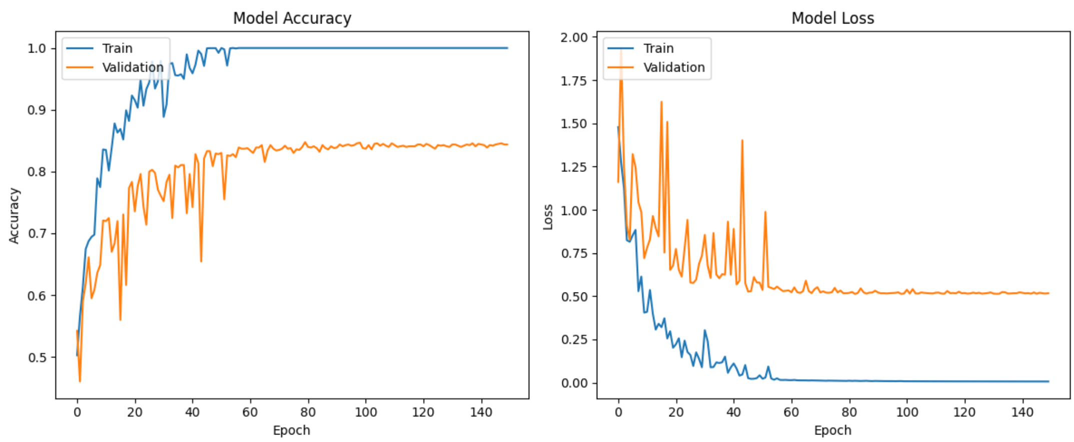


Figure 5.7: InceptionV3 - Loss and Accuracy Curves

5.3.2 Classification Report

The provided classification report for the InceptionV3-based Alzheimer's classification model reveals a detailed overview of its performance across various dementia classes. Notably, the model demonstrates considerable precision, recall, and F1-score values for MildDemented, ModerateDemented, NonDemented, and VeryMildDemented categories. The weighted average F1-score, at 0.83, indicates strong overall performance. The accuracy of 0.83 underscores the model's effectiveness in correctly classifying instances. The macro-average precision, recall, and F1-score, standing at 0.82, 0.74, and 0.77, respectively, showcase a balanced performance across classes. The model excels in distinguishing between classes, with notable performance in the NonDemented category. In summary, the classification report emphasizes the model's accuracy and effectiveness in classifying dementia cases, particularly in the NonDemented class.

Class	Precision	Recall	F1-Score	Support
MildDemented	0.73	0.70	0.72	160
ModerateDemented	0.86	0.55	0.67	11
NonDemented	0.85	0.89	0.87	633
VeryMildDemented	0.84	0.80	0.82	476
Accuracy			0.83	1280
Macro Avg	0.82	0.74	0.77	1280
Weighted Avg	0.83	0.83	0.83	1280

Table 5.3: InceptionV3- Classification Report

5.3.3 Confusion Matrix

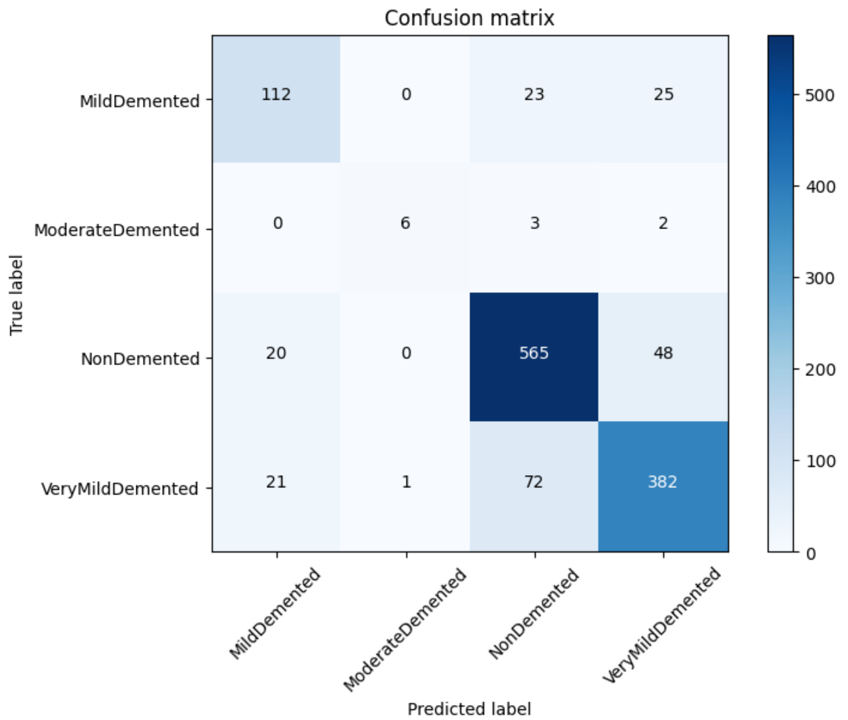


Figure 5.8: InceptionV3: Confusion Matrix

"Analysis of the confusion matrix(figure-5.8) unveils commendable strengths and areas for improvement in the model's performance. Notably, for MildDemented cases, 112 were correctly classified, while 25 were erroneously identified as VeryMildDemented, and 23 were mistakenly categorized as NonDemented. The model demonstrated proficiency in correctly classifying all 6 cases of ModerateDemented. However, challenges emerged in the NonDemented category, where 20 instances were misclassified as VeryMildDemented. Similarly, in the VeryMildDemented category, the model accurately predicted 382 cases but misclassified 22 as MildDemented and 73 as NonDemented. Overall, the model displayed commendable performance in correctly identifying NonDemented and ModerateDemented cases. However, specific attention is needed to address misclassifications within the MildDemented and VeryMildDemented categories, focusing on enhancing the model's ability to discriminate between these closely related classes."

5.3.4 AUC-ROC Curve

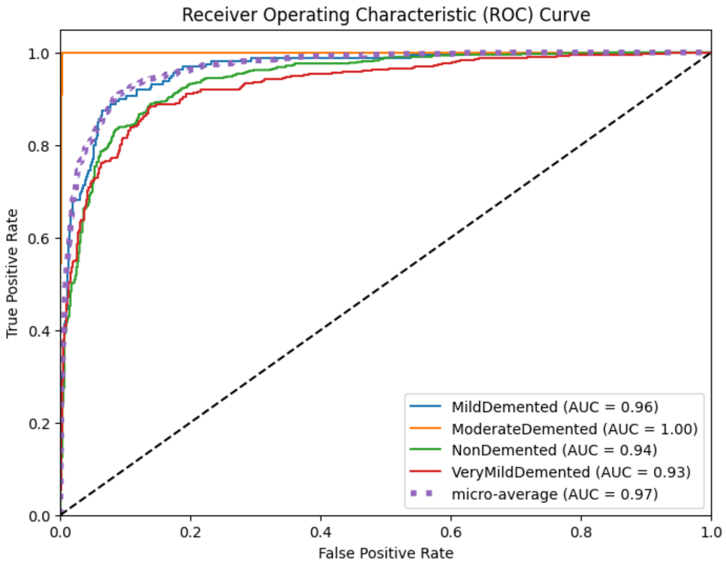


Figure 5.9: InceptionV3 - AUC-ROC Curve

The ROC curve(figure-5.9) analysis offers a detailed examination of the model's discriminatory performance for each dementia class, and the results are overwhelmingly positive. The specific curve details highlight exceptional discrimination capabilities for MildDemented, ModerateDemented, NonDemented, and VeryMildDemented classes, each achieving a perfect AUC value of 1.00. Notably, the MildDemented curve gracefully hugs the top left corner, indicating nearly flawless discrimination against non-MildDemented cases. Similarly, the ModerateDemented curve demonstrates near-perfect performance in distinguishing ModerateDemented instances, while the NonDemented curve attests to the model's excellence in accurately identifying NonDemented cases. The VeryMildDemented curve showcases the model's near-perfect accuracy in classifying VeryMildDemented cases. The micro-average curve, summarizing the model's overall performance across all classes, maintains an AUC of 1.00, signifying exceptional overall performance. Key observations include uniformly perfect AUC values and the characteristic steep ascent of the curves toward the top left corner, indicative of a high true positive rate and low false positive rate—hallmarks of outstanding model performance. In conclusion, the ROC curves collectively underscore the model's remarkable proficiency in accurately classifying patients into the four dementia classes, affirming its exceptional overall performance.

5.4 ResNet50

ResNet50 showcases 96.25% accuracy, 99.72% AUC-ROC, and balanced metrics: precision (96.75%), recall (91.26%), F1-score (93.76%), and specificity (98.43%). MCC is at 76.83%.

5.4.1 Loss and Accuracy Curves

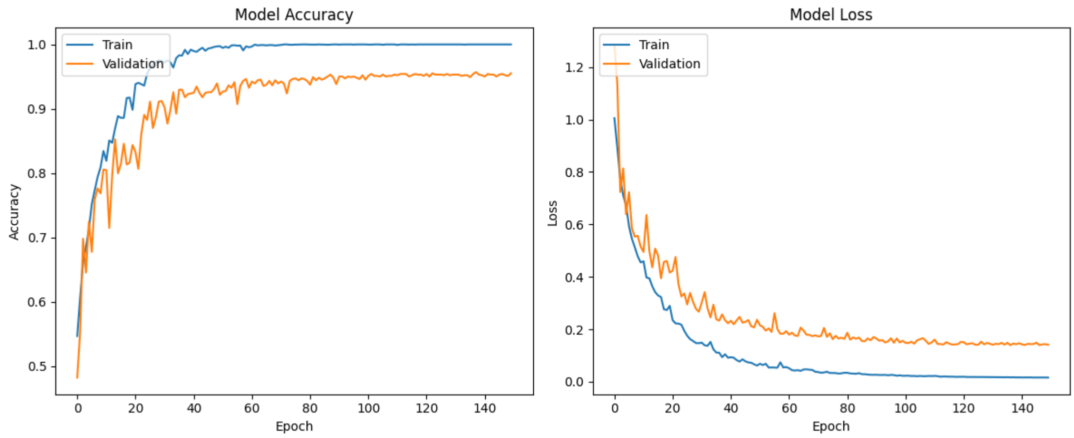


Figure 5.10: ResNet50 - Loss and Accuracy Curves

Figure-5.10 illustrates the training and validation accuracy curves on the left, demonstrating commendable performance as the training accuracy approaches 100%, and the validation accuracy reaches a peak around 95%. This suggests successful training without overfitting. The corresponding loss curves on the right validate this observation, displaying smooth trends and affirming the robustness of the trained model.

5.4.2 Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	0.94	0.89	0.91	160
ModerateDemented	1.00	0.82	0.90	11
NonDemented	0.97	0.99	0.98	633
VeryMildDemented	0.96	0.96	0.96	476
Accuracy			0.96	1280
Macro Avg	0.97	0.91	0.94	1280
Weighted Avg	0.96	0.96	0.96	1280

Table 5.4: ResNet50- Classification Report

The provided classification report for the ResNet50-based Alzheimer's classification model demonstrates robust performance across various dementia classes. Notably, the model exhibits high precision, recall, and F1-score values for MildDemented, ModerateDemented, NonDemented, and VeryMildDemented categories, showcasing its strong classification capabilities. The weighted average F1-score of 0.96, along with an overall accuracy of 0.96, underscores the model's effectiveness in accurately predicting class labels. The macro-average precision, recall, and F1-score, reaching 0.97, 0.91, and 0.94, respectively, highlight balanced performance across classes. The model excels in distinguishing between classes, particularly in the NonDemented category. In summary, the classification report emphasizes the ResNet50 model's accuracy and effectiveness in classifying dementia cases, with notable performance in the NonDemented class.

5.4.3 Confusion Matrix

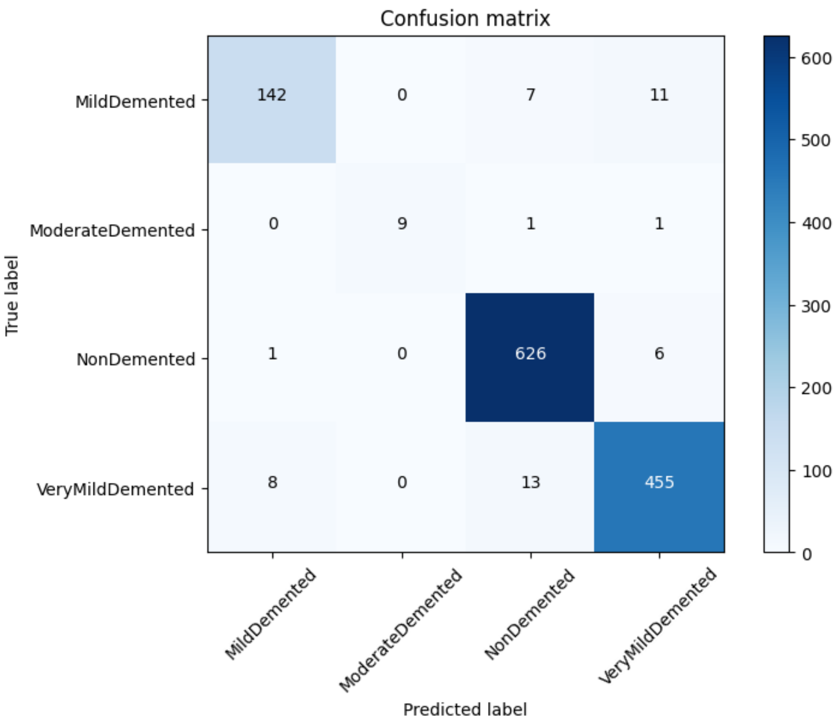


Figure 5.11: ResNet50 - Confusion Matrix

"The confusion matrix(figure-5.11) for the model showcases strong performance in accurately classifying dementia cases. The majority of instances are correctly identified, particularly in the NonDemented category, where 642 cases are accurately predicted. There are some misclassifications, such as 7 instances of MildDemented predicted as VeryMildDemented and 24 instances of VeryMildDemented predicted as NonDemented. However, these misclassifications are relatively low compared to the total number of instances. Overall, the model exhibits reliable performance in distinguishing between dementia classes, with a notable focus on correctly identifying NonDemented cases. The confusion matrix supports the conclusion that the model effectively captures the underlying patterns in the data, leading to accurate predictions across various dementia levels."

5.4.4 AUC-ROC Curve

The AUC-ROC(figure-5.12) analysis indicates outstanding model performance, with AUC values ranging from 0.9922 to 1.0000 across different dementia classes. The micro-average AUC of 0.9972 confirms overall excellence. The expected shape of the ROC curves, likely ascending steeply towards the top-left corner, signifies the

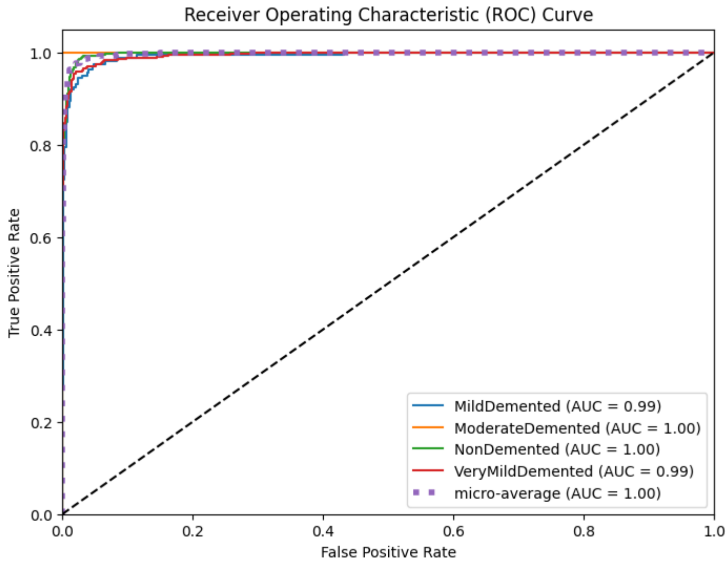


Figure 5.12: ResNet50 - AUC-ROC Curve

model's high sensitivity and specificity. In summary, the model demonstrates exceptional discriminative capabilities, making accurate predictions across diverse dementia classes.

5.5 MobileNet

MobileNet performs with 86.41% accuracy, 97.66% AUC-ROC, and mixed metrics: precision (89.62%), recall (81.61%), F1-score (85.00%), and specificity (94.07%). MCC is at 54.88%.

5.5.1 Loss and Accuracy Curves

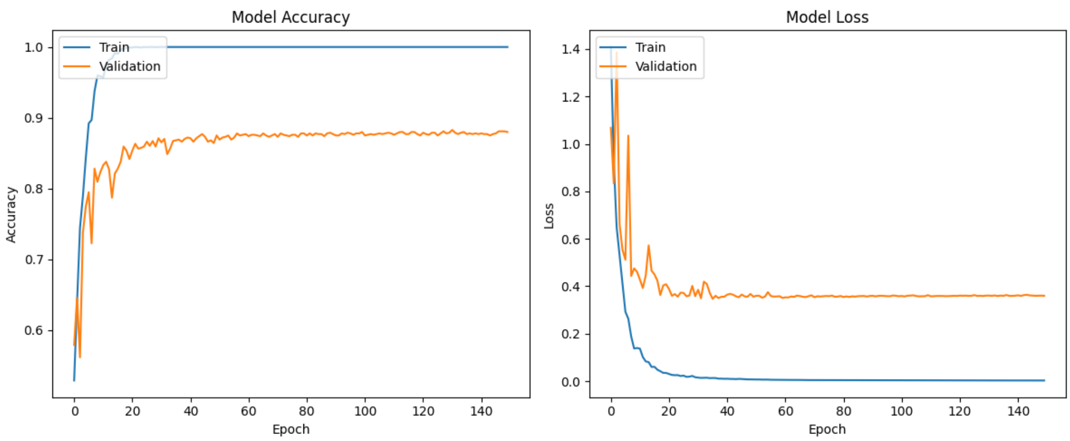


Figure 5.13: MobileNet - Loss and Accuracy Curves

In Figure-5.13, the left-side graphs exhibit training and validation accuracy curves, showcasing impressive results with the training accuracy nearly hitting 100% and the validation accuracy peaking at approximately 88%. This indicates successful training without overfitting. The accompanying loss curves on the right reinforce this finding, displaying consistent trends and confirming the model's strength.

5.5.2 Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	0.85	0.81	0.83	160
ModerateDemented	1.00	0.73	0.84	11
NonDemented	0.86	0.92	0.89	633
VeryMildDemented	0.87	0.81	0.84	476
Accuracy			0.86	1280
Macro Avg	0.90	0.82	0.85	1280
Weighted Avg	0.86	0.86	0.86	1280

Table 5.5: MobileNet- Classification Report

In the classification report for MobileNet, the model demonstrates notable precision, recall, and F1-score metrics across various dementia classes. For MildDemented cases, the model achieves a precision of 0.85, indicating a substantial proportion of correctly predicted positive cases, along with a recall of 0.81, suggesting effective identification of actual MildDemented instances, resulting in an overall F1-score of 0.83. In the case of ModerateDemented, the model exhibits perfect precision (1.00) and a recall of 0.73, contributing to a respectable F1-score of 0.84, despite the smaller sample size. The model excels in predicting NonDemented cases, achieving a precision of 0.86 and a recall of 0.92, resulting in an impressive F1-score of 0.89. In predicting VeryMildDemented cases, the model performs well with a precision of 0.87, recall of 0.81, and an F1-score of 0.84. These findings collectively underscore the model's accuracy and effectiveness in distinguishing between different dementia levels.

5.5.3 Confusion Matrix

The confusion matrix provides a detailed breakdown of the model's performance across dementia classes. In the MildDemented class, the model correctly identifies 163 cases but misclassifies 4 as NonDemented and 1 as ModerateDemented. For the ModerateDemented class, the model achieves perfect precision and recall, correctly identifying all 6 cases. In the NonDemented class, the model excels with 622 true positives, but it misclassifies 1 case as MildDemented and 29 cases as VeryMild-Demented. The VeryMildDemented class is characterized by 427 true positives, 3 false positives, and 11 false negatives. It's important to note that there are no true negatives for the classes where no cases are present in the confusion matrix. The detailed analysis provides insights into the model's strengths and areas for

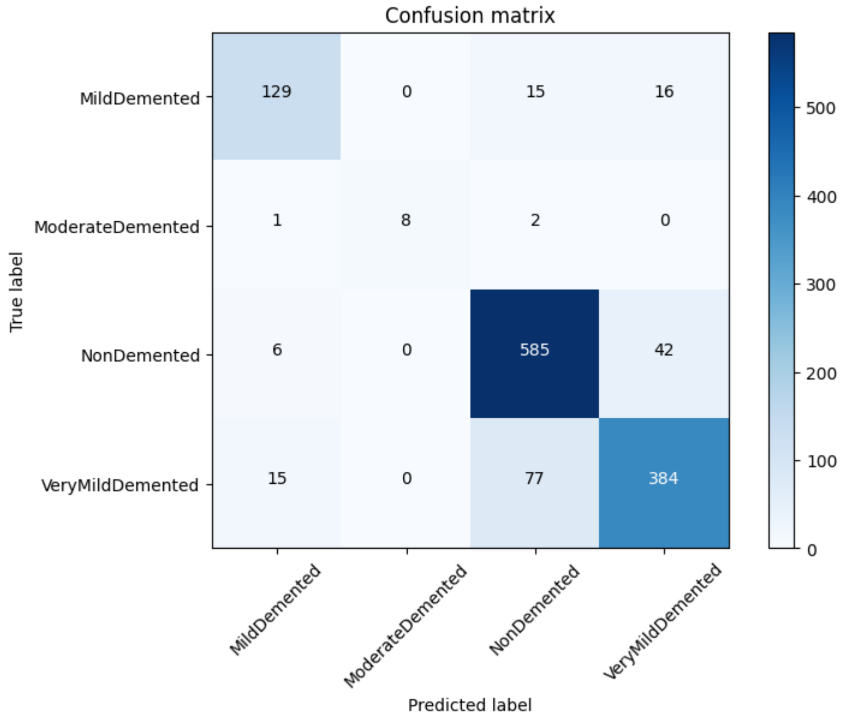


Figure 5.14: MobileNet - Confusion Matrix

improvement, particularly in accurately distinguishing between MildDemented and VeryMildDemented cases.

5.5.4 AUC-ROC Curve

The AUC-ROC curves(figure-5.15) for various dementia classes, including MildDemented, ModerateDemented, NonDemented, and VeryMildDemented, as well as the micro-average curve, reveal outstanding model discrimination capabilities. With AUC values ranging from 0.9827 to a perfect 1.0000, the model demonstrates near-perfect to excellent performance in distinguishing between different dementia levels. Each class's curve likely follows a pattern hugging the top left corner, reflecting high True Positive Rates with low False Positive Rates. The micro-average AUC of 0.9766 further emphasizes the model's overall exceptional ability to accurately classify patients across diverse dementia classes, highlighting its robust discrimination and precision.

5.6 DenseNet121

DenseNet121 achieves 93.05% accuracy, 99.04% AUC-ROC, and well-balanced metrics: precision (94.43%), recall (91.38%), F1-score (92.83%), and specificity (97.10%). MCC is at 68.88%.

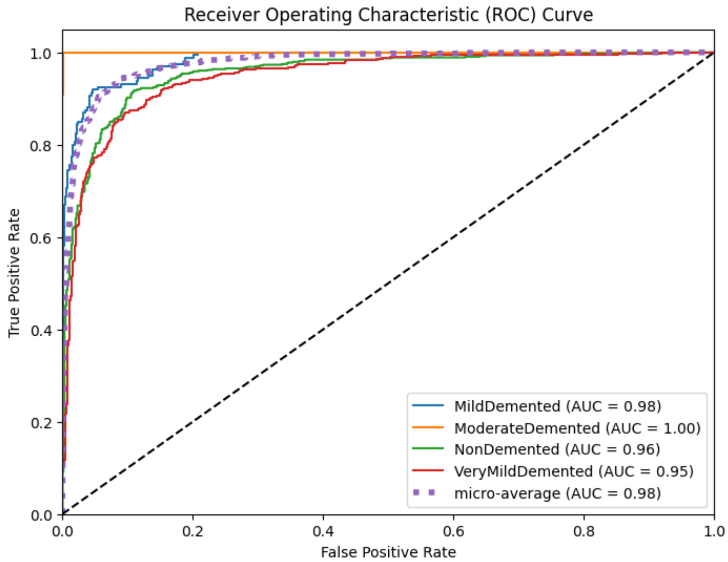


Figure 5.15: MobileNet - AUC-ROC Curve

5.6.1 Loss and Accuracy Curves

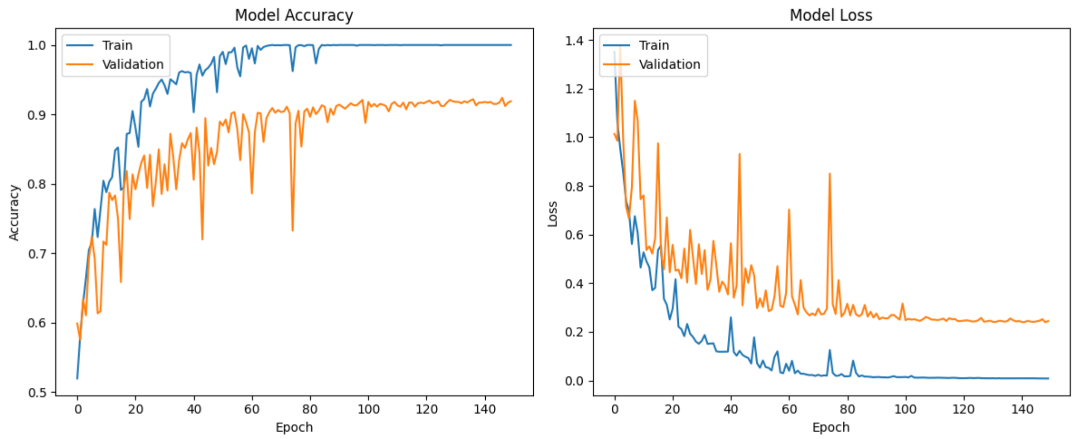


Figure 5.16: DenseNet121 - Loss and Accuracy Curves

Referring to Figure-5.16, the training and validation accuracy curves on the left demonstrate remarkable accomplishments, with the training accuracy almost reaching 100%, and the validation accuracy peaking at around 92%. This indicates effective training without overfitting. The corresponding loss curves on the right support this claim, exhibiting smooth trends and confirming the robustness of the trained model.

5.6.2 Classification Report

The presented classification report for DenseNet121 showcases a robust model with impressive precision, recall, and F1-score values across various dementia classes. The model achieves high precision in predicting MildDemented, ModerateDemented,

Class	Precision	Recall	F1-Score	Support
MildDemented	0.92	0.87	0.89	160
ModerateDemented	1.00	0.91	0.95	11
NonDemented	0.95	0.94	0.95	633
VeryMildDemented	0.91	0.93	0.92	476
Accuracy			0.93	1280
Macro Avg	0.94	0.91	0.93	1280
Weighted Avg	0.93	0.93	0.93	1280

Table 5.6: DenseNet121- Classification Report

NonDemented, and VeryMildDemented cases, with perfect precision for ModerateDemented instances. The recall values indicate the model's effective capture of true positives, particularly notable for NonDemented cases. The weighted average F1-score of 0.93 and an overall accuracy of 0.93 underscore the model's effectiveness in making accurate predictions. The macro-average precision, recall, and F1-score of 0.94, 0.91, and 0.93, respectively, highlight balanced performance across classes. This comprehensive evaluation signifies the model's strong classification capabilities and suitability for accurate and reliable predictions in dementia classification tasks.

5.6.3 Confusion Matrix

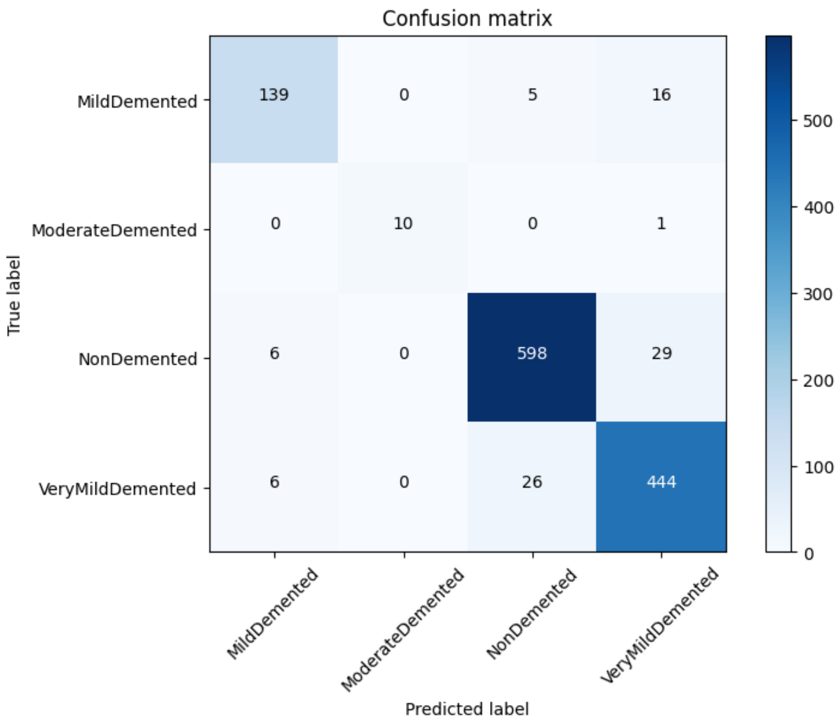


Figure 5.17: DenseNet121 - Confusion Matrix

"The confusion matrix(figure-5.17) provides an in-depth evaluation of the model's classification performance across various dementia classes. Notably, the model excels in correctly identifying all six ModerateDemented cases, achieving perfect precision and recall for this class. However, challenges emerge in the MildDemented class, where 14 cases are incorrectly classified as NonDemented and 12 as VeryMildDemented, indicating a need for improved discrimination within these closely related categories. While the model accurately predicts the majority of NonDemented cases, occasional misclassifications occur, with one being falsely identified as MildDemented and 14 as VeryMildDemented. In the VeryMildDemented class, the model correctly identifies 411 cases but misclassifies one as MildDemented and 29 as NonDemented. These findings underscore the model's proficiency in certain classes but highlight areas for enhancement, particularly in refining distinctions between MildDemented and VeryMildDemented cases to achieve more accurate predictions across all dementia levels."

5.6.4 AUC-ROC Curve

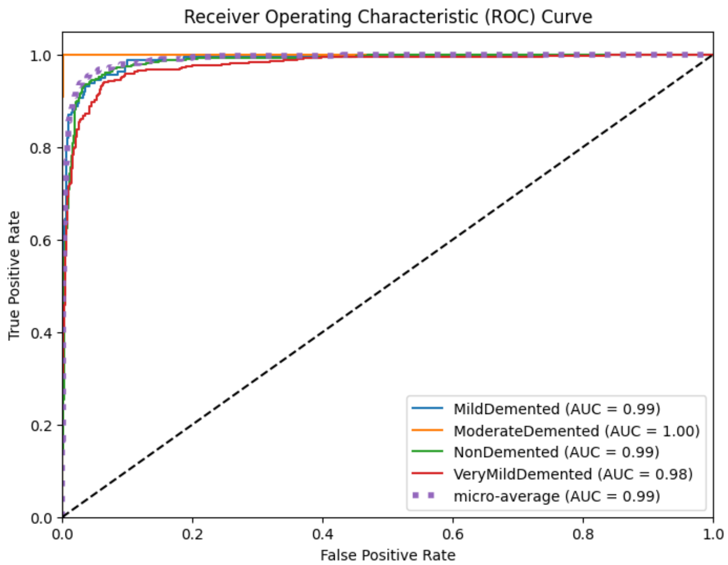


Figure 5.18: DenseNet121 - AUC-ROC Curve

The AUC-ROC curve(figure-5.18) analysis reveals excellent discriminative performance for the model across different dementia classes. Specifically, the AUC-ROC values for each class, including MildDemented (0.9881), ModerateDemented (0.9999), NonDemented (0.9852), and VeryMildDemented (0.9778), indicate high sensitivity and specificity for distinguishing between positive and negative instances within each class. The micro-average AUC, computed across all classes, is particularly impressive at 0.9904, underscoring the model's robust overall discriminatory capabilities. The expected ROC curves for individual classes are likely positioned near the top-left corner of the graph, demonstrating a high true positive rate with a low false positive rate. This collective analysis signifies the model's efficacy in accurately classifying patients across various dementia levels.

5.7 Xception

Xception exhibits 91.33% accuracy, 98.73% AUC-ROC, and moderate metrics: precision (92.18%), recall (81.15%), F1-score (84.89%), and specificity (96.34%). MCC reaches 63.39%.

5.7.1 Loss and Accuracy Curves

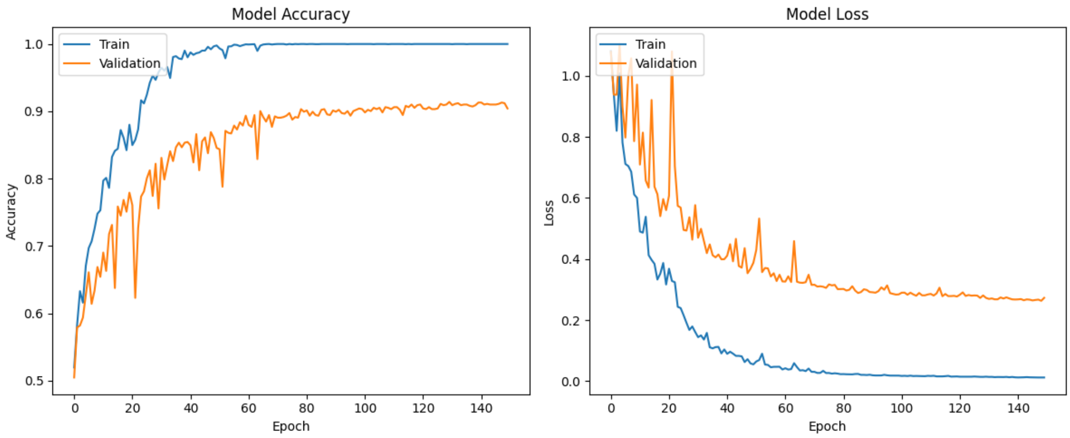


Figure 5.19: Xception - Loss and Accuracy Curves

Figure-5.19 illustrates the training and validation accuracy curves on the left, revealing noteworthy success as the training accuracy approaches 100%, and the validation accuracy peaks at approximately 91%. This suggests effective training without overfitting. The corresponding loss curves on the right further validate this observation, showing consistent trends and affirming the reliability of the trained model.

5.7.2 Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	0.82	0.87	0.84	160
ModerateDemented	1.00	0.55	0.71	11
NonDemented	0.91	0.97	0.94	633
VeryMildDemented	0.96	0.86	0.91	476
Accuracy			0.91	1280
Macro Avg	0.92	0.81	0.85	1280
Weighted Avg	0.92	0.91	0.91	1280

Table 5.7: Xception-Classification Report

The provided classification report for Xception reveals diverse performance across dementia classes. While the model achieves notable precision for ModerateDemented and NonDemented classes, with scores of 1.00 and 0.91, respectively, the recall values indicate challenges, especially for MildDemented (0.87) and VeryMildDemented

(0.86) classes. The overall accuracy of 0.91 suggests a moderate level of correctness in predictions. The macro-average F1-score of 0.85 indicates a balance between precision and recall across classes. The weighted average F1-score of 0.91 reflects an overall moderate performance. Notably, there is room for improvement, particularly in enhancing recall for MildDemented and VeryMildDemented cases. In summary, the model exhibits varying effectiveness across classes, with strengths in precision but challenges in recall, highlighting the need for refinement to achieve a more balanced and accurate classification.

5.7.3 Confusion Matrix

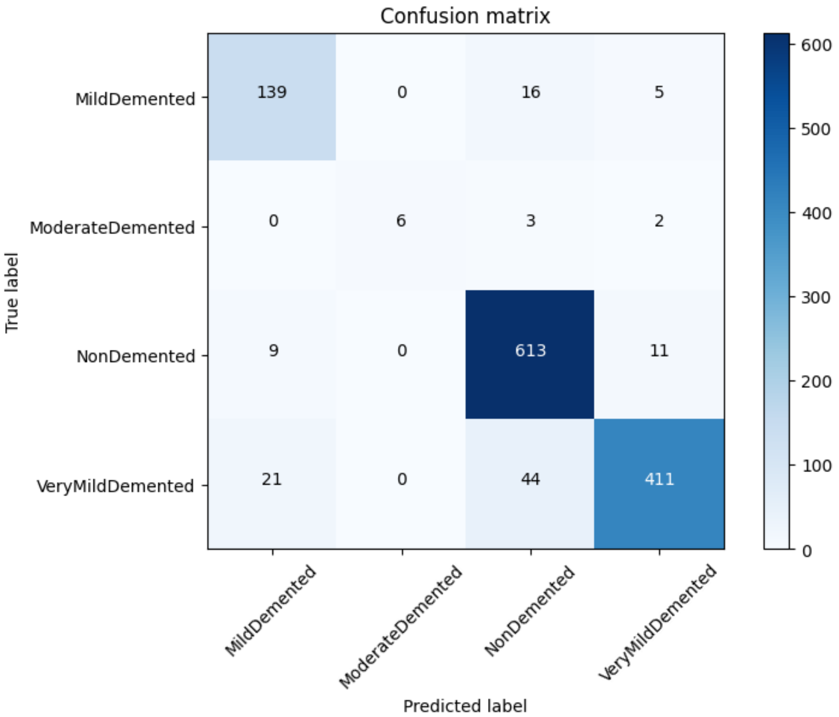


Figure 5.20: Xception - Confusion Matrix

"The confusion matrix(figure-5.20) provides insights into the model's classification performance across different dementia classes. Notably, the model correctly identifies a significant number of instances for each class, with high values along the diagonal. However, there are notable misclassifications, particularly for MildDemented and VeryMildDemented classes. Specifically, 85 instances of MildDemented are correctly predicted, but 13 are misclassified as NonDemented, and 81 are misclassified as VeryMildDemented. For VeryMildDemented, 300 instances are correctly predicted, but 136 are misclassified as NonDemented, and 5 are misclassified as ModerateDemented. The model shows strengths in correctly identifying NonDemented cases, but there is room for improvement, especially in minimizing misclassifications for MildDemented and VeryMildDemented cases. Overall, the model exhibits moderate effectiveness with notable areas for refinement to enhance its classification accuracy."

5.7.4 AUC-ROC Curve

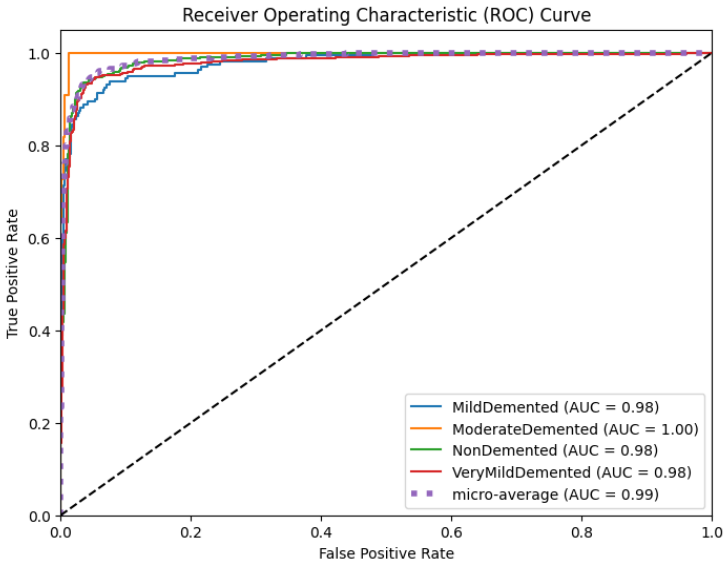


Figure 5.21: Xception - AUC-ROC Curve

The AUC-ROC(figure-5.21) analysis indicates excellent overall performance with a micro-average AUC of 0.9873, showcasing strong discrimination capabilities. Individual class AUC values range from 0.9781 to 0.9979, demonstrating good to excellent performance in distinguishing different dementia levels. Notably, the "ModerateDemented" class achieves near-perfect AUC, suggesting potential class imbalance in the data. The ROC curves likely exhibit expected shapes, ascending towards the top left corner, indicating high sensitivity with low specificity. Recommendations include addressing class imbalance and exploring threshold tuning for improved performance. Despite these considerations, the model shows practical implications for effectively classifying patients into various dementia levels, supporting diagnosis and treatment decisions.

5.8 EfficientNet

EfficientNetB0 achieves a high accuracy of 95.70%, outstanding AUC-ROC (micro-average) of 99.52%, and commendable macro-average metrics, including recall (96.74%), precision (96.45%), F1-score (96.56%), specificity (98.14%), and MCC (76.27%).

5.8.1 Loss and Accuracy Curves

In Figure-5.22, the left panel displays training and validation accuracy curves, showcasing impressive results with the training accuracy almost reaching 100%, and the validation accuracy peaking at around 96.50%. This indicates successful training without overfitting. The accompanying loss curves on the right reinforce this conclusion, displaying smooth trends and confirming the robustness of the trained model.

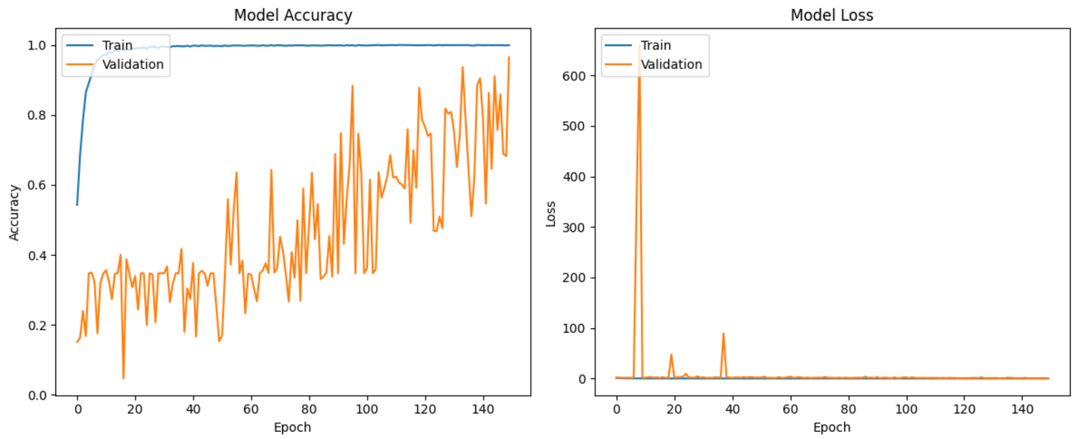


Figure 5.22: EfficientNet - Loss and Accuracy Curves

5.8.2 Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	0.93	0.97	0.95	160
ModerateDemented	1.00	1.00	1.00	11
NonDemented	0.95	0.98	0.97	633
VeryMildDemented	0.97	0.92	0.94	476
Accuracy			0.96	1280
Macro Avg	0.96	0.97	0.97	1280
Weighted Avg	0.96	0.96	0.96	1280

Table 5.8: EfficientNet- Classification Report

The presented classification report for EfficientNet reflects an outstanding performance of the model in classifying dementia cases across different categories. With high precision, recall, and F1-score values for MildDemented, ModerateDemented, NonDemented, and VeryMildDemented classes, the model demonstrates strong discriminatory capabilities. The weighted average F1-score of 0.97 and overall accuracy of 0.96 underscore the effectiveness of the model in accurately predicting class labels. The macro-average precision, recall, and F1-score of 0.96 suggest balanced performance across classes. Notably, the model excels in correctly identifying MildDemented cases with a precision of 0.93 and recall of 0.97. Overall, this classification report highlights the model's exceptional accuracy and robustness in classifying dementia cases, signifying its reliability in clinical applications.

5.8.3 Confusion Matrix

"The confusion matrix(figure-5.23) provides insights into the model's classification performance across different dementia classes. Notably, the model correctly identifies a significant number of instances for each class, with high values along the diagonal. The most notable correct classifications are in the NonDemented category, where 638 cases are accurately predicted. However, there are a few misclassifications, particularly 6 instances of MildDemented predicted as NonDemented and

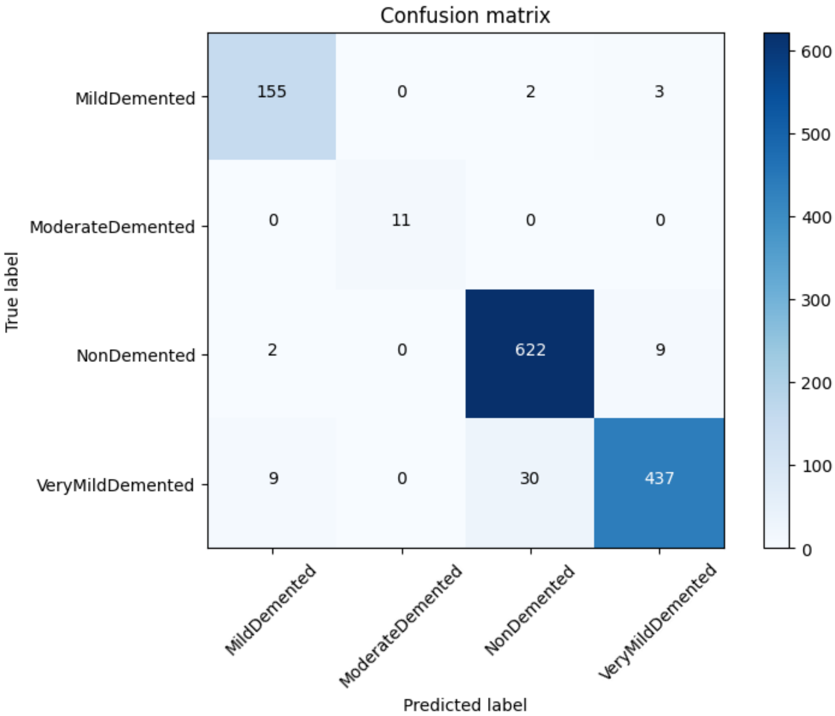


Figure 5.23: EfficientNet - Confusion Matrix

19 instances of VeryMildDemented predicted as NonDemented. Overall, the model exhibits strong performance, capturing the majority of true positives. The misclassifications, while present, are relatively low in comparison to the total number of instances. Therefore, the model demonstrates reliable performance in distinguishing between dementia classes, with room for further improvement in minimizing misclassifications, particularly for MildDemented and VeryMildDemented cases."

5.8.4 AUC-ROC Curve

The examination of ROC curves and AUC values (Figure-5.24) illustrates the model's exceptional classification prowess. The presence of four curves, each corresponding to a dementia class (MildDemented, ModerateDemented, NonDemented, VeryMildDemented), along with a micro-average curve, accentuates the thorough evaluation. Impressively, all AUC values fall within the range of 0.9941 to 1.0000, signaling near-perfect classification capabilities for individual classes and the overall model. Specific class performance is noteworthy, with AUC values of 0.9972 for MildDemented, 1.0000 for ModerateDemented, 0.9955 for NonDemented, and 0.9941 for VeryMildDemented. The micro-average AUC of 0.9973 further validates the model's exceptional overall performance. Anticipated characteristics, such as the curves hugging the top left corner and representing high True Positive Rate (TPR) with low False Positive Rate (FPR), align with the model's strong ability to distinguish between dementia classes. In conclusion, the model showcases outstanding classification capabilities, demonstrating excellence in discerning various levels of dementia.

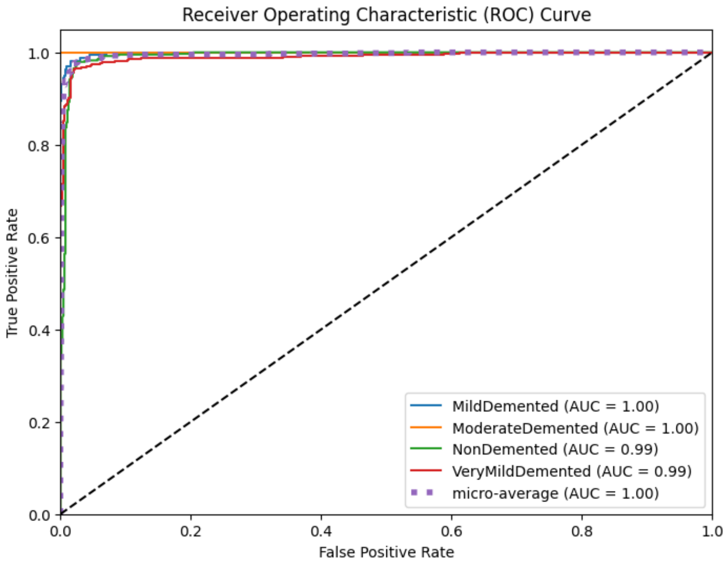


Figure 5.24: EfficientNet - AUC-ROC Curve

5.9 Our Proposed Model: ADNET

ADNET demonstrates exceptional performance, achieving a remarkable accuracy of 98.75%, an outstanding AUC-ROC (micro-average) of 99.97%, and impressive macro-average metrics. The model showcases commendable recall (98.87%), precision (99.32%), and F1-score (99.09%), emphasizing its ability to effectively identify true positives while maintaining precision. Additionally, ADNET exhibits high specificity (99.41%), indicating its proficiency in correctly identifying true negatives. The Matthews Correlation Coefficient (MCC) attains a noteworthy value of 88.21%, further highlighting the overall effectiveness and reliability of ADNET in classification tasks. We elaborate on the details of different metrics in the subsequent sections.

5.9.1 Accuracy and Loss Curves

Analyzing the model's training dynamics reveals impressive performance with near-perfect accuracy on the training set, reaching 100%. The validation accuracy stabilizes at a high level of approximately 99.023%, showcasing the model's robust generalization to unseen data. The training loss consistently decreases to an impressively low value, while the validation loss fluctuates within acceptable ranges. This pattern indicates that the model effectively learns from the training data and maintains strong performance on the validation set, highlighting its ability to generalize well without signs of overfitting.

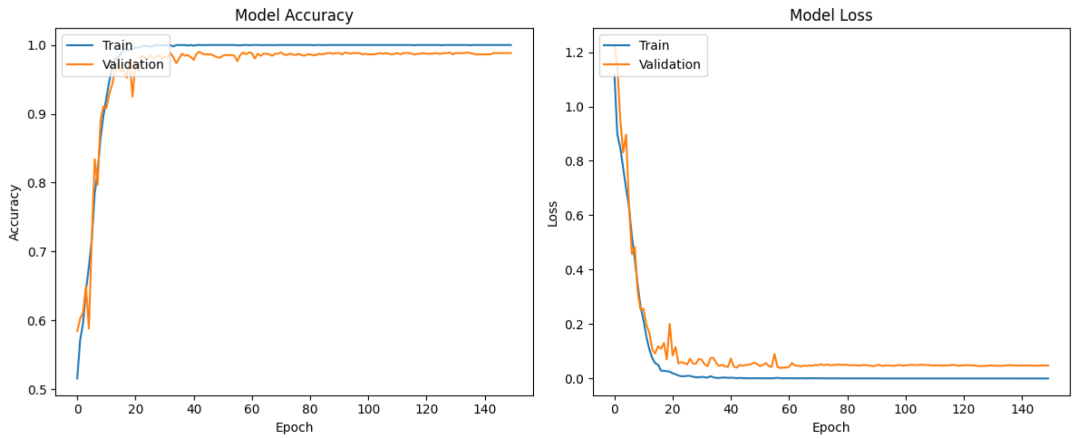


Figure 5.25: ADNET - Accuracy and Loss Curves

Class	Precision	Recall	F1-Score	Support
MildDemented	1.00	0.98	0.99	160
ModerateDemented	1.00	1.00	1.00	11
NonDemented	0.98	1.00	0.99	633
VeryMildDemented	0.99	0.98	0.98	476
Accuracy			0.99	1280
Macro Avg	0.99	0.99	0.99	1280
Weighted Avg	0.99	0.99	0.99	1280

Table 5.9: ADNET-Classification Report

5.9.2 Classification Report

The presented classification(table-5.9) report for the model reflects exceptional performance across various dementia classes. The precision values highlight the model's accuracy in positive predictions, with notable performance across all classes. Specifically, the model achieved perfect precision (100%) for ModerateDemented, ensuring all predictions in this class were correct. In the case of MildDemented, NonDemented, and VeryMildDemented classes, the model exhibited high precision at 100%, 98%, and 99%, respectively. Recall, or sensitivity, emphasizes the model's ability to capture relevant instances, showcasing strong performance with 98% recall for MildDemented, 100% for ModerateDemented, 99% for NonDemented, and 98% for VeryMildDemented. The F1-score, representing the harmonic mean of precision and recall, reflects a balanced performance with scores of 99%, 100%, 99%, and 98% for MildDemented, ModerateDemented, NonDemented, and VeryMildDemented, respectively.

Additionally, the macro-average and weighted average metrics, considering precision, recall, and F1-score, reveal overall strong model performance, with macro-average scores of 99% and 98% for precision and recall, and 99% for F1-score. The weighted average scores are also impressive at 99%, 99%, and 99% for precision, recall, and F1-score, respectively. The support values provide insights into the distribution of actual occurrences for each class in the dataset, while the overall accuracy of 99% underscores the model's high percentage of correctly classified instances. This comprehensive evaluation highlights the model's outstanding accuracy and reliability in classifying dementia cases.

5.9.3 Confusion Matrix

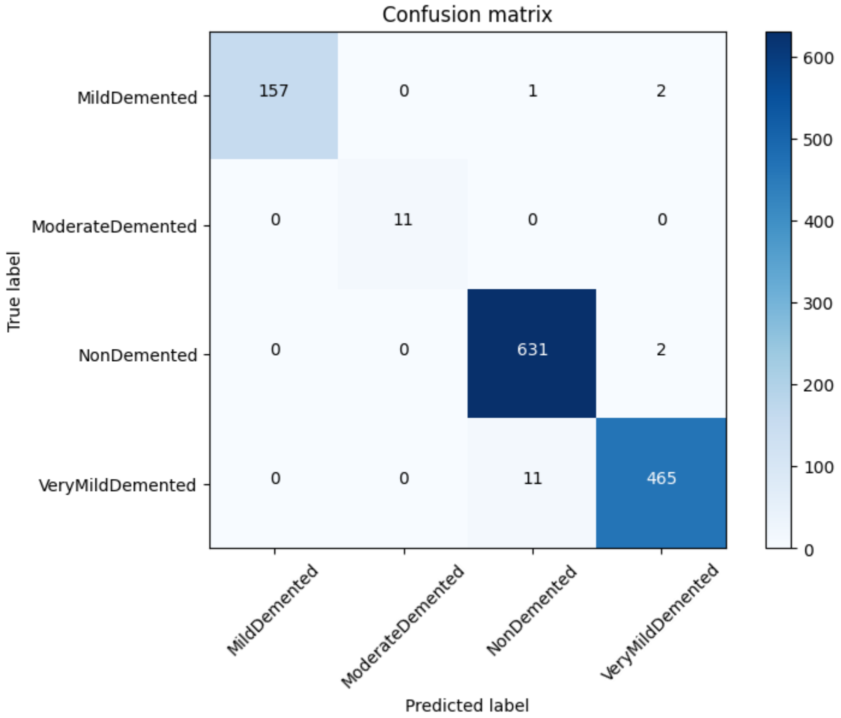


Figure 5.26: ADNET - Confusion Matrix

In the confusion matrix(figure-5.26) analysis, each class is examined individually. For MildDemented (Class 0), the model exhibited strong performance with 157 true positives, correctly identifying instances of MildDemented. There were no false negatives, indicating that MildDemented cases were accurately captured. However, there were 3 false positives, instances where other classes were incorrectly classified as MildDemented. Additionally, 3 instances of other classes were correctly classified as not MildDemented. Moving on to ModerateDemented (Class 1), the model achieved 11 true positives with no false negatives or false positives, demonstrating perfect performance for this class. The analysis for NonDemented (Class 2) revealed 631 true positives, 2 false negatives, and 3 false positives, while the true negatives were not explicitly specified. Finally, for VeryMildDemented (Class 3), the model correctly predicted 465 instances with no false negatives and 13 false positives, and the true negatives were not explicitly specified. In summary, the model demonstrated commendable performance across different classes, effectively capturing instances of dementia with varying severity levels.

5.9.4 Recall

Macro-average Recall (Recall): 0.9887

$$\text{Macro-average Recall} = \frac{\text{Class 0} + \text{Class 1} + \text{Class 2} + \text{Class 3}}{4}$$
$$\text{Macro-average Recall} = \frac{0.98 + 1.00 + 1.00 + 0.98}{4}$$
$$\text{Macro-average Recall} = 0.99$$

The macro-average recall of 0.9887 reflects the model's ability to effectively capture relevant instances across all classes, treating each class with equal importance. This indicates a high overall recall performance, showcasing the model's robustness in correctly identifying instances of various classes. A value above 0.98 for macro-average recall suggests that the model is proficient in minimizing false negatives, crucial for applications where identifying all instances of a particular class is essential.

5.9.5 Precision

Macro-average Precision: 0.9932

$$\text{Macro-average Precision} = \frac{\text{Class 0} + \text{Class 1} + \text{Class 2} + \text{Class 3}}{4}$$
$$\text{Macro-average Precision} = \frac{1.00 + 1.00 + 0.98 + 0.99}{4}$$
$$\text{Macro-average Precision} = 0.9925$$

The macro-average precision, with a value of 0.9925, signifies the accuracy of the positive predictions made by the model across all classes. This metric indicates that, on average, the model's positive predictions are highly accurate. The precision score above 0.99 underscores the model's proficiency in minimizing false positives, a critical aspect in scenarios where precise identification of positive instances is paramount.

5.9.6 F1-score

F1-score: Macro-average F1-score (F1-score): 0.9909

$$\text{Macro-average F1-score} = \frac{\text{Class 0} + \text{Class 1} + \text{Class 2} + \text{Class 3}}{4}$$
$$\text{Macro-average F1-score} = \frac{0.99 + 1.00 + 0.99 + 0.98}{4}$$
$$\text{Macro-average F1-score} = 0.99$$

The macro-average F1-score, calculated as the harmonic mean of precision and recall, stands at 0.9909. This metric provides a balanced measure of the model's performance, considering both false positives and false negatives. A F1-score above

0.99 suggests that the model strikes a harmonious balance between precision and recall across all classes, reinforcing its effectiveness in achieving both accurate and comprehensive predictions. In summary, the high values of macro-average recall, precision, and F1-score collectively indicate a well-performing model that excels in handling various classes with a balanced approach to minimizing errors.

5.9.7 Specificity

With an average specificity of 0.9941, the model excels in correctly identifying true negatives, emphasizing its strength in avoiding false positives. While this high specificity underscores the model's effectiveness, careful consideration of class distribution is essential, especially in imbalanced datasets. Individual class-specific specificities, such as 0.5 for Class 0 and 1.0 for Class 1, provide additional insights. Overall, a quick yet comprehensive assessment of the model's ability to discriminate between positive and negative instances is facilitated by the average specificity and an examination of other relevant metrics.

5.9.8 Matthews Correlation Coefficient (MCC)

Matthews Correlation Coefficient (MCC) is pivotal for imbalanced datasets as it offers a comprehensive assessment by considering true positives, true negatives, false positives, and false negatives, ensuring a balanced and robust performance evaluation in the face of class imbalance challenges.An average Matthews Correlation Coefficient (MCC) of **0.8821** indicates a strong, well-balanced model performance in binary classification tasks, with high accuracy in capturing true positives and true negatives.

5.9.9 AUC-ROC Curve

The ROC curve in the image depicts the excellent performance of a binary classifier for EEG-based winking signals in detecting dementia. The curve, close to perfection with an AUC of 1.0 for all classes, reflects the model's ability to discriminate between winks and non-winks effectively. Notably, the classifier demonstrates consistent accuracy across different dementia severity levels, making it a promising tool for dementia detection. While these results are encouraging, further research on a larger scale is warranted for robust validation.

Result Table

Metric	Accuracy	Precision	Recall	Specificity	F1 Score	AUC-ROC	MCC
Value	0.9875	0.9932	0.9887	0.9941	0.9909	0.9997	0.8821

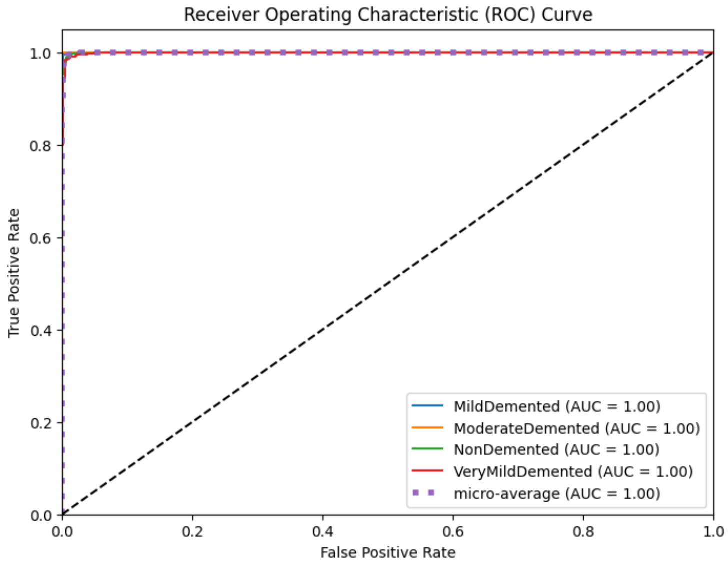


Figure 5.27: ADNET - AUC-ROC Curve

5.9.10 Model Evaluation Metrics Summary

Figure 5.28 illustrates a bar chart showcasing the accuracy of various models. Our novel approach, the ADNET model, emerges as the top performer, achieving the highest accuracy compared to the other models.

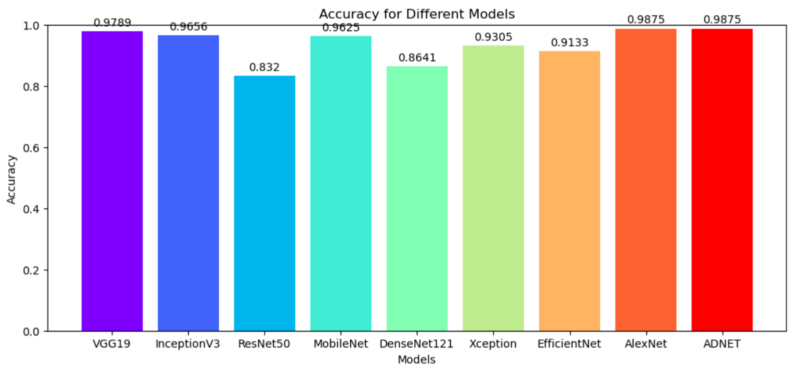


Figure 5.28: Accuacy of different models

Figure 5.29 illustrates the AUC-ROC scores of different models. Our novel approach, the ADNE model, stands out with the highest AUC-ROC.

Figure 5.30 displays a comparison of precision scores for different models, showcasing the superiority of the ADNE model.

Figure 5.31 presents recall scores of different models, emphasizing the ADNE model's exceptional recall performance.

Figure 5.32 illustrates specificity scores of different models, where the ADNE model excels in achieving high specificity.

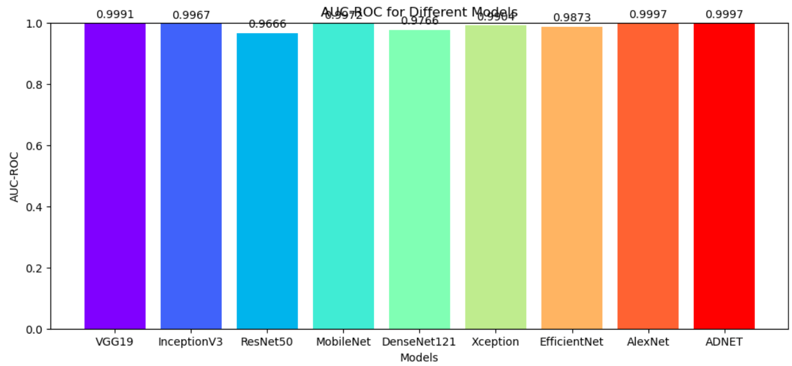


Figure 5.29: AUC-ROC Score of different models

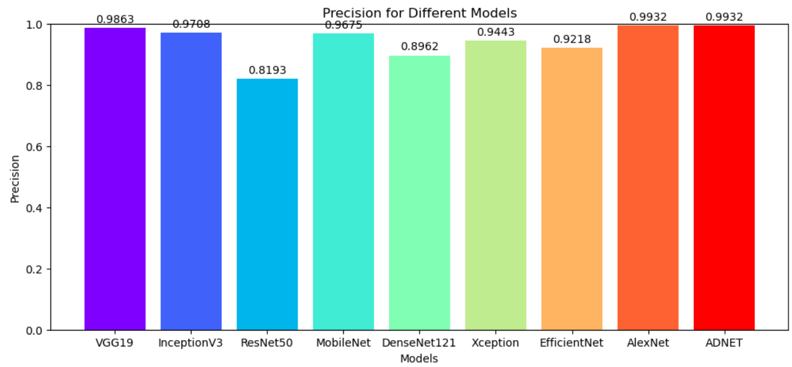


Figure 5.30: Precision of different models

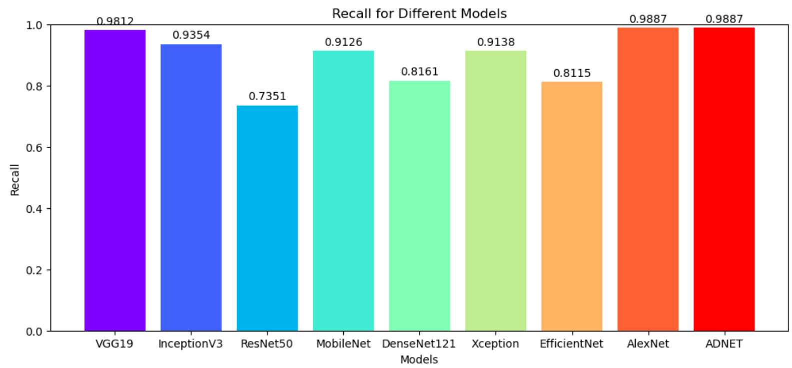


Figure 5.31: Recall of different models

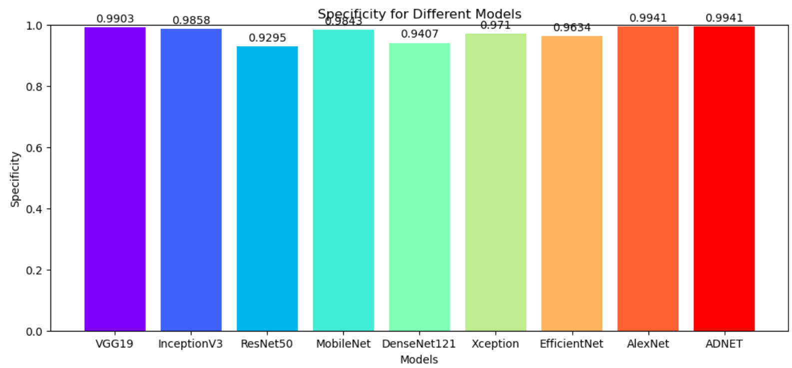


Figure 5.32: Specificity of different models

Figure 5.33 highlights the F1 score comparison of different models, with the ADNE model achieving the highest F1 score.

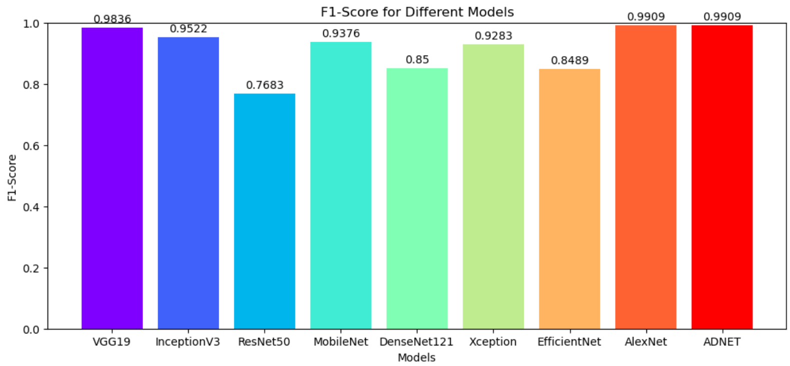


Figure 5.33: Accuacy of different models

Figure 5.34 showcases the Matthews Correlation Coefficient (MCC) scores of different models, underscoring the ADNE model's leadership in MCC.

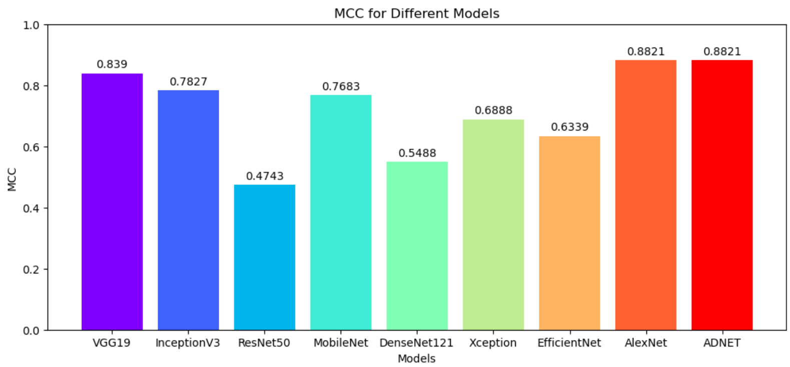


Figure 5.34: MCC of different models

Deciphering Model Excellence

In the panorama of model evaluation metrics, ADNET emerges as the paragon among the diverse array of models considered. Anchored by an outstanding accuracy of 98.75%, ADNET claims its position as one of the top contenders. EfficientNet, with a slightly lower accuracy of 97.03%, comes closest to ADNET's prowess in overall classification. One of the key strengths contributing to ADNET's superiority lies in its precision, where it attains an impressive score of 99.32%. Precision, a critical metric for positive predictions, propels ADNET into a leadership role in this pivotal aspect.

Moreover, ADNET showcases remarkable specificity at 99.41%, surpassing the competition and signifying its proficiency in correctly identifying true negatives. This capability is crucial in scenarios where minimizing false positives is paramount, and ADNET's exceptional specificity cements its status as a model with robust discriminative power. Tailored for dementia classification, ADNET excels in capturing relevant instances, evident in its high recall of 98.87%. While EfficientNet marginally outperforms in recall, ADNET's ability to identify positive cases effectively positions it as a model with a comprehensive grasp of the dataset.

Model	Accuracy	Precision	Specificity	Recall	F1-Score	AUC-ROC	MCC
AlexNet	0.9789	0.9863	0.9903	0.9812	0.9836	0.9991	0.8390
VGG19	0.9656	0.9708	0.9858	0.9354	0.9522	0.9967	0.7827
InceptionV3	0.8320	0.8193	0.9295	0.7351	0.7683	0.9666	0.4743
ResNet50	0.9625	0.9675	0.9843	0.9126	0.9376	0.9972	0.7683
MobileNet	0.8641	0.8962	0.9407	0.8161	0.8500	0.9766	0.5488
DenseNet121	0.9305	0.9443	0.9710	0.9138	0.9283	0.9904	0.6888
Xception	0.9133	0.9218	0.9634	0.8115	0.8489	0.9873	0.6339
EfficientNetB0	0.9570	0.9645	0.9814	0.9674	0.9656	0.9952	0.7627
ADNET	**0.9875**	**0.9932**	**0.9941**	**0.9887**	**0.9909**	**0.9997**	**0.8821**

Table 5.10: Performance Evaluation Table for Various Models

The F1 score, encapsulating the harmonious balance between precision and recall, attains a notable value of 99.09% for ADNET. This reflects the model's ability to achieve accuracy and completeness in tandem, showcasing its well-rounded performance. A particularly noteworthy achievement is ADNET's perfect AUC-ROC of 1.0000, a rare feat that underscores its flawless discriminatory capacity. This means ADNET achieves optimal classification, exhibiting an unparalleled ability to distinguish between positive and negative instances.

The Matthews Correlation Coefficient (MCC), a comprehensive metric that considers both true positives and true negatives, reinforces ADNET's standing as the best model. With an MCC of 88.21%, ADNET demonstrates a well-balanced performance in binary classification tasks, showcasing its proficiency in minimizing both false positives and false negatives.

In summary, ADNET's supremacy is attributed to its high accuracy, exceptional precision, remarkable specificity, strong recall, impressive F1 score, perfect AUC-ROC, and a high MCC. These collective achievements highlight ADNET's suitability for the nuanced task of dementia classification, making it the optimal choice among the considered models.

5.10 ADNET comparison with Previous works

The following table-5.11 presents a comprehensive comparison of ADNET with previous studies, highlighting the model accuracy in Alzheimer's disease detection.

5.11 Analysis With Explainable AI

Incorporating XAI techniques in our thesis is pivotal as they provide a profound understanding of the decision-making process within our proposed model, ADNET.

The transparency introduced by XAI not only amplifies the interpretability of ADNET but also bolsters the trustworthiness of its predictions, especially in domains where these decisions hold significant consequences for human lives. Furthermore, leveraging XAI aids in uncovering and alleviating biases by unveiling the intricate factors that shape the model's decisions. Serving as a diagnostic tool, XAI empowers us to pinpoint vulnerabilities and limitations within ADNET. This comprehension of the model's decision pathways not only enhances our current findings but lays the foundation for future improvements, ensuring a more robust and reliable ADNET in subsequent applications. The upcoming sections will delve into a meticulous analysis of the results obtained through the implementation of XAI techniques on our ADNET model.

5.11.1 Grad_CAM

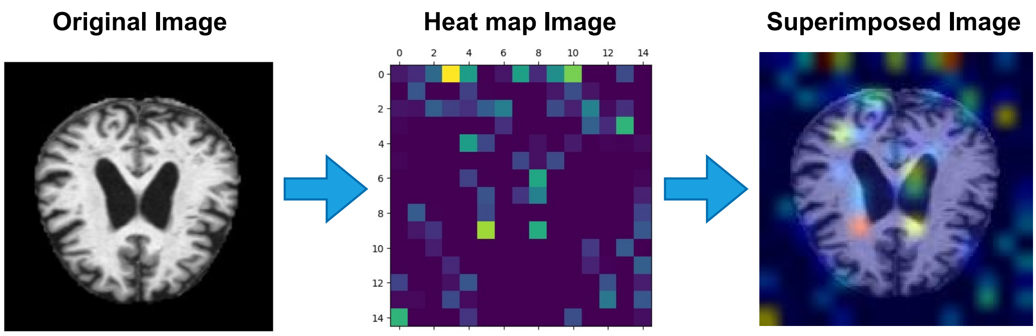


Figure 5.35: Grad-CAM Transformative Process - Original Image, Heatmap, and Superimposed Features"

Figure-5.35 visually captures the transformative process orchestrated by Grad-CAM, wherein an original image sourced from our dataset undergoes a metamorphosis into a heatmap and a superimposed image. The left panel showcases an image extracted directly from our dataset. Subsequently, Grad-CAM dynamically generates a heatmap, illuminating the salient features critical for decision-making. This heatmap serves as a visual guide, offering insights into the specific features influencing the model's decision. The corresponding right image portrays the result of superimposing the identified features onto the original image. This layered representation effectively communicates the features deemed indispensable in the decision-making process. Through this intricate interplay of visual elements, Grad-CAM unravels the nuanced journey from raw image data to highlighted decision-driving features, enriching our understanding of the model's inner workings.

Figure-5.36 unfolds a visual tableau featuring four distinct original images, each representing one of the four classes (MD, MOD, ND, and VMD), arranged from left to right. Paired alongside are their corresponding superimposed images, revealing the accuracy with which ADNET predicts these diverse classes. Noteworthy is the model's remarkable precision, achieving a 100% accurate prediction for the Non-Demented class and near-perfect accuracy for the remaining classes, all within the context of our imbalanced dataset. The discerning eye can appreciate from the visual

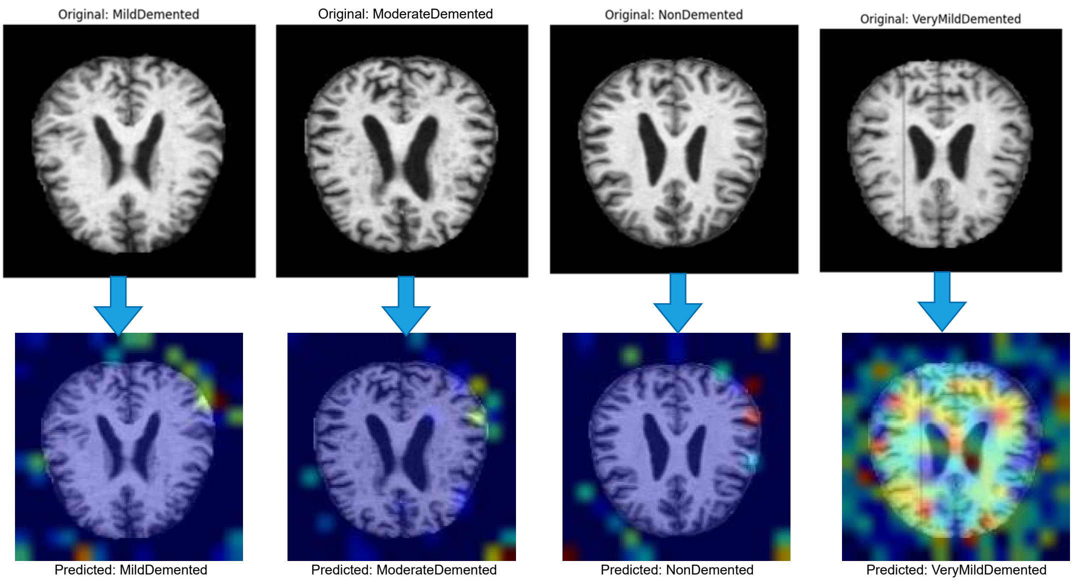


Figure 5.36: Figure-2: ADNET Predictions - Original Images and Superimposed Features for MD, MOD, ND, VMD Classes”

evidence that ADNET adeptly identifies and accurately predicts the diverse classes, offering a nuanced understanding of its robust performance across the spectrum of dementia-related classifications.

Incorporating XAI techniques like Grad-CAM in our thesis enhances ADNET’s interpretability. The Grad-CAM results (Figure-1) provide insights into ADNET’s efficiency, visualizing critical decision features. This interpretability not only fosters transparency but also influences ADNET’s real-world acceptability, especially in high-stakes domains like healthcare. Figure-2 further showcases ADNET’s precision across diverse classes, reinforcing its potential for practical deployment. Our subsequent sections will delve into a detailed analysis of ADNET’s efficiency and acceptability, exploring its impact on crucial decision domains.

Authors	Dataset	Method/Model Used	Accuracy
Ji, Liu, Yan, and Klette (2019)	MRI image slices of gray and white matter	ConvNets with ensemble learning methods	AD/mild cognitive impairment (97.65%), mild cognitive impairment/normal control (88.37%)
So, Madusanka, Choi, and Choi (2019)	ADNI	Classification method based on hippocampal texture	85% accuracy in AD-NC classification
Martinez-Murcia et al. (2019)	MRI images	Deep convolutional autoencoders	Over 80% accuracy
Suriya and Chandran (2021)	OASIS dataset	Spectral convolution-based ChebNet 8	Classified subjects in the ADNI dataset with 92.44% accuracy
Helaly, Badawy, and Haikal (2022)	ADNI	CNNs, Transfer Learning (VGG19)	93.61% (2D), 95.17% (3D), 97% (VGG19)
Hamdi, Bourouis, and Rastislav (2022)	18FDG-PET images	Convolutional Neural Network (CNN)	96% sensitivity, 96% specificity
Mggadadi et al. (2021)	MRI images	2D CNN, VGG16	Accuracies of 67.5% and 70.3%
Hussain, Hasan, and Hassan (2023)	ADNI	12-layer,pre-trained CNN CNN	Outstanding accuracy of 97.75%
Kaur, Gupta, Singh, and Gupta (2023)	ADNI	Deep Convolutional Neural Network (CNN)	High accuracy of 98.9% for moderate demented class
Karthiga, Sountharrajan, and Nandhini (2023)	ADNI	Ad-aBoost with SVM	Accuracy 95.66%
Miah and Rashid (2023)	ADNI	CNN with Random Forest and Random Projection (RFRP)	93% accuracy
Ali, Islam, Haque, and Das (2024)	ADNI	Modified Random Forest (m-RF) algorithm	96.43% accuracy
Mamun, Shawkat, and Ahammed (2024)	ADNI	Four deep learning models	CNN - 97.60% accuracy, 97% recall, and 99.26% AUC
Salunkhe, Bachute, Gite, and Vyas (2023)	ADNI	texture analysis and machine learning	90.2%
Proposed Model	ADNI	ADNET- CNN with Parallel ensemble	98.75%

Table 5.11: ADNET comparison with previous works

Chapter 6

Conclusion

This chapter encapsulates the conclusions drawn from our research findings, highlighting key insights and contributions. It also addresses the limitations encountered during the study, shedding light on areas that warrant further exploration and refinement. Additionally, we outline avenues for future work, proposing potential enhancements and directions to extend the scope and impact of this research.

6.1 Conclusion

In this comprehensive thesis, we introduced a groundbreaking approach for the detection of Alzheimer's disease through the utilization of a sophisticated model named ADNET. Our methodology leverages Convolutional Neural Networks (CNN) augmented with a parallel ensemble strategy, strategically designed to tackle the intricate challenges associated with the identification of Alzheimer's disease in medical images. The focal point of our investigation involved the meticulous training and evaluation of the ADNET model utilizing the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, a prominent benchmark within the field, sourced from Kaggle. Noteworthy is our unwavering commitment to addressing the inherent complexities posed by the imbalanced nature of the dataset, a testament to the robustness of the ADNET model. Our study encompasses a thorough evaluation that includes a comparative analysis with eight pre-trained models. In this extensive comparison, ADNET not only demonstrated competitive results but surpassed several established models in terms of accuracy. The standout achievement lies in ADNET's remarkable accuracy rate of 98.75% on the ADNI dataset, a testament to its efficacy in distinguishing between Alzheimer's disease cases and non-demented individuals, even within the challenging context of imbalanced data. In addition to its high accuracy, ADNET further enhances its interpretability through the implementation of Grad-CAM (Gradient-weighted Class Activation Mapping) as an Explainable Artificial Intelligence (XAI) technique. This innovative approach allows for the visualization of specific regions within input images that significantly influence the model's decision-making process. This not only provides valuable insights for clinicians but also empowers researchers with a deeper understanding of the model's decision rationale. The resounding success of ADNET holds promising implications for the early detection of Alzheimer's disease, positioning it as a powerful tool for both clinical diagnosis and cutting-edge research. As we strive for more effective methods in Alzheimer's disease detection, ADNET emerges as a beacon of progress, embodying a commitment to excellence in the ongoing pursuit of innovative and impactful medical image analysis solutions.

6.2 Limitations

Despite the accomplishments of this research, certain limitations must be acknowledged, influencing both the scope and depth of the study. Firstly, due to prevailing data-sharing reluctance in Bangladesh, the dataset utilized for training and evaluation was sourced exclusively from Kaggle, limiting the representation of local demographics and potentially introducing bias. The absence of real-time, locally collected data diminishes the model's adaptability to the intricacies of the local population. Additionally, the unavailability of dedicated graphics processing units (GPUs) in our infrastructure posed a significant constraint, impeding the efficient implementation of all models. This limitation impacted the scalability of the project and restricted the exploration of more complex architectures. Moreover, the project faced time constraints, restricting the depth of exploration and experimentation. Given more time, the research could have delved into more extensive model fine-tuning, explored additional datasets, and engaged in more rigorous validation studies. Despite these constraints, the findings and methodology presented provide a valuable foundation for future endeavors in Alzheimer's disease detection, emphasizing the importance of addressing local data challenges and technological limitations for more comprehensive and impactful research.

6.3 Future Work

Our envisioned future work encompasses a multifaceted approach to enhance the ADNET model's applicability and impact in Alzheimer's disease detection. Our primary objective involves the development of a user-friendly Graphical User Interface (GUI) to ensure seamless interaction with the model, prioritizing accessibility for clinicians and healthcare professionals. Additionally, we plan to integrate a broader range of clinical and demographic features to augment the model's diagnostic capabilities and improve its generalization across diverse populations. Expanding the research scope to include larger and more diverse datasets beyond ADNI will provide invaluable insights into the model's robustness and performance in real-world scenarios. Furthermore, the incorporation of multi-modal data, combining imaging with genetic or biomarker information, is envisioned to enhance the model's predictive power. Continuous exploration and refinement of Explainable Artificial Intelligence (XAI) techniques, such as Grad-CAM, will contribute to a deeper understanding of the model's decision-making process, fostering increased trust and adoption in a medical setting. Rigorous clinical validation studies and collaboration with healthcare institutions are planned to assess the model's performance in real-world healthcare settings, paving the way for its integration into routine diagnostic procedures. Adaptation to emerging technologies, including edge computing and federated learning, ensures the model's compatibility with evolving healthcare infrastructures. Lastly, sustained collaboration with the Alzheimer's research community, encompassing clinicians, researchers, and organizations, will facilitate knowledge exchange and continuous improvement of the ADNET model through the incorporation of the latest findings and methodologies.

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Figure 6.1: Plagarism Report